

Ocular biomarker profiling after complement factor I gene therapy in geographic atrophy secondary to age-related macular degeneration

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Abstract

Objective

Complement biomarker analysis in ocular fluid samples from subjects with geographic atrophy (GA) in a Phase I/II clinical trial of subretinal AAV2 complement factor I (*CFI*; FI) gene therapy, PPY988 (formerly GT005), to understand target pharmacokinetics/pharmacodynamics. Clinical findings were subsequently utilized to investigate the therapeutic dose in an *in vitro* complement activation assay.

Design, setting and participants

Biomarker data were evaluated from 28 subjects in FOCUS, a Phase I/II clinical trial evaluating the safety and efficacy of three ascending doses of PPY988.

Main outcomes and measures

Vitreous humor (VH), and aqueous humor (AH) from subjects before surgery and at serial timepoints (week 5 or 12, 36, 96) were evaluated for changes in levels of intact complement

factors I, B and H (FI, FB, FH) components C3, C4, and C1q and breakdown products (Ba, C3a, C3b/iC3b, C4b) using validated assays and OLINK[®] proteomics.

A modified *in vitro* assay of complement activation modelling VH complement concentrations was used to compare PPY988 potency to the approved intravitreal C3 inhibitor pegcetacoplan (Apellis) and complement Factor H (FH).

Results

An average 2-fold increase in VH FI was observed post-treatment at week 36 and week 96. This correlated with a marked post-treatment reduction in VH concentration of the FB breakdown product Ba and Ba:FB ratio, but minimal changes in C3a and C3b/iC3b levels. Variable concordance in complement biomarker levels in VH versus AH suggest AH is not a reliable proxy for VH for complement activation. During the experimental comparison of doses, a 2-fold increase of FI achieved in the vitreous had only a minor effect on the complement amplification loop *in vitro*, indicating limited impact [IC50: 1229nM]. Pegcetacoplan completely blocks C3a generation at concentrations much lower than the estimated trough level for monthly intravitreal injections [IC50: 2nM]. Supplementation with FH in the assay revealed similar potency to pegcetacoplan [IC50: 6nM].

Conclusions and relevance

PPY988 subretinal gene therapy may not have provided sufficient FI protein to meaningfully modulate complement activation to slow GA growth. Reviewing VH biomarkers is important for understanding target expression, pathway engagement, and determining optimal dose, thereby informing future clinical development.

eLife assessment

This **important** work advances our understanding of factors influencing efficacy assessments and biomarker viability for complement-directed gene therapy against age-related macular degeneration. The data presented is **convincing** and offers insights and teachings for the design of gene therapy and complement-targeted therapeutics in the eye and more broadly for future ocular biomarker studies.

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Introduction

Age-related macular degeneration (AMD) is a leading cause of blindness, particularly in developed countries. In the United States (US) alone, there are approximately 18 million individuals aged 40 and above with early-stage AMD, and 1.5 million with late-stage AMD, which includes geographic atrophy (GA) and/or neovascular AMD¹. GA is a condition characterized by the progressive loss of photoreceptors, retinal pigment epithelium (RPE), and underlying choriocapillaris, resulting in a central scotoma². Reduced vision can significantly impact activities of daily living and quality of life^{3,4}. There have been two recent Food and Drug Administration (FDA)-approved therapies that inhibit complement activity, with a modest slowing of GA growth^{5,6}. However,

they require life-long monthly or bimonthly intravitreal (IVT) injections and have so far exhibited limited functional efficacy that present challenges for long term compliance⁷. Therefore, a one-time gene therapy that provides long-lasting efficacy equivalent to or better than repeated IVT therapies is a promising avenue for exploration.

One such gene therapy, called PPY998 (formerly GT005), is a subretinal gene therapy that supplements complement factor I (*CFI*; FI) protein in the retinal layers where atrophy occurs⁸. A Phase II study (HORIZON, NCT04566445) was initiated, however a decision to stop the PPY998 program was taken by Novartis based on a recommendation from the independent Data Monitoring Committee following an overall benefit-risk assessment of available data from the program studies. [Press release: Novartis AG; September 11th 2023; <https://www.novartis.com/news/gt005-ppy988-development-program-geographic-atrophy>]. This raises important questions about whether *CFI* was sufficiently expressed after subretinal gene therapy, whether there was evidence of modulation of intraocular complement activity, or if this modulation was sufficient to potentially alter the disease course. Such information is crucial to the retina community, as there are many companies developing complement modulating therapies for AMD⁹. Sharing the learnings from the PPY998 program would improve the risk-benefit analysis for these programs and help understand if new approaches differentiate from PPY998, justifying further clinical study of complement targets for GA.

The complement system, specifically the alternative pathway (AP), plays an important role in the development of AMD, and increasing local FI concentration could be a promising approach to regulate ocular complement activity. The AP is a part of the complement system that constantly surveys for foreign cells and debris and requires tight regulation to prevent inflammation and self-cell damage¹⁰. Regulatory proteins like factor H (FH), decay-accelerating factor (DAF; CD55), membrane cofactor protein (MCP; CD46), complement receptor 1 (CR1; CD35), and FI help control the AP, with FI being the only regulatory enzyme¹¹. FI, expressed from the *CFI* gene, cleaves C3b by forming a trimolecular complex with cofactors FH for C3b, and MCP and CR1 for C3b and C4b^{12–18}. Studies have shown that carriers of rare variants (RV) in *CFI* have increased risk of developing advanced AMD and more rapid disease progression^{19–23}. *CFI* RV can result in low levels of serum FI (Type I variants) or normal levels with functional impairment (Type II variants)^{24,25}. Research has also suggested that doubling serum FI levels may help control complement overactivation driven by other AMD risk genetic factors mapping to complement genes²⁶. Based on these findings, clinical trials were conducted to explore the potential of sub-retinal delivery of AAV2-*CFI* as a gene therapy for treating dry AMD⁸.

The FOCUS study is a Phase I/II open-label study evaluating the safety, and dose response of three doses of PPY998 AAV2-*CFI* subretinal gene therapy in GA. To assess transgene pharmacokinetics and pharmacodynamics, vitreous humor (VH) and aqueous humor (AH) samples were collected at baseline (BL) and four time points up to 96 weeks after dosing and were evaluated for exploratory complement biomarkers. This report shares findings regarding change from baseline in FI and complement protein levels and activation, over time after vitrectomy and PPY998 administration. Levels of FI achieved in VH from FOCUS subjects were then used in an *in vitro* simulated human vitreous complement activation assay, to gain insights on the therapeutic potencies of FI, C3 inhibitor pegcetacoplan (Syfovre, Apellis), and FH. We also offer critical learnings for other AMD complement gene therapy programs and a guide for future analyses of complement biomarker data in ocular samples.

Methods

Ethics

The FOCUS study was conducted in accordance with the tenets of the Declaration of Helsinki and all other local regulations, in accordance with the International Conference on Harmonization E6 Guidelines for Good Clinical Practice and with applicable local, state, and federal laws.

Institutional review board or ethics committee approval was obtained, and the study was registered at [ClinicalTrials.gov](https://clinicaltrials.gov) (reference: NCT03846193) and EudraCT (reference: 2017-003712-39). All subjects provided written informed consent to participate, and an independent data monitoring committee provided ongoing oversight.

Study design

FOCUS is an open-label first in human Phase I/II multicentre clinical trial to evaluate the safety, and dose response of PPY988 (AAV2-*CFI*)⁸, administered as a single subretinal injection in subjects with GA secondary to AMD, followed for up to 240 weeks (5 years) post-dosing. Subjects were administered one of three doses to the study eye under local anaesthesia by a qualified vitreoretinal surgeon: 2E10 viral genome (vg; low), 5eE10 vg (medium) and 2E11 vg (high). At screening, a saliva sample was taken from subjects and extracted DNA was analysed for *CFI* RV using targeted next generation sequencing²¹. Genotyping results were not required at screening if already available through participation in a previous Gyroscope sponsored study. The term ‘rare’ was defined as variants that changed the *CFI* coding sequence, had $\leq 1\%$ minor allele frequency in GnomAD non-Finnish Europeans (V2.1.1, accessed <http://gnomad.broadinstitute.org>), and were previously associated with FI haploinsufficiency¹¹. A detailed description of study design, route of administration, dose, number of subjects, biomarker matrices, sampling timing and methodology are provided in Supplementary Methods and **Supplementary Table 1**.

Complement protein and total protein quantification

AH and VH samples of subjects were taken on the treated eye immediately prior to subretinal surgery at baseline (BL) and at 5 or 12 weeks, 36, and 96 weeks post-treatment, as described in detail in **Supplementary Table 1**. Samples were rapidly frozen at -80°C and transported frozen for analysis. Prior to analysis, samples were thawed and subjected to centrifugation at 17,000g for 10 minutes at 4°C . Samples were kept on ice during sample preparation.

Complement factors C2, C4b, C5, C5a, factor D, mannose binding lectin (MBL), FI, C1q, C3b/iC3b, C3, FH, FB and C4 were quantified in AH and VH using the MILLIPLEX Human Complement Panel 1 and 2 (cat. no. HCMP1MAG-19K and HCMP2MAG-19K, EMD Millipore Corporation, Billerica, MA) following extensive quality control and validation of calibration curves. Plates were read using the Luminex MagPix and xPONENT software (Luminex Corporation, Austin, TX). C5a was omitted from further analysis as measurements fell under the lower limit for quantification.

Complement factors C3a and Ba were quantified in AH and VH using MicroVue EIA kits (cat. no. A032 and A034, QuidelOrtho Corporation, San Diego, CA) following manufacturer instructions. Plates were read using a Tecan Sunrise absorbance microplate reader using the Magellan Tracker data analysis software (Tecan Trading AG, Switzerland).

Each assay was validated and considered fit for purpose. Sample processing and analyses were conducted in accordance with Good Clinical Laboratory Practice, the analytes or measurements that did not meet acceptance criteria (coefficient of variation of 20%, or 30% when results are below the lower limit of quantification but above limit of detection) were excluded from the final analysis. The number of successfully sampled subjects for each biomarker at each time-point is reported in **Supplementary Table 2**.

Total protein was quantified in AH and VH using the Quant-iT protein assay kit (cat. no. Q33210, Invitrogen, Thermo Fisher Scientific, Waltham, MA) as per the manufacturer's instructions. Plates were read using a Varioskan LUX fluorescent microplate reader using the SkanIt data acquisition software (Thermo Fisher Scientific, Waltham, MA).

***In vitro* modelling the ocular complement system and potency assay**

The assay was performed using reagents from the Wieslab AP assay kit (cat. no. COMPLAP330, Svar Life Science AB, Sweden) and purified human serum complement proteins (Complement Technology Inc., Texas USA). All preparation steps were performed on ice. Serial dilutions of the test analytes FI, FH and pegcetacoplan were separately prepared and added to a mix of the complement components C3, FB, FH, FI and Properdin simulating the final concentrations measured in the VH baseline of FOCUS subjects (*CFI* RV negative population). The simulated vitreous complement mix was transferred on the lipopolysaccharide (LPS) coated Wieslab plate on ice. FD was separately added to the mix, the plate was immediately put at 37°C and incubated for 1 hour. At the end of the incubation, the solution was transferred into a dilution plate and kept on ice, EDTA was added to each well to obtain a final 20mM concentration. Ba and C3a generated in the assay were measured by using the MicroVUE EIA kits (cat. no. A032 and A034, QuidelOrtho Corporation, San Diego, CA) following the manufacturer instructions. The ELISA plates were scanned using the Varioskan plate reader and analysed using the SkanIT software (Thermo Fisher Scientific, Waltham, MA). The *in vitro* FI potency comparability assay is described in Supplementary Methods. The trough level of pegcetacoplan in the VH after 27 days was calculated by halving the treatment dose of 15mg/eye 6 times (pegcetacoplan half-life 4.5 days), the number obtained was converted to µg/mL using an estimated volume of VH of 4 mL ²⁷ [↗](#).

Statistics

All subject sample biomarker analyses were performed in R version 4.1.0. The analysis set consisted of all subjects who received subretinal delivery of PPY998 from cohorts 1 to 4 and had at least one non-missing biomarker assessment performed at BL and either week 36 or week 96. Given sparsity of data at week 5 and week 12, the data were combined, and the average value was used in instances where subjects had non-missing data from both visits. Concentrations values above the upper limit of quantification (ULOQ) was set to ULOQ and values below the lower limit of detection (LOD) were set to LOD. Please see Supplementary Tables 3-4 for limits.

Significance testing for the change in protein level from BL to week 36 or week 96 were performed on the differences of log-transformed data with the non-parametric Wilcoxon rank-sum test. Means and standard error bars are shown in the respective figures. Correlations between VH and AH sampling, as well as between FI and other proteins, were summarized with the Spearman's Rank correlation coefficient. As analyses were exploratory in nature and hypothesis-generating, no adjustments for multiple testing were performed.

For the *in vitro* VH modelling assay, to calculate IC50s, 4-parameter logistic curves were generated using GraphPad Prism version 10.0.0 for Windows, GraphPad Software, Boston, Massachusetts USA, www.graphpad.com [↗](#).

Results

Biomarker detection in FOCUS subjects that received PPY988

In FOCUS Part 1 and 2 (cohorts 1 to 4), 31 subjects received one of three doses of PPY988 via the standard subretinal delivery approach (**Supplementary Table 1** [↗](#)). A total of 97% of the ocular samples expected under the protocol were obtained prior to early termination of the study which

triggered cessation of further ocular fluid sampling. A total of 88 VH, 63 AH and 45 plasma samples from 30 subjects were obtained at various timepoints including BL, and 5 or 12, 36 and 96 weeks post-treatment (Supplementary Methods and **Supplementary Tables 1** [↗](#) and **2** [↗](#)).

In FOCUS, PPY988 was well tolerated with no serious ocular AEs, some dose-related RPE changes in the bleb area were evident and there was no clinically significant vector-related inflammation. A full account of the FOCUS study safety and clinical data, as well as full Phase II PPY988 safety, efficacy and biomarker data will be reported separately following study completion.

Vitrectomy for subretinal AAV-*CFI* gene therapy procedure results in sustained loss of protein from the VH

To test impact of total vitrectomy performed during transvitreal subretinal delivery of PPY988 (**Figure 1A** [↗](#) and B) influencing global proteomic matrix changes and biomarker analyses/interpretation, total protein (TP) measurements were obtained at BL and each sampling timepoint (according to sample availability). In VH, a ~30% reduction in TP from baseline was observed at either week 5 or 12 and week 36 (NS, $P=0.162$), with a significant >50% reduction identified at week 96 ($P=0.033$), while there were minimal changes in AH TP at week 5/12, 36 or 96 (**Figure 1C** [↗](#)). Because VH TP was reduced post-operatively, we performed TP normalisation to compare the results with non-normalised data interpretation for VH but not AH.

AAV2-*CFI* delivery results in a sustained increase in FI protein level in the VH and AH

Transgene expression was measured in VH and AH samples at weeks 5 or 12, 36, and 96 after treatment with PPY988. On average, we identified a 2-fold increase in VH FI at week 36 compared to baseline, with non-TP normalized baseline concentrations of 500 ng/mL compared to 1000 ng/mL at week 36 (non-normalised $P<0.001$; TP normalised increase ~2-fold ($P=0.002$)) (**Figure 2A, 2B** [↗](#)). The increase in FI concentration was sustained, with a ~2.5-fold increase in FI observed at week 96. (mean ~1250 ng/mL) (non-normalised $P<0.001$; TP normalised increase ~3-fold ($P=0.002$)). *CFI* RV (n=5) and non-*CFI* RV ('broad') (n=19) genotype groups demonstrate comparable magnitudes of absolute FI elevation post-treatment, with the overall *CFI* RV cohort FI starting concentration ~50% reduced (NS). Elevated FI expression in AH samples at week 36 ($P<0.001$) were sustained out to week 96 ($P<0.001$) (**Figure 2C, 2D** [↗](#)). Lastly, there was a variable response to the three PPY988 doses in both VH and AH with no significant dose effect observed (**Figure 2C, 2D** [↗](#)).

Increased FI expression results in a reduction in Ba and Ba:FB in the VH

To test for FI target engagement at the level of formation of the alternative pathway C3 convertase, we measured Ba and FB concentrations at each sampling week after subretinal delivery of PPY988. A reduction in Ba in the VH is indicative of a reduction in C3 convertase formation in the ocular environment. We observed that Ba levels were reduced by ~2-fold compared to baseline at week 36 and 2.5-fold at week 96 in non-normalised data (week 36: $P=0.004$; week 96: $P=0.003$) (**Figure 3A** [↗](#)) and TP normalised data (week 36: $P=0.002$; week 96: $P=0.038$) (**Figure 3B** [↗](#)). To add confidence to this read-out by accounting for fluctuations in endogenous FB biosynthesis, ratios of intact complement proteins and their breakdown products were assessed as in previous studies [28](#)–[30](#) [↗](#). Ba:FB ratio changes post-treatment reflected the raw data. When reviewing biomarker levels from both non-normalized and TP normalized datasets, a significant reduction in Ba:FB was observed at week 36 (both $P<0.001$) and at week 96 (non-TP normalized ($P=0.001$); TP normalized ($P=0.033$)) (**Figure 3C, 3D** [↗](#)).

Figure 1.

Surgical delivery of PPY988 AAV2-*CFI*.

(A) PPY988 vector design. FI protein is secreted by cells targeted by AAV2 serotype and containing the cDNA sequence for wild-type *CFI* (NM_000204.4) driven by a CAG promoter⁸. (B) Subretinal delivery of PPY988. (C) Vitrectomy during SR delivery of PPY988 results in loss of total protein (TP) in the vitreous humor. TP concentration (y-axis, log₂ μg/μL) was measured at baseline and 5 or 12, 36, and 96 weeks post-surgery (y-axis). Mean $\pm$ SE is shown by blue bars and lines whilst all subjects are shown by black lines for both vitreous (left) and aqueous (right) humor. A significant reduction in TP was observed at week 96 in vitreous humor (**P*=0.033). Significance is defined as **P*<0.05. BL = Baseline; WK= week.

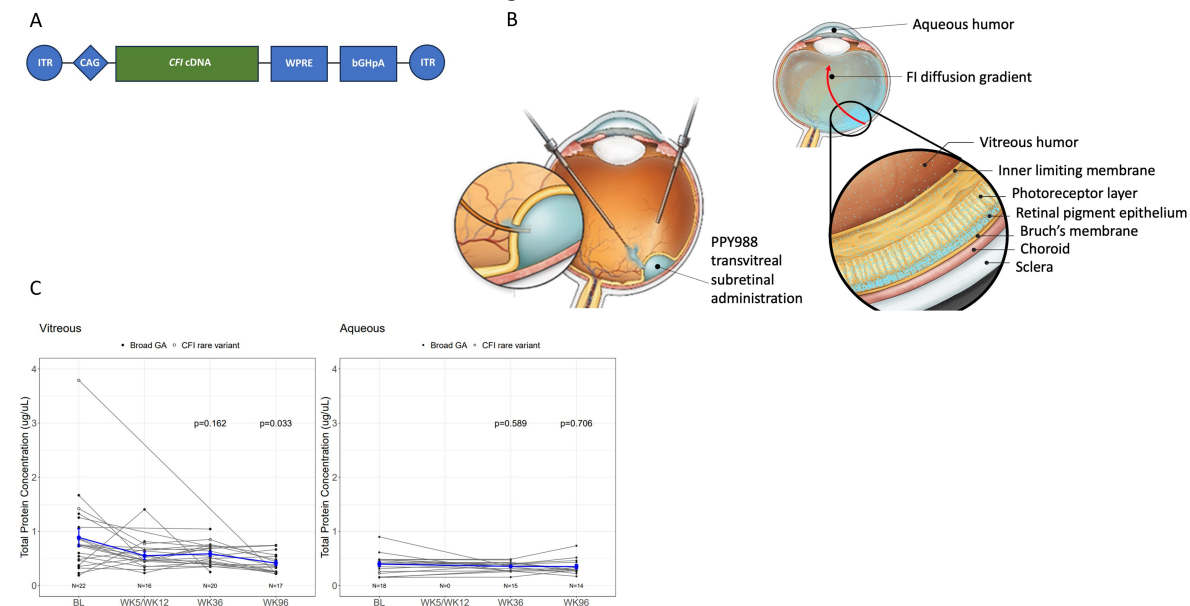
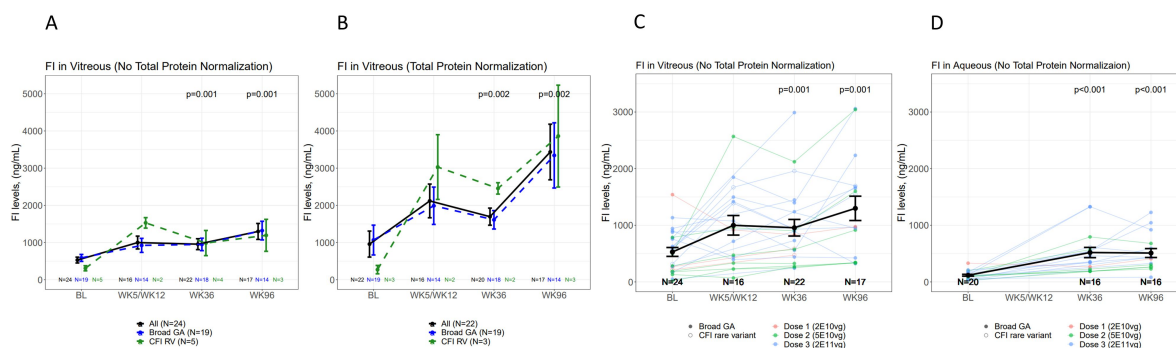


Figure 2.

Temporal expression of factor I (FI) in vitreous humor (VH) and aqueous humor (AH) after PPY988 delivery.

(A,B) FI levels (y-axis, ng/mL) in VH were significantly increased at week 36 and 96 post-surgery (x-axis) both when non-normalised (A) (*P*=0.001 for both timepoints) and total protein-normalised (B) (*P*=0.002 at both timepoints). Mean $\pm$ SE is shown by bars for all (black lines) broad (blue lines) and CFI RV (green lines) groups. (C,D) FI level (y-axis, ng/mL) for each subject in the cohort is tracked by week (x-axis) and separated by dose (2E10 vg: orange lines; 5E10 vg: green lines; 2E11 vg: blue lines) in VH (C) and AH (with significant increase in AH FI at week 36 and 96 (both *P*<0.001)) (D), where mean $\pm$ SE shown by black bars. Significance is defined as **P*<0.05; ***P*<0.01, ****P*<0.001. N numbers are shown above X-axis. BL = Baseline; WK= week.



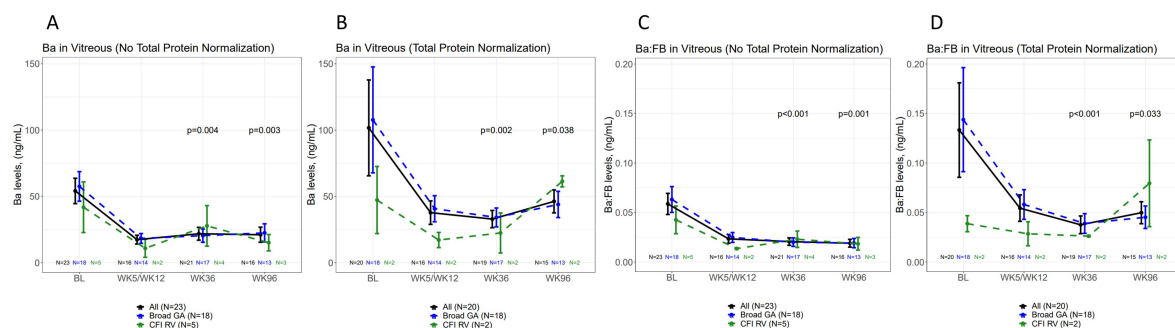


Figure 3.

Modulation of Ba and Ba:FB by PPY988 delivery.

(A,B) Ba concentrations (Y-axis, ng/mL) in vitreous humor (VH) samples by week post-surgery (X-axis). Mean $\pm$ SE is shown by bars for all (black lines) broad (blue lines) and CFI RV (green lines) groups. Significant reductions in Ba were observed at week 36 for both non-normalised ($P=0.004$) (A) and total protein (TP) normalised ($P=0.002$) (B) data and week 96 for non-normalised data ($P=0.003$) and TP normalised ($P=0.038$). (C,D) Ratios of Ba:FB concentration (y-axis, ng/mL) were calculated for each subject at each sampling post-surgery week (x-axis). Mean $\pm$ SE is shown by bars for all (black lines) broad (blue lines) and CFI RV (green lines) groups. Significant reductions were identified at week 36 for both non-normalised ($P<0.001$) (C) and TP normalised ($P<0.001$) (D) Ba:FB ratios, with a reduction at week 96 for non-normalized ($P=0.001$) and TP normalized ($P=0.033$) ratios. Significance is defined as * $P<0.05$; ** $P<0.01$, *** $P<0.001$. N numbers are shown above X-axis. BL = Baseline; WK= week.

Increased FI expression results in modest alterations in VH and AH complement biomarker profiles

To investigate the effect of PPY988 delivery on complement AP and classical pathway (CP) activity, a panel of biomarkers including FI, C3, FB, FH, C1q, Ba, C3a, C3b/iC3b, C4 and C4b were measured in the VH and AH (in addition to plasma) at BL, week 5 or 12, week 36 and week 96. The complete set of concentrations at BL are reported in the [Supplementary Table 3](#) (VH), [Supplementary Table 4](#) (AH), and [Supplementary Table 5](#) (plasma). In addition, Ba:FB, C3a:C3 and C3b/iC3b:C3 ratios were calculated at each time-point. Baseline values for each biomarker with descriptive statistics are reported in [Supplementary Table 3](#) (VH) and [Supplementary Table 4](#) (AH), with baseline and post-treatment pairwise correlations between FI and biomarkers presented in [Supplementary Figure 1](#) (VH) and [Supplementary Figure 2](#) (AH). Fold changes in complement biomarker concentration from baseline are presented for each biomarker or ratio at each post-surgery sampling week in VH non-normalised data ([Figure 4A](#)), VH TP normalised data ([Figure 4B](#)) and AH non-normalised data ([Figure 4C](#)). In the non-normalised VH dataset, we observed reductions in Ba and Ba:FB, with a reduction in C1q possibly indicative of reduced ocular immune cell (e.g. microglial) activation ³¹ ([Figure 4A](#)). We detected no modulation of VH C3a and C4b. Although levels of C3b/iC3b per unit of C3 fall substantially at week 96 in both non-normalized ($P=0.002$) and TP-normalized ($P=0.012$) analyses ([Supplementary Figure 3](#)), pairwise correlation analyses reveal that no biomarkers or ratios were negatively correlated with increasing FI level at week 36 or 96 post-treatment ([Supplementary Figures 1](#) and [2](#)). Minimal reduction of CP activity was indicated by C4b:C4 ratio analysis, however, this was not observed when normalizing for TP ([Supplementary Figure 4](#)). VH FB and FH concentrations do not fluctuate substantially from BL values after treatment with PPY988. Overall, TP normalisation did not substantially alter the interpretation of VH biomarkers with broadly similar trends observed ([Figure 4B](#)).

AH sampling is not reflective of VH complement activation profile

Complement biomarker profiling in AH demonstrated increasing concentrations of a majority of biomarkers ([Figure 4C](#)), including for Ba, C1q, and C3a (at week 96) which was discordant with VH sampling data. Whether these components (C1q) and breakdown products (Ba, C3a) are generated locally, systemically, or diffusely from the retina or VH is unclear. A similar trend to VH was observed when considering the increase in AH C3 and associated fall in C3a:C3 and C3b/iC3b:C3 ratios at week 96 in particular. However, in the case of AH, the increase in C3 was also observed at week 36, suggesting compartmentalization of response of the AH vs the VH. In line with these data, pairwise correlations between VH and AH levels of Ba, C3a and C3b/iC3b post-treatment are modest and variable ([Supplementary Figure 5](#)), with only FI consistently well correlated between the two ocular compartments.

Proteomic analysis on VH using Olink provides supportive evidence for target expression and modest modulation of the complement system

Olink proteomic analysis of the VH samples identified a 2-fold median increase in week 36 FI levels relative to baseline (Supplementary Methods, [Supplementary Figure 6a](#)). Further review of differentially expressed proteins revealed trends for elevation of additional complement components including factor B, properdin and C2 ([Supplementary Figure 6b](#), [Supplementary Table 6](#)).

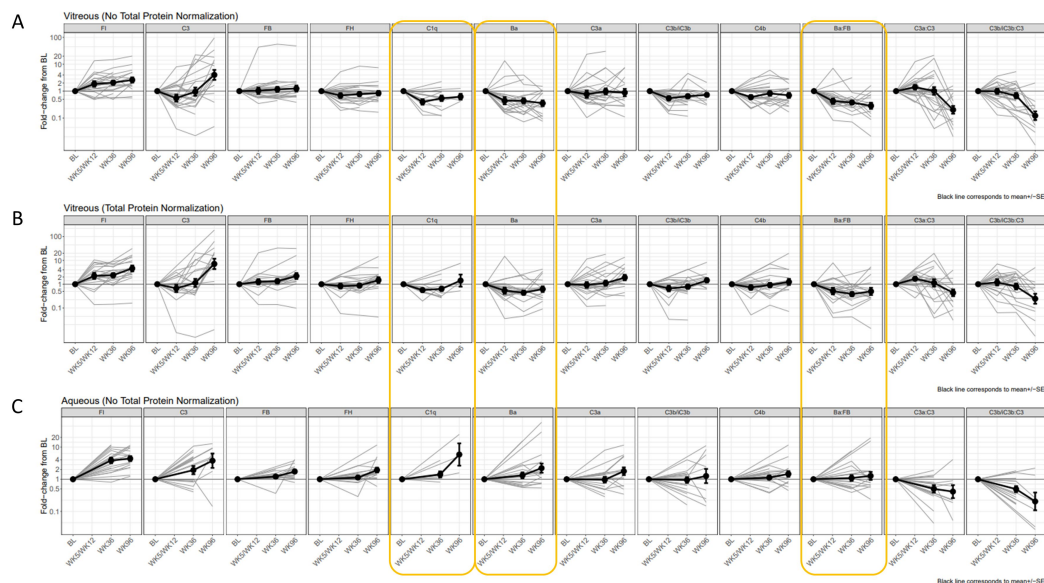


Figure 4.

Modulation of complement AP and CP biomarkers by PPY988 delivery.

Fold-changes (y-axis) in complement biomarker [FI, C3, FH, FB, C1q, Ba, C3a, C3b/iC3b, C4b, Ba:FB, C3a:C3, C3b/iC3b:C3] concentrations from baseline at each post-surgery week (VH: week 5/12, 36 and 96; AH: week 36 and 96; X-axis) are shown in VH with no normalisation (A), VH with total protein normalisation (B), and AH with no normalisation. Mean is shown by thick black lines, $\pm$ SE is shown by bars. Individual subject changes are shown by grey lines. Yellow boxes highlight disparate trends in VH vs AH biomarkers. BL = Baseline; WK= week.

VH modelling of complement activation reveals inferior potency of FI vs pegcetacoplan

By modelling the components of the VH based on median measurements of C3, FB, FH, FI (broad cohort measurements) and properdin (all measurements) at baseline in the FOCUS study, we compared the potency of FI, FH and the C3 inhibitor pegcetacoplan for their effectiveness in controlling the generation of C3a on an LPS-coated plate in a simulated ocular environment context (**Figure 5** [↗](#)). In this assay, C3 is directly activated by O-antigens on LPS thus triggering activation to drive the AP amplification loop. A wide range of FI, FH and pegcetacoplan concentrations were titrated through an artificial VH matrix so that comparisons could be made to clinically relevant levels of both FI and pegcetacoplan. Serum purified FI was revealed to be of similar potency to FI expressed from Good Manufacturing Practice (GMP) manufactured PPY988 (**Supplementary Figure 7** [↗](#)). The trough level of pegcetacoplan at 27 days was estimated using injection dose and half-life information available on the Syfovre™ prescribing information datasheet. No diffusion data in human was available so this was not counted in the trough calculation. The concentration obtained was similar to that reported in the pegcetacoplan rabbit pharmacokinetics (PK) study and PK/PD prediction study published by Apellis [32](#) [33](#) [34](#). Pegcetacoplan vitreal trough level was compared to the 1st and 3rd quartile increase (at week 36 vs BL) in FI associated with PPY988 delivery. Pegcetacoplan exhibited greater potency than FI, with complete inhibition of the AP amplification loop and C3a generation at predicted trough level (~59 µg/mL), whilst FI was observed to have minimal effect at levels of expression associated with PPY988 after 36 weeks. Overall curves generated by logistic regression revealed an IC₅₀ of 1229 nM for FI, whilst FH [IC₅₀: 6 nM] was similar in potency to pegcetacoplan [IC₅₀: 2 nM]. The estimated vitreous trough level of pegcetacoplan was approximately 100-fold greater than that required for complete inhibition of the AP in this assay. On the other hand, FI was observed to have minimal effect on AP activation at the levels of PPY988 measured in VH after 36 weeks of treatment.

Discussion

Gene therapy has been suggested as a promising treatment approach for slowing macular degeneration in GA secondary to AMD. This manuscript describes a comprehensive examination of the pharmacokinetic / pharmacodynamic properties of complement biomarkers in the context of the PPY988 GA subretinal gene therapy program and provides technical considerations for analyzing biomarker data from treated subjects. Despite the early termination of the clinical development program, the PPY988 study achieved notable milestones in the field, including the implementation of subretinal delivery of gene therapy for GA and the use of serial vitreous sampling. Herein, we present for the first time an analysis of VH complement biomarkers at baseline and after surgical delivery of a subretinal gene therapy. Baseline levels were similar to a previous study of VH complement [34](#) with potentially confounding loss of total protein also observed in our study, highlighting the need to analyze data in the context of changing total protein levels post-vitreotomy. Our observations reveal consistent elevation of VH FI levels over a 96-week period, accompanied by reduced detection of the FB activation product Ba and ratios of Ba:FB, consistent with sustained modulation of complement AP activity in the vitreous humor compartment. These data confirm that AAV gene therapy has a consistent and lasting effect. Furthermore, although unpowered due to low N, the effects of FI gene therapy on complement breakdown product biomarker profiles (Ba (**Figure 2A** [↗](#)) and C3b/iC3b (**Supplementary Figure 3A** [↗](#)) appear to be comparable in both CFI RV positive and negative sub-populations, which is counter to the original hypothesis of a more dramatic effect in a sub-population with lower baseline FI levels [19](#) [20](#) [35](#). However, due to small sample size this finding should be interpreted with caution. Although the impact on C3 breakdown products was minimal at week 36, it became significant at week 96. Elevation of FI and modulation of the broader complement system was also

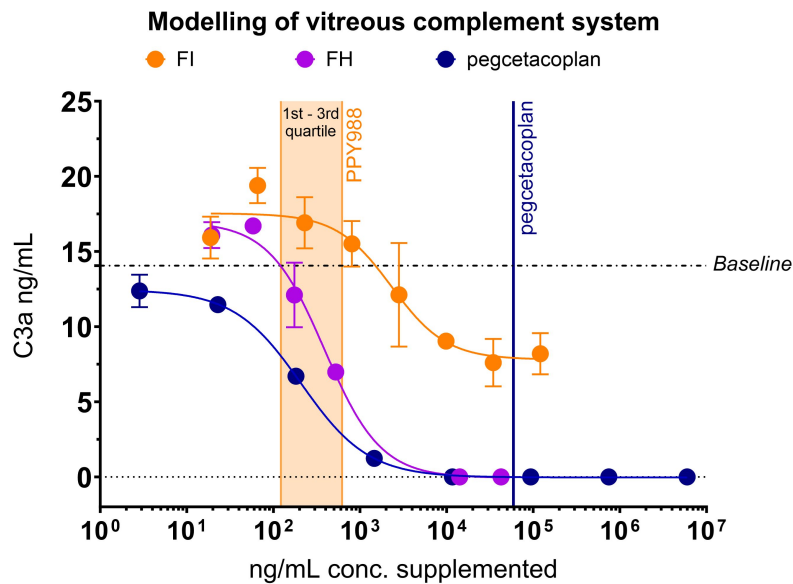


Figure 5.

***In vitro* modelling of vitreous humor complement activation inhibition by FI vs pegcetacoplan and additional complement regulators.**

FI (orange), FH (purple), and pegcetacoplan (blue) were titrated (x-axis, ng/mL) through artificial vitreous complement matrix on LPS-coated Weislab™ plates and C3a generation was measured by ELISA (y-axis, ng/mL). C3a is a marker of convertase activity in the complement activation cascade, the potency of complement inhibitors or regulators is reflected in lower amount of C3a generated in the assay when testing comparable concentrations. Mean is shown with +/-SD represented by bars and 4-parameter fit logistic regression curves are shown by lines. The 1st-3rd quartile of increased FI expression from baseline after 36 weeks of PPY988 delivery is shown by orange area. The calculated trough concentration of pegcetacoplan in eyes 27 days after IVT injection is shown by the blue vertical line.

observed in Olink proteomic profiling of VH samples (**Supplementary Figure 6**), supporting the complement biomarker findings in the validated panel. Elevated levels of FI were detected in both AH and VH. PPY988 efflux from the sub-retinal injection may have transduced cells in the anterior chamber of the eye resulting in an increase in AH FI concentration. However, the correlation of other complement biomarkers between VH and AH was inconsistent, in particular, Ba and C1q exhibited disparate post-treatment trends (**Figure 4**), indicating that AH alone may not be a reliable biomarker matrix for assessing the activity of complement biomarkers at the target tissues. This cautionary advice should be considered in context of previous reports, suggesting AH could be a broadly reliable proxy for VH³⁵, albeit that only full length complement proteins were studied therein. To ensure quality control, protein level normalization was added to the analysis comparing biomarker levels before and after vitrectomy. Interestingly, the overall conclusions remained consistent regardless of protein adjustment, which strengthens confidence in the study's findings. After determining the futility of the Phase II study (HORIZON, NCT04566445), an *in vitro* assay using clinically relevant levels of complement proteins was conducted to investigate a possible biomarker-related explanation. The results showed that pegcetacoplan had higher potency than FI in reducing C3 convertase activity *in vitro* suggesting that limited *in-vitro* potency of FI may have contributed to the lack of clinical success at Phase II. Although the translatability of the VH modelling assay to GA remains to be demonstrated, FI's lower potency suggests that further increasing FI concentration on its own by gene therapy may not achieve clinically meaningful levels of C3 inhibition. It should however be noted that intravitreal pegcetacoplan will be at its highest concentration in the mid-vitreous whereas FI will most likely be at the highest level in the subretinal space where it is being produced. Hence directly comparing vitreous levels of each compound may underestimate the activity of FI compared with pegcetacoplan at the level of the subretinal space.

The lack of efficacy in the PPY988 Phase II study does not detract from the viability of complement activation as an effective target for intervention in GA, given the therapeutic effect observed for complement inhibitors pegcetacoplan and avacincaptad pegol (Izervay; Iveric Bio)^{5,6}, and this study's observation that pegcetacoplan achieves a much greater potency than FI *in vitro*. FI had a disproportionately larger effect upon Ba compared to C3a and C3b/iC3b, which may be due to the lability of C3a, differences in diffusion capabilities of the molecules from the retina, and possible contribution of C3 concentration from the choriocapillaris as well as biosynthesis by local immune cells. In addition, we hypothesize FI's capacity to regulate the production of Ba without causing significant changes in the levels of C3b and C3a in the fluid phase could be contributed to by regulation of AP tickover, resulting from binding and activation of FB on water-hydrolysed C3 (C3H₂O) followed by release of Ba¹⁰. Similarly, the intravitreal administration of a neutralizing factor D Fab, which was found to be ineffective in Phase III trials for GA, was also able to reduce ocular Ba levels²⁸. However, it did not have a significant impact on the generation of VH C3 breakdown products. In this study, by week 96, there was an observed rise in the levels of full-length C3 and a corresponding decrease in the ratios of breakdown products to full length. While the specific implications of this change are uncertain, it potentially suggests a reduction in C3 consumption, which aligns with the known activity of FI, but this may be confounded with upregulated biosynthesis of C3 given its acute phase nature³⁶. C3 breakdown products only partially diffuse due to cell-surface adherence. Based on this understanding, we hypothesize that changes in C3 breakdown products may only become detectable when a certain threshold of modulation in tissue complement activity is reached. Overall, the data suggests that fully functional FI is being secreted into the VH, but the regulatory effects on the level of Ba may be representative of convertase formation in the VH and not the macula retina/RPE nor the choroid. To validate this hypothesis, one approach would be to conduct vitreal sampling using an effective drug targeting C3 for GA in a larger cohort. Further investigations are needed to gain a better understanding of these possibilities.

The study has several limitations that should be considered when interpreting the results. Firstly, the study did not include a control group, which means that the observed changes in complement biomarkers may not be solely attributed to transgene expression. Other factors, such as the impact of subretinal gene therapy administration triggering local inflammation or the effects of total vitrectomy on the protein environment, could also contribute to these changes. During the study, no significant safety concerns or adverse events were reported aside from dose-responsive incidents of RPE changes in the area of the subretinal injection. However, increased levels of antibodies to the vector and transgene were observed following treatment³⁷. It is important to note that these antibody levels were transient and resolved shortly after treatment, indicating that the observed antibody responses were not persistent or associated with any significant clinical consequences. It is worth noting that the subjects included in this early clinical trial had more advanced disease compared to the typical clinical trial GA population which may impact the generalizability of the findings to the broader GA population. Additionally, the study had a relatively small sample size, which increases the risk of bias. Furthermore, the biomarker conclusions drawn from the study subjects may not be representative of the subjects treated in the subsequent Phase II study. Therefore, caution should be exercised when generalizing the findings to larger populations. Despite convincing and robust genetic links to AMD reported for Type I *CFI* variants^{19–22}, this study observed no difference in biomarker response when comparing *CFI* RV carriers to the broad population. Interestingly, emerging data have revealed more modest ORs for *CFI* RVs in progression from early to late AMD²³ and macular thinning was observed in UK Biobank eyes of carriers <60 years old³⁸. Moreover, the long-term effects of complement therapy on the progression of GA are still uncertain. Another limitation to consider is the translatability of the *in vitro* VH modelling assay to the disease processes occurring in GA, and the relevance of the assay to cell-surface complement activation occurring in the area of GA. The LPS-coated surface used for the VH modelling assay would most likely result in more acute complement activation than in the GA retina. Finally, this is a relative potency assay and does not take into consideration the biodistribution of different modalities and routes of administration. Despite these limitations, the conclusions drawn from this study are compelling and hold significant implications for other gene therapy programs. The introduction of the innovative *in vitro* complement activation assay serves as a valuable benchmark for gene therapy programs.

The superior potency of FH compared to FI highlights the importance of conducting potency comparisons using *in vitro* assays that reflect physiologically relevant levels of complement proteins to guide decision-making. This finding opens promising avenues for further investigation, with potential for engineering of more active RCA protein truncates and chimeras including CR1, MCP, DAF and FH, FHRs. To advance gene therapy programs, future research should focus on building upon the pharmacokinetic / pharmacodynamic findings. It is crucial to establish tissue concentrations and estimates of the target required for complement modulation before moving into clinical stages. Pre-clinical assays that can be effectively translated to GA remain elusive. Overall, the role of complement in GA and the potential benefits of a safe one-time gene therapy in reducing AMD progression suggest the need for further clinical development of complement gene therapy.

Data Availability

All data produced in the present study are available upon reasonable request to the authors

Conflict of interest

The authors declare the following competing interests: Drs T Hallam, E Gardenal, A Jones, S Ellis, E Pekle, K Carney, and L Drage and Ms C Wenden, Ms H Beadsmoore and Mr T Haye are employees of Gyroscope Therapeutics Limited, a Novartis company. Drs Poor, McBlane, Kaiser, Lu, Kassem, Cho, Ferraro, Rangaswamy, Obeidat, Saint-Geniez, and Grosskreutz are employees of Novartis. T Hallam is an author on a pending patent for factor I trimolecular complex building/screening of complement changes.

Claire L Harris (In previous three years: consultant: Q32 Bio, Biocryst; Grant funding: Ra Pharmaceuticals). CLH was an employee of Gyroscope Therapeutics Limited, a Novartis Company.

Professor David H Steel (Consultant: Alcon, Gyroscope, BVI, DORC, Eyepoint, Complement therapeutics, Alimera; Grant funding: Bayer, Alcon, Roche, DORC, BVI, Boehringer, Gyroscope Therapeutics Ltd).

Professor Robert E MacLaren (Consultant: Biogen, AGTC, and Beacon Therapeutics; Grant funding: Biogen). REM is also a named inventor on the Biogen BIIB112 patent licensed by the University of Oxford.

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Abbreviations

- AEs: Adverse events
- AH: Aqueous humor
- AMD: Age-related macular degeneration
- AP: Alternative Pathway
- BL: Baseline
- C3: Complement component 3
- cDNA: Complementary deoxyribonucleic acid
- CR1: Complement receptor 1
- CS: Complement system
- DAF: Decay-accelerating factor
- EDTA: Ethylenediaminetetraacetic acid
- FAF: Fundus autofluorescence
- FB: Factor B
- FD: Factor D
- FDA: Food and Drug Administration
- fdr: False discovery rate
- FH: Factor H
- FI: Factor I
- GA: Geographic atrophy
- GMP: Good Manufacturing Process
- IVT: Intravitreal
- LLOQ: Lower limit of quantification
- LOD: Limit of detection

- LPS: Lipopolysaccharide
- MBL: Mannose binding lectin
- MCP: Membrane cofactor protein
- NPX: Normalized protein expression
- PK: Pharmacokinetics
- RCA: Regulators of complement activation
- RPE: Retinal pigment epithelium
- RV: Rare variant
- SDS: Subretinal delivery system
- TP: Total protein
- TVSI: Transvitreal surgical intervention
- UK: United Kingdom
- ULOQ: Upper limit of quantification
- US: United States
- VH: Vitreous humor

Supplementary Methods

Study design

FOCUS was conducted at 13 sites over two countries and was sponsored by Gyroscope Therapeutics, a Novartis company. Key inclusion criteria included a clinical diagnosis of GA secondary to AMD in the study eye as determined by a retinal specialist, and a diagnosis of AMD in the contralateral eye (except if subject was monocular). Study eye GA lesion(s) total size was $\geq 1.25\text{mm}^2$ and $\leq 17.5\text{mm}^2$ (cohort 1 to 6) or $\geq 1.25\text{mm}^2$ (cohort 7), and GA lesions resided completely within the fundus autofluorescence (FAF) fundus image. Key exclusion criteria included presence of the following in the study eye; CNV, non-proliferative diabetic retinopathy, glaucomatous optic neuropathy, or cataract. Subjects were also excluded if they had an active malignancy or a history of vitrectomy, sub-macular surgery, macular photocoagulation or previously receiving a gene or cell therapy in the study eye.

As outlined in **Supplementary Table 1**, the study had four parts, part 1 (cohorts 1-3); dose escalation, Part 2 (cohort 4); dose expansion, Part 3 (cohorts 5-6); dose escalation with the Orbit Sub-retinal Delivery System (SDS), and Part 4 (cohort 7); dose-expansion with the Orbit SDS.

PPY988 gene therapy aimed to transduce RPE and PR cells, considered a key site for AMD pathology and FI production. Parts 1 and 2 (cohorts 1-4) delivered PPY988 via standard transvitreal subretinal injection (TVSI), previously used for subretinal gene therapy delivery^{1,2}. The surgeon then performed the TVSI procedure involving a standard 3-port pars plana vitrectomy. The retina was partially detached through a cannula using buffered salt solution for bleb formation, and a fixed volume of 0.1mL containing the gene therapy dose was injected into the bleb, which was expected to flatten after 24 hours.

Subjects in Parts 3 and 5 (cohorts 5-7) received PPY988 subretinally via suprachoroidal cannulation using Orbit SDS, a 510(k) cleared device in the US (K200325, 14-Jul-2020, US FDA), developed to improve the safety profile of subretinal injections as compared with TVSI³. However, because VH samples were not collected in Parts 3 to 5, analysis of biomarker data from subjects treated in these cohorts is not discussed herein.

VH, AH and plasma samples were collected from cohort 1 to 4 subjects at BL prior to surgery, at weeks 12, 36 and 96 after dosing, except for sample collection at week 5 (not week 12) in cohorts 1 and 2, and AH (not VH) was collected for cohort 4 at Week 12.

Approximately 25 to 100µL of AH and VH samples were collected under microscopic visualization. AH samples were collected using a 30G needle attached to a 1mL syringe. The 30G needle was entered into the anterior chamber through the temporal limbus using a clear corneal approach. VH samples were collected at BL using a non-primed 23G vitrectomy probe connected to a 1mL syringe, and post-BL using a 30G needle through the *pars plana*. Immediately following collection VH and AH samples were transferred to a sterile collection vial and frozen at -80°C within 30 minutes.

All subjects were assessed for the occurrence of adverse events (AEs) at each follow-up visit and underwent functional visual and retinal imaging/anatomical assessments and biological sampling for safety and immunogenicity monitoring at periodic timepoints up to week 48, considered the primary endpoint timepoint of the study.

It was intended that subjects would be invited to consent to an additional long-term follow-up assessment until week 240 (5 years). At the time of Phase II study termination, FOCUS was fully enrolled. As a consequence of the early termination of the Phase II study, all ocular sampling in FOCUS from that point onward was terminated, and subjects who had not completed the study assessment window to week 48 were transferred onto a safety monitoring regimen.

Olink proteomics

VH samples from 17 subjects were analyzed using the OLINK® Target 96 platform (OLINK® Proteomics, Uppsala, Sweden) to identify broader pathway changes affected by subretinal delivery of PPY988. A single measurement was performed for each sample. The following 4 panels were assessed by Olink: inflammation and inflammation II, cardiometabolic and neurology. Protein levels were measured on a relative scale and presented as the normalized protein expression (NPX), which is an arbitrary unit on a log2 scale.. Principal component analysis was performed to detect outlier samples. Olink provides a limit of detection (LOD) for each protein based on the background noise generated in negative controls (buffer run as a normal sample). Nonetheless, the Olink protein expression data reported NPX values even for measurements below the LOD. Low abundance proteins were excluded from the statistical analyses. To identify the proteins changing over time relative to BL, a linear mixed effect model was used. The model included the following covariates: timepoint, subject age and sex, sample total protein content and *CFI* RV status (subject ID as random effect). All analyses were carried out using R version 4.2.3. Nominal p-values were adjusted using false discovery rate.

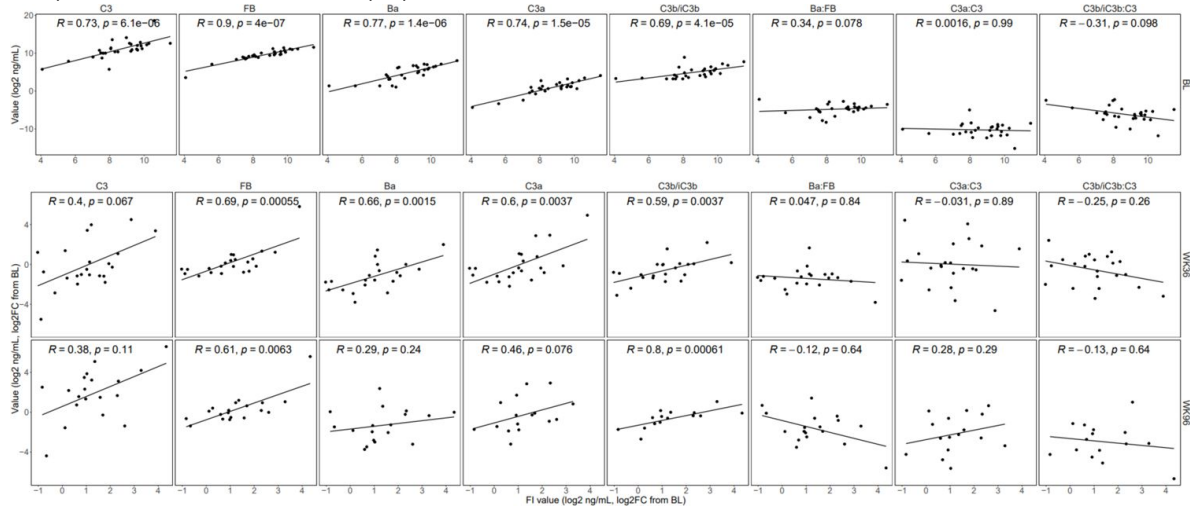
***In vitro* FI potency comparability assay**

HEK293 adherent cells with the *CFI* and *CFH* genes knocked out by CRISPR-Cas9 were transduced with PPY988 (AAV2-*CFI*) vector material produced under good manufacturing practice (GMP) at 10 different MOIs. After a 24-hour incubation (37°C at 5% CO₂) with the vector, the media was replaced and left for a further 48 hours. Supernatants were collected and stored at -20°C. Supernatants were thawed, centrifuged (5000xG for 15 mins) then mixed with a fixed volume of human purified C3b and FH (A114 and A137, Comptech) and incubated for 10 minutes at 37°C. The mixture was then reduced and labelled by incubating for 5 minutes at 90°C with a mixture of DTT, SDS buffer and the LIF dye Chromeo P503 (30693, Sigma). A standard curve of Purified FI (Comptech, Tyler, TX) was made starting at a final concentration of 4000 ng/mL down to 11 ng/mL. This was diluted in non-transduced control sample media to ensure the same background matrix for both transduction supernatants samples and Purified FI standard curve samples. Proteins were separated by CE-SDS using LIF detection and an adapted version of a published method [4](#). The areas of the peak corresponding 114 kDa alpha prime subunit (normalised to the 75 kDa beta subunit) are collected for both the Reference (Comptech Purified FI) and the Test sample (AAV2-*CFI* GMP). The depletion of the peak of interest is plotted using QuBAS 3.0 Statistical Software (Quantics Biostatistics) which performs a parallel line comparison of the Test compared to the reference and produces a Relative Potency value corresponding to this analysis. Outliers are statistically identified by a Grubbs Test and removed automatically from the analysis prior to calculating the Relative Potency.

Supplementary Figure 1.

VH complement biomarker pairwise correlation analysis.

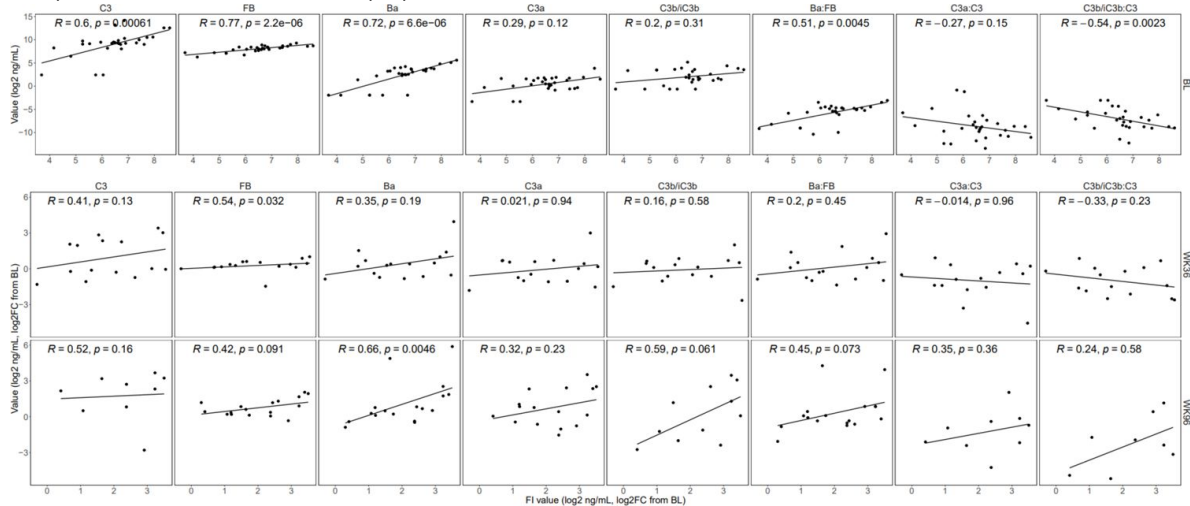
Pairwise correlation analysis of FI with intact complement proteins and breakdown products and ratios in the vitreous at baseline, week 36 and week 96 from subjects in cohorts 1 to 4. FI concentrations are shown by the x-axis while biomarker concentrations and ratios are shown by the y-axis. Correlations at BL are determined using log₂ ng/mL concentrations, correlations at weeks 36 and 96 are determined using log₂ fold change from BL. BL = Baseline; WK = week; FC = Fold Change; R = Spearman Correlation Coefficient; p = p-value.

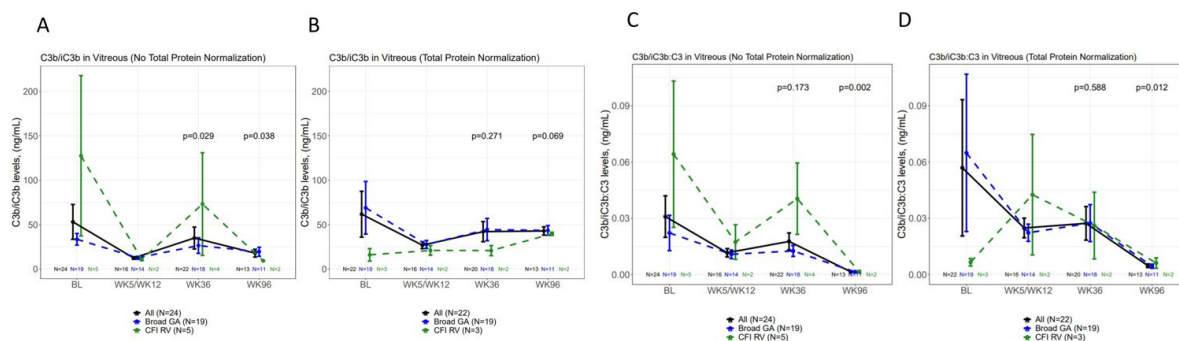


Supplementary Figure 2.

AH complement biomarker pairwise correlation analysis.

Pairwise correlation analysis of FI with intact complement proteins and breakdown products and ratios in the aqueous humor at baseline, week 36 and week 96 from subjects in cohorts 1 to 4. FI concentrations are shown by the X-axis while biomarker concentrations and ratios are shown by the y-axis. Correlations at BL are determined using log₂ ng/mL concentrations, correlations at weeks 36 and 96 are determined using log₂ fold change from BL. BL = Baseline; WK = week; FC = Fold Change; R = Spearman Correlation Coefficient; p = p-value.

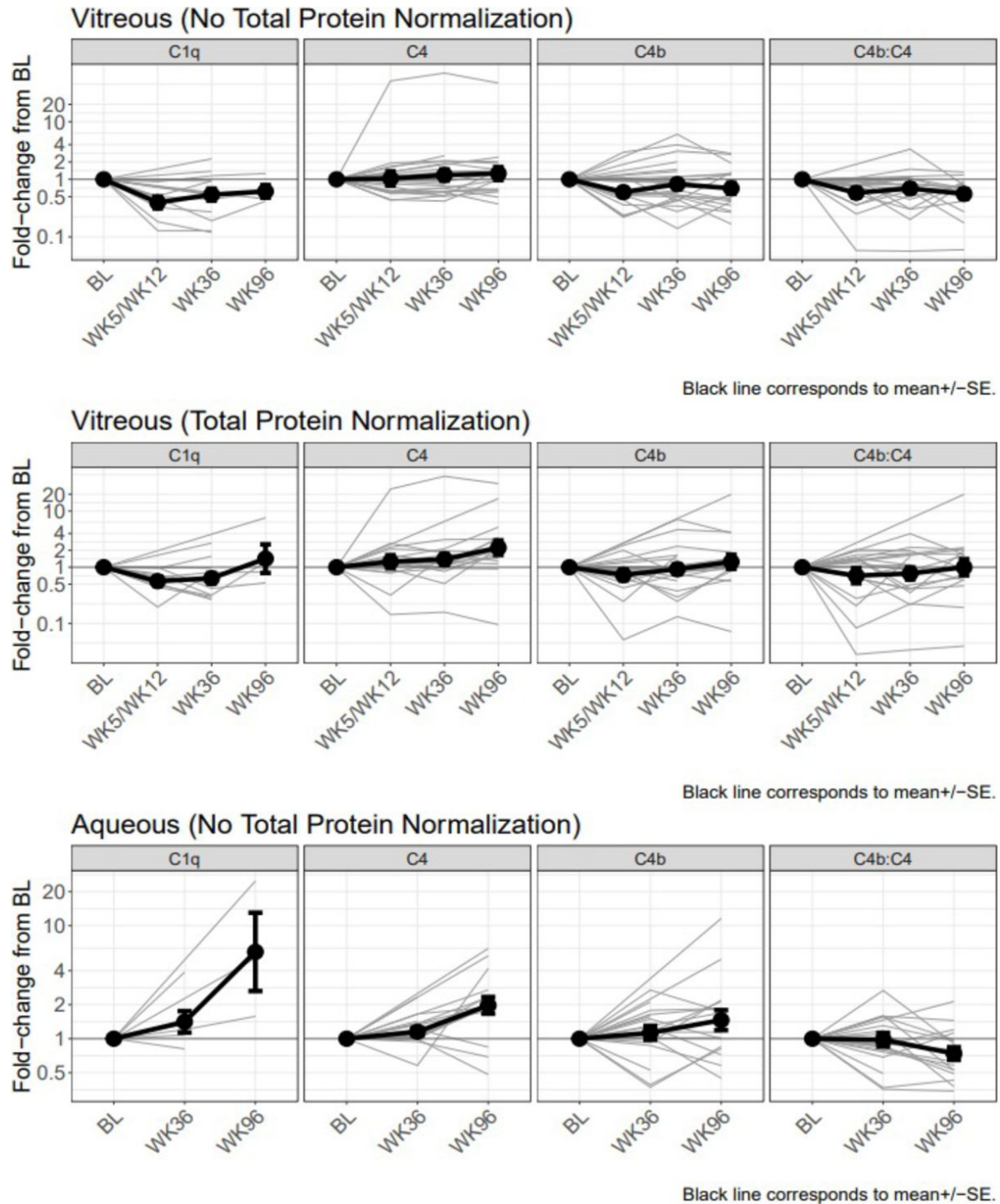




Supplementary Figure 3.

Modulation of C3b/iC3b and C3b/iC3b:C3 by PPY988 delivery.

(A,B) C3b/iC3b concentrations (y-axis, ng/mL) in VH samples by week post-surgery (x-axis) from subjects in cohorts 1 to 4. Mean $\pm$ SE is shown by bars for all (black lines) broad (blue lines) and CFI RV (green lines) groups. Significant reductions in C3b/iC3b were observed at week 36 for both non-normalised (P=0.029) (A) but not total protein (TP) normalised (P=0.271) (B) data, and at week 96 for non-normalised (P=0.038) but not TP normalised (P=0.069) data. Notably these reductions are modest compared to for Ba modulation. (C,D) Ratios of C3b/iC3b:C3 concentration (Y-axis, ng/mL) were calculated for each subject at each sampling post-surgery week (X-axis). Mean $\pm$ SE is shown by bars for all (black lines) broad (blue lines) and CFI RV (green lines) groups. No reductions were observed at week 36 for either non-normalised (P=0.173) (C) or TP normalised (P=0.588) (D) C3b/iC3b:C3 ratios. However, reductions were observed at week 96 for non-normalized (P=0.002) and TP normalized (P=0.012) C3b/iC3b:c3 ratios. Significance is defined as *P<0.05; **P<0.01, ***P<0.001. N numbers are shown above X-axis. BL = Baseline; WK= week.

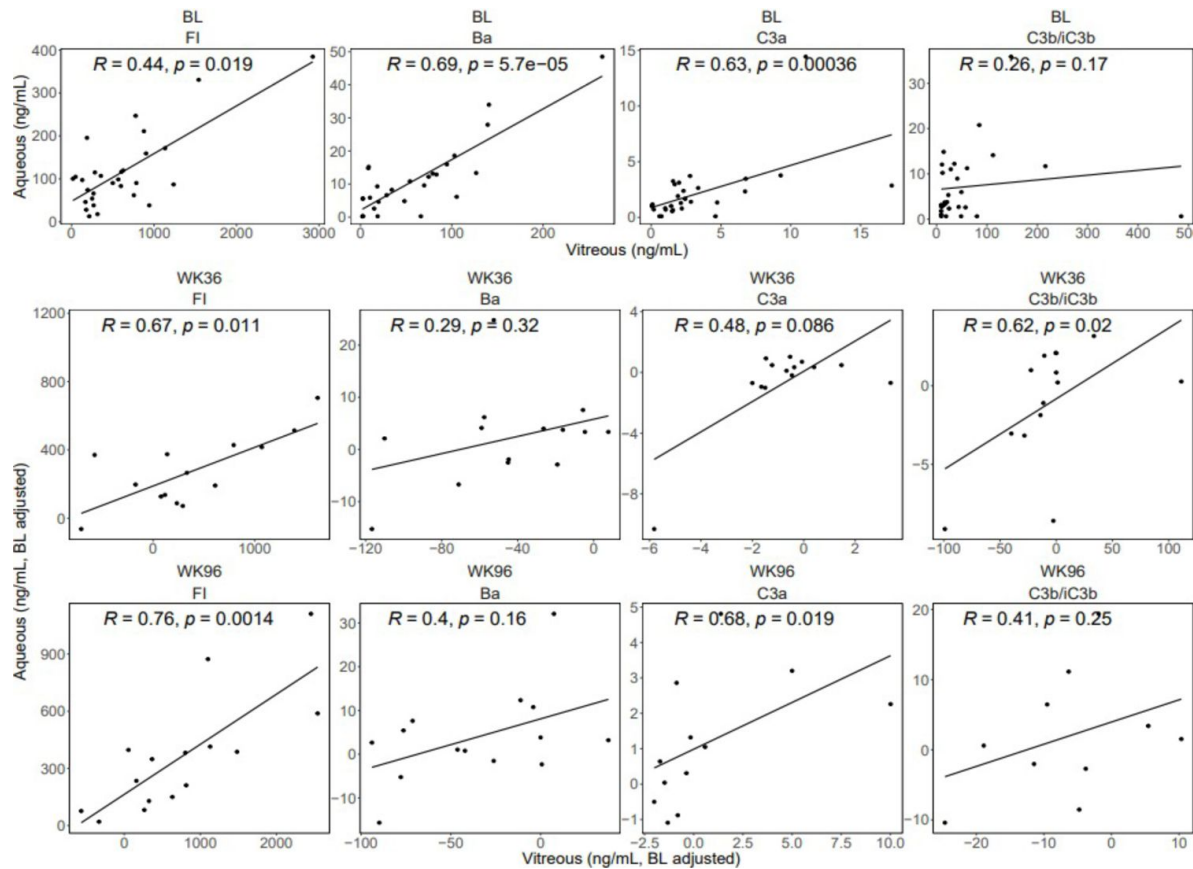


Supplementary Figure 4.

Classical pathway (CP) engagement post-treatment with PPY988.

Fold-changes (y-axis) in CP complement biomarkers [C1q, C4, C4b and C4b:C4] concentrations from baseline at each post-surgery week (VH: week 5/12, 36 and 96; AH: week 36 and 96; X-axis) are shown in VH with no normalisation (A), VH with total protein normalisation (B), and AH with no normalisation. Mean is shown by thick black lines, $\pm$ SE is shown by bars.

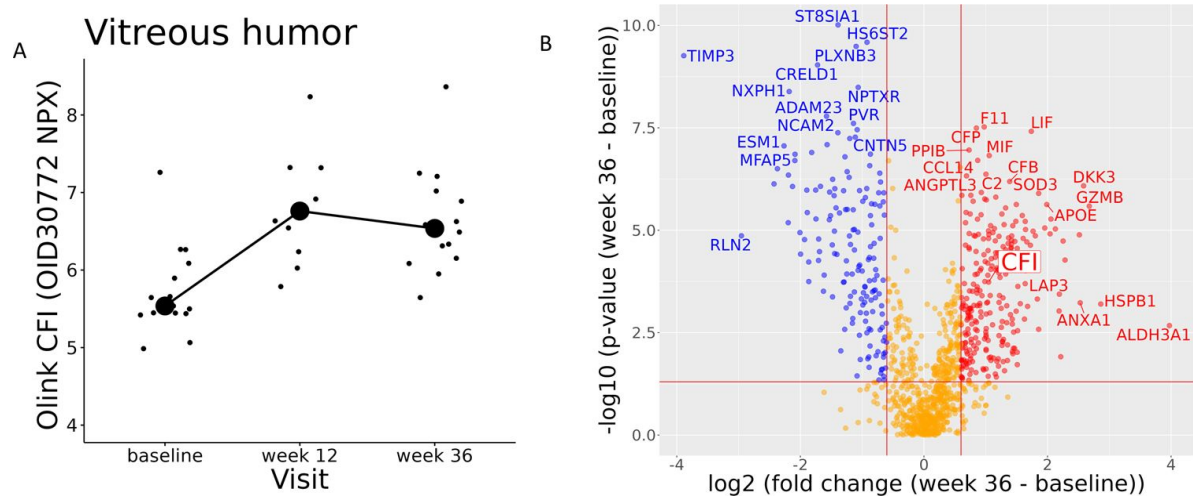
Individual subject changes are shown by grey lines. BL = Baseline; WK= week.



Supplementary Figure 5.

Pairwise correlation analysis of vitreous versus aqueous humor complement biomarkers at baseline and post-treatment from subjects in cohorts 1 to 4.

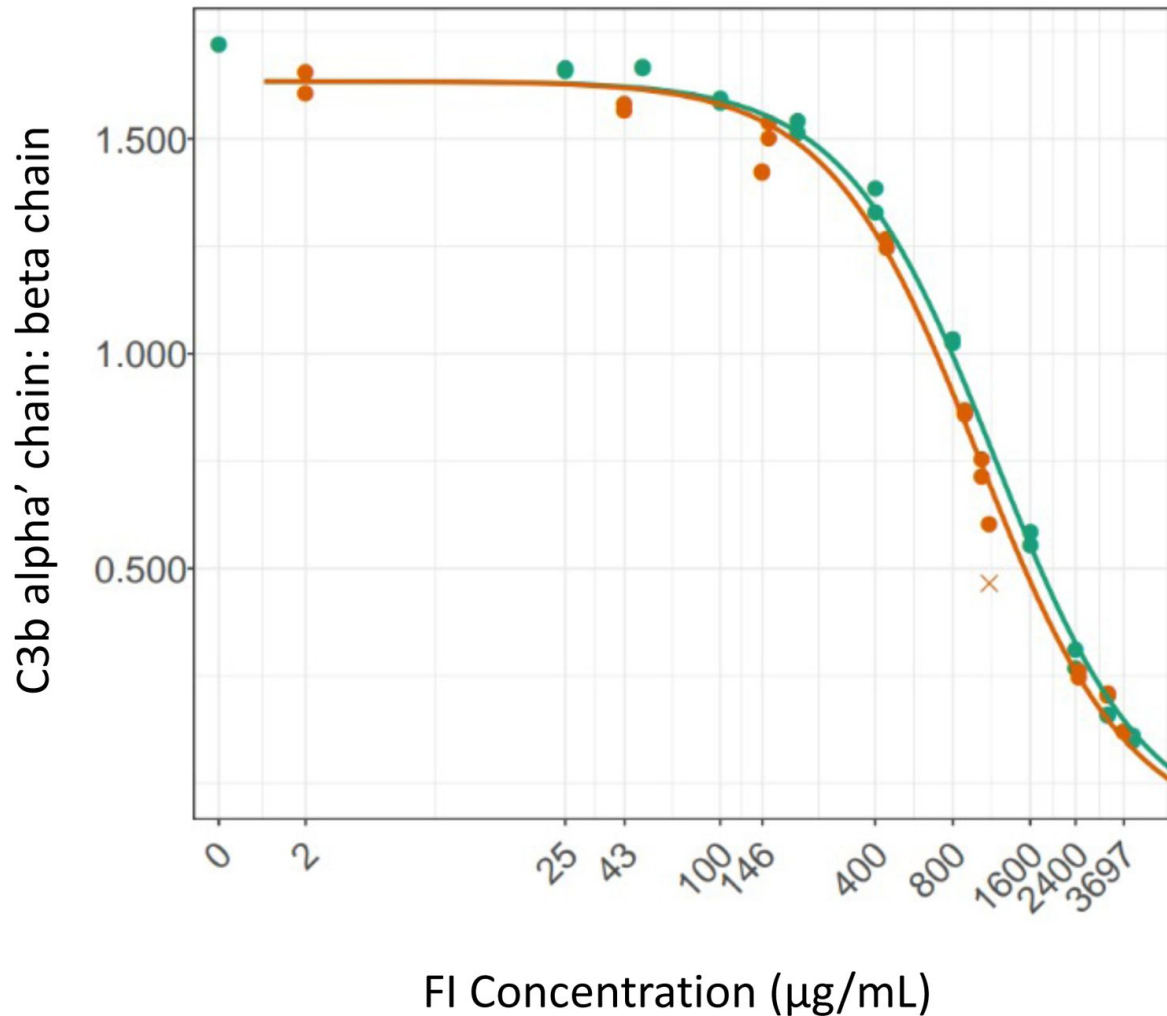
VH concentrations on x-axis and AH concentrations on y-axis of FI, Ba, C3a and C3b/iC3b at baseline, week 36 and week 96 post-surgery. Vitreous and aqueous concentrations of FI and C3a significantly correlate at all time points, Ba concentrations correlate only at baseline, while C3b/iC3b concentrations only correlate significantly at week 36 post-treatment. BL = Baseline; WK = week; R = Spearman Correlation Coefficient; p = p -value.



Supplementary Figure 6.

Proteomic analysis of vitreous humor using Olink.

(A) Change in FI levels (Olink OID30772) over time. Median FI values measured at baseline, week 12, and week 36 are linked. Sample numbers were baseline; n=17, week 12; n=14, week 36; n=17. **(B)** Volcano plot of differential protein expression (week 36 relative to baseline). Minimum protein fold-change was set at absolute 1.5. Data points with a nominal p-value of less than 0.05 were considered statistically significant (red dots indicate significantly upregulated proteins, blue dots indicate significantly downregulated proteins and orange dots represent no significant change on a log2 scale).



Supplementary Figure 7.

Comparability of commercial FI to AAV2-CFI potency.

C3b alpha' chain breakdown normalised to C3b intact beta chain (y-axis) produced by increasing concentrations of PPY988 (GMP#3) in orange compared to commercial FI (Complement Technology (CompTech), Inc) in green. The serum purified CompTech FI protein was diluted in non-transduced control sample so that both FI (reference sample) and PPY988 FI (test sample) were in the same background matrix to ensure comparability. No difference in potency was observed between PPY988 and CompTech FI Protein. Relative Potency of PPY988: 1.151; 95% CI range: 1.065-1.244; Precision: 1.168. See supplemental methods for further details.

Supplementary Table 1.

FOCUS study overview.

FOCUS consisted of four parts of either PPY988 dose escalation or expansion, divided by seven cohorts varying according to route of gene therapy administration, dose (vg; viral genomes), and biomarker matrix. Only Parts 1 and 2 (cohorts 1 to 4) are described given subretinal gene therapy delivery permitted vitreous humour (VH) sampling. VH, aqueous humour (AH) and plasma biomarker samples were collected from subjects prior to surgery at baseline (BL), and subsequent timepoints up to week 96 post-treatment. Emerging data and analysis on cohorts 1 and 2 indicated week 5 sampling from the VH was too early to evaluate an impact post-vitrectomy, so the first post-treatment evaluation for cohorts 3 and 4 was moved to a later time-point in the study (week 12), when it was expected that the proteome would have stabilised, and the transduced cells achieved maximal expression.

Part (name)	Cohort	SR delivery method	PPY988 dose [vg/eye]	Subjects (n)	Biomarker matrix; sampling timepoints
1 (dose escalation)	1	Standard	low, 2E10	3	VH; BL, week 5, 36, 96 AH; BL, week 36, 96 Plasma: BL, week 5, 36, 96
	2		mid, 5E10	4	VH; BL, week 5, 36, 96 AH; BL, week 36, 96 Plasma: BL, week 12, 36, 96
	3		high, 2E11	4	VH, BL, week 12, 36, 96 AH; BL, week 36, 96 Plasma: BL, week 12, 36, 96
2 (dose expansion)	4	Standard	low, 2E10 (n=3); mid, 5E10 (n=6); high, 2E11 (n=11)	20	VH, BL, week 12, 36, 96 AH; BL, week 36, 96 Plasma: BL, week 12, 36, 96

Supplementary Table 2.

Number of FOCUS samples analysed longitudinally across study.

VH and AH samples from subjects in cohorts 1 to 4 that passed acceptance criteria taken at baseline (BL) and at each timepoint post treatment are provided. Only those subjects with sampling at both BL and at least one of week 36 or 96 time-points are included.

Vitreous

Visit	FI	C3	FB	FH	C1q	Ba	C3a	C3b/iC3b	C4	C4b	Ba:FB	C3a:C3	C3b/iC3b:C3	C4b:C4	C5	FD	MBL
BL/pre-dosing	24	24	24	24	15	23	24	24	24	24	23	24	24	24	23	24	24
Week 5/12	16	16	16	16	8	16	16	16	16	16	16	16	16	16	15	16	16
Week 36	22	22	22	22	14	21	22	22	22	22	21	22	22	22	21	22	22
Week 96	17	17	17	17	4	16	15	13	17	16	16	15	13	16	16	17	17

Aqueous

Visit	FI	C3	FB	FH	C1q	Ba	C3a	C3b/iC3b	C4	C4b	Ba:FB	C3a:C3	C3b/iC3b:C3	C4b:C4	C5	FD	MBL
BL/pre-dosing	20	19	20	20	8	20	20	19	20	20	20	19	19	20	19	20	20
Week 5/12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Week 36	16	15	16	16	6	16	16	15	16	16	16	15	15	16	16	16	16
Week 96	16	8	16	16	3	16	15	10	16	16	16	8	7	16	15	16	16

Plasma

Visit	FI	C3	FB	FH	C1q	Ba	C3a	C3b/iC3b	C4	C4b	Ba:FB	C3a:C3	C3b/iC3b:C3	C4b:C4	C5	FD	MBL
BL/pre-dosing	15	16	16	16	9	20	13	12	16	17	13	13	12	16	17	15	17
Week 5/12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Week 36	15	16	16	16	9	20	13	12	16	17	13	13	12	16	17	15	17
Week 96	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Supplementary Table 3.

Descriptive statistics of vitreous humour baseline complement biomarkers concentrations from subjects in cohorts 1 to 4.

The number of samples correspond to the total number which could be measured and had a value within acceptance criteria. All concentrations shown are expressed in ng/mL.

Biomarker	n	Mean	SD	Median	25 th Percentile	75 th Percentile	Minimum	Maximum	LOD	N LOD	ULOQ	N ULOQ
FI	29	608.88	587.42	499.91	215	785.62	16.91	2924.77	4.625	0	12500	0
C3	29	17350.67	74622.03	2163.41	1104.46	5344.56	53.7	404817.6	53.7	2	60000	0
FB	29	1083.14	817.41	765	504.85	1402.1	11.81	3037.52	5.8	0	6000	0
FH	29	466.7	427.59	273.56	181.66	652.73	20.7	1430.34	20.7	1	30000	0
C1q	19	140.45	106.43	126.07	70.85	179.48	18.48	478.34	18.48	2	7200	0
Ba	28	58.07	60.43	41.3	10.06	86.07	2.08	264.86	2.6	4	416	0
C3a	28	3.36	3.83	2.06	1.34	3.68	0.05	17.23	0.05	1	125	0
C3b/iC3b	29	59.01	94.58	24.97	13.68	57.56	9.4	485.39	9.405	4	15000	0
C4	29	2488.05	1998.79	1929.44	810.08	3472.82	31.51	7185.16	28.8	0	40000	0
C4b	29	130.76	137.09	112	17.49	180.12	3.77	640.53	15.55	3	25000	0
Ba:FB	28	0.05	0.05	0.04	0.02	0.06	0	0.22	NA	0	NA	0
C3a:C3	28	0	0.01	0	0	0	0	0.03	NA	0	NA	0
C3b/iC3b:C3	29	0.04	0.06	0.01	0.01	0.03	0	0.21	NA	0	NA	0
C4b:C4	29	0.06	0.09	0.04	0.02	0.07	0	0.49	NA	0	NA	0
C5	27	322.23	412.28	137	137	320.05	137	2022	137	14	100000	0
FD	29	98.45	116.77	59.43	41.42	113.73	4.81	600.14	0.8625	0	625	0
MBL	29	0.83	0.44	0.62	0.62	0.86	0.3	2.31	0.625	15	1250	0

Supplementary Table 4.

Descriptive statistics of aqueous humour baseline complement biomarkers concentrations from subjects in cohorts 1 to 4.

The number of samples correspond to the total number which could be measured and had a value within acceptance criteria. All concentrations shown are expressed in ng/mL.

Biomarker	n	Mean	SD	Median	25 th Percentile	75 th Percentile	Minimum	Maximum	LOD	N LOD	ULOQ	N ULOQ
FI	30	117.22	86.85	98.04	62.91	149.36	12.9	384.48	7.24	0	10000	0
C3	29	1881.16	3753.1	626.81	509.15	1016.37	5.34	18027.98	5.34	3	60000	2
FB	30	292.98	125.62	293.88	210.25	373.01	77.29	620.46	0.82	0	600	0
FH	30	81.97	53.39	70.97	45.81	116.55	0.61	225.46	0.61	1	3000	0
C1q	18	12.81	12.07	13.43	1.64	19.5	1.64	42.29	1.644	8	720	0
Ba	30	10.58	10.71	8.78	4.68	13.3	0.26	48.5	0.26	6	41.6	0
C3a	30	2.23	2.59	1.53	0.8	3.08	0.1	14.43	0.1	3	62.5	0
C3b/iC3b	29	7.17	7.72	3.7	2.35	11.25	0.64	35.85	0.6425	5	1500	0
C4	30	533.3	309.05	494.6	273.48	739.33	133.86	1304.22	2.25	0	4000	0
C4b	30	28.12	19.1	22.46	13.96	42.28	1.1	74.92	13.96	11	20000	0
Ba:FB	30	0.03	0.03	0.03	0.02	0.04	0	0.12	NA	0	NA	0
C3a:C3	29	0.04	0.13	0	0	0	0	0.56	NA	0	NA	0
C3b/iC3b:C3	29	0.02	0.04	0.01	0	0.02	0	0.12	NA	0	NA	0
C4b:C4	30	0.06	0.03	0.06	0.03	0.09	0	0.12	NA	0	NA	0
C5	30	185.77	290.99	109.6	109.6	109.6	56.89	1424	109.6	21	80000	0
FD	30	49.39	36.58	39.08	27.62	55.55	13.97	169.61	0.69	0	500	0
MBL	30	0.77	0.35	0.81	0.81	0.81	0.18	2.3	0.81	24	1000	0

Biomarker	n	Mean	SD	Median	25 th Percentile	75 th Percentile	Minimum	Maximum	LOD	N LOD	ULOQ	N ULOQ
FI	17	49870.41	24289.4	39078.61	33021.08	72494.63	22808.29	103375.3	69.4	0	100000	0
C3	18	962012.1	1043101	604199	505331.4	892754.6	283085.1	4823003	2220	0	2400000	0
FB	16	300852.6	121657.1	337809.1	199806.7	405761.4	105039	509413.1	328	0	240000	0
FH	18	533381	232796.5	486689	369826.8	558282.9	306135.1	1182989	820	0	1200000	0
C1q	11	277162.5	190015.1	230509.1	136774.3	362309.1	38599.86	601720.6	739.2	0	288000	0
Ba	24	1463	843.96	1309.74	940.7	1693.52	572.56	3994.84	40	0	8320	0
C3a	13	82.78	39.61	72.61	58.45	106.75	34.61	167.01	0.8	0	500	0
C3b/iC3b	12	2304.74	1409.21	1849.01	1280.33	2464.01	1068.35	5007.02	174	0	600000	0
C4	18	844393.7	548014.1	772567.2	319323.1	1220485	213627.3	1984159	1300	0	1600000	0
C4b	19	92538.56	92460.5	39370.39	35871.43	134493.1	32750.46	294918	157	0	200000	0
Ba:FB	13	0.01	0.01	0	0	0.01	0	0.02	NA	0	NA	0
C3a:C3	13	0	0	0	0	0	0	0	NA	0	NA	0
C3b/iC3b:C3	12	0	0	0	0	0	0	0	NA	0	NA	0
C4b:C4	18	0.12	0.06	0.12	0.07	0.16	0.02	0.22	NA	0	NA	0
C5	19	184245.6	58160.91	166647.7	147884.4	214571.5	112160.5	300772.6	864.8	0	800000	0
FD	17	3488.28	1266.75	3895.65	2414.58	4350.97	1676.85	6036.23	4.5	0	5000	0
MBL	19	2083.38	2020.93	915.63	707.42	3337.35	101.59	7415.09	3.3	0	10000	0

Supplementary Table 5.

Descriptive statistics of plasma baseline complement biomarkers concentrations from subjects in cohorts 1 to 4.

The number of samples correspond to the total number which could be measured and had a value within acceptance criteria. All concentrations shown are expressed in ng/mL.

Supplementary Table 6.

Temporal change in protein levels as evaluated using the Olink platform.

Spreadsheet listing Olink (<https://olink.com/>) protein identifiers (IDs) that passed quality control alongside metrics for differential expression in vitreous humor comparing baseline to post-treatment at week 12 and week 36 timepoints. fdr = false discovery rate; NPX = normalised protein expression.

OlinkID	symbol	Olink panel	median_N PX (baseline)	median_N PX (week 12)	median_N PX (week 36)	p_value (week 12 - baseline)	fdr (week 12 - baseline)	fold change (week 12 - baseline)	p_value (week 36 - baseline)	fdr (week 36 - baseline)	fold change (week 36 - baseline)
OID20051	HNRNPK	Cardiomet abolic	-3.12295	-2.0972	-2.64035	0.014699	0.028414	2.049611	0.37648	0.456654	1.219128
OID20052	CEBPB	Cardiomet abolic	-1.0125	-0.21365	-0.249	0.013955	0.027176	1.266546	0.058387	0.096841	1.705684
OID20055	CRHR1	Cardiomet abolic	-0.4056	0.00055	0.06065	0.327746	0.399406	1.1776	0.185553	0.254671	1.381514
OID20062	AKR1C4	Cardiomet abolic	-0.78755	-0.81395	-0.2486	0.214993	0.282709	1.453368	0.20042	0.270678	1.751693
OID20064	SERPINB5	Cardiomet abolic	-2.53095	0.23315	-0.08395	0.000126	0.000502	5.307932	0.004613	0.012001	2.44037
OID20065	HK2	Cardiomet abolic	-0.34135	-0.24105	-0.3366	0.241496	0.310132	1.234605	0.941123	0.954158	1.08666
OID20067	VSTM2L	Cardiomet abolic	1.9692	1.30905	1.37605	0.056673	0.089278	-1.59842	0.051732	0.086853	-1.70385
OID20068	TIA1	Cardiomet abolic	-1.2432	0.0282	-0.154	0.001264	0.003452	2.119318	0.006469	0.015962	2.399943
OID20069	CNPY2	Cardiomet abolic	0.2917	0.75625	0.60015	0.045243	0.074366	1.404786	0.136512	0.197483	1.453469
OID20072	SUSD1	Cardiomet abolic	-4.56985	-4.345	-4.42105	0.315394	0.386499	1.1776	0.338239	0.419646	1.534002
OID20074	TNF	Cardiomet abolic	-0.6919	0.1163	0.19355	0.090972	0.134905	1.225822	0.059468	0.09819	2.008544
OID20079	ACTA2	Cardiomet abolic	-1.5104	-0.8014	-0.41595	0.013183	0.026128	1.375446	0.001313	0.004267	1.897434
OID20081	EDIL3	Cardiomet abolic	0.7179	0.54985	0.6915	0.300225	0.372628	-1.05457	0.261678	0.338425	1.335056
OID20083	PTN	Cardiomet abolic	6.1776	6.6878	6.71405	0.240719	0.309781	1.423014	0.040851	0.070748	1.400022
OID20086	COMT	Cardiomet abolic	-3.1683	-1.76115	-2.2563	0.001018	0.002926	2.291036	0.032182	0.057832	2.227615
OID20087	MMP7	Cardiomet abolic	-1.2438	-0.97405	-0.93775	0.67172	0.728083	1.079789	0.237966	0.311407	1.398713
OID20089	ADAMTS1 6	Cardiomet abolic	-1.056	-0.8707	-0.74315	0.225748	0.292579	-1.072	0.712485	0.767722	-1.00298
OID20090	GP2	Cardiomet abolic	-5.38955	-4.51705	-4.8175	0.12129	0.172733	1.764611	0.292659	0.373217	1.965573
OID20094	SLITRK6	Cardiomet abolic	0.9046	-0.36395	-0.107	2.92E-08	1.52E-06	-2.23814	4.27E-06	5.58E-05	-1.55833
OID20095	GPR37	Cardiomet abolic	4.3167	1.52145	1.69925	1.24E-06	1.69E-05	-4.6491	7.4E-07	2.03E-05	-5.37532
OID20096	GZMH	Cardiomet abolic	-5.98065	-3.7922	-3.7076	1.63E-05	9.78E-05	4.466981	1.83E-05	0.000149	4.765244
OID20098	THPO	Cardiomet abolic	-6.9717	-5.87555	-6.47275	0.007501	0.015914	1.858257	0.098794	0.150199	1.493762
OID20099	CLEC1A	Cardiomet abolic	-4.38635	-4.08795	-4.02765	0.460643	0.535224	1.031004	0.356279	0.436601	-1.05709
OID20100	ENTPD6	Cardiomet abolic	-0.0159	-0.59585	-0.3971	0.000272	0.000951	-1.37368	0.001534	0.004839	-1.37183
OID20101	IL6	Cardiomet abolic	0.13915	0.85985	1.24275	0.160015	0.219895	1.680686	0.015771	0.032725	1.608928
OID20102	CEACAM8	Cardiomet abolic	-5.1184	-4.35085	-4.2898	0.091042	0.134905	1.897829	0.161371	0.225427	1.775961
OID20103	S100P	Cardiomet abolic	-3.2768	-1.6713	-2.3663	0.004923	0.011032	3.313716	0.012443	0.027325	2.11367

Supplementary Table 6. (continued)

OID20104	ENTPD5	Cardiometabolic	-2.18205	-1.62835	-1.61445	0.036847	0.062339	1.602528	0.015249	0.031831	1.526682
OID20105	ACE2	Cardiometabolic	-3.3173	-2.6121	-2.78	0.002155	0.005403	1.948886	0.008034	0.018971	2.008544
OID20106	DCN	Cardiometabolic	-0.83115	-1.35895	-1.06985	0.057028	0.08958	-1.21348	0.139964	0.20089	-1.17985
OID20107	GLO1	Cardiometabolic	0.65635	1.86205	1.60855	4.02E-06	3.8E-05	2.06401	5.8E-05	0.000353	1.987424
OID20108	PAG1	Cardiometabolic	-2.50415	-2.18245	-3.03555	0.347596	0.42033	1.249802	0.156596	0.220157	-1.38453
OID20109	ADAM15	Cardiometabolic	-2.8568	-3.27655	-3.19475	0.001101	0.003099	-1.33432	0.004023	0.010696	-1.18308
OID20110	USP8	Cardiometabolic	-4.1697	-3.72955	-3.58785	0.032681	0.056157	1.324502	0.098246	0.149618	1.452764
OID20113	CTSH	Cardiometabolic	0.77825	-0.8898	-0.5457	1.71E-08	1.17E-06	-3.62744	9.66E-07	2.31E-05	-2.71726
OID20115	MSTN	Cardiometabolic	-0.38575	-1.6902	-1.60615	2.52E-06	2.83E-05	-2.77733	9.77E-06	0.000101	-2.21223
OID20116	BOC	Cardiometabolic	-1.82555	-1.92875	-1.57635	0.920032	0.932775	-1.00553	0.008776	0.020372	1.198891
OID20121	CCL27	Cardiometabolic	-4.6039	-5.76355	-5.74235	1.09E-05	7.33E-05	-2.16168	9.89E-05	0.000527	-1.99917
OID20122	AK1	Cardiometabolic	-2.2553	-1.22635	-1.238	0.001549	0.004109	2.038559	0.008574	0.020031	2.008544
OID20123	ANXA4	Cardiometabolic	-1.37655	-0.12485	-0.4641	0.000545	0.001739	2.481393	0.015035	0.031626	2.120199
OID20125	NTproBNP	Cardiometabolic	1.8125	0.76945	1.0228	0.001989	0.005068	-1.56901	0.018138	0.036788	-1.44683
OID20126	CLC	Cardiometabolic	-3.16675	-2.9296	-3.2115	0.248412	0.318166	1.170358	0.730422	0.784739	-1.07777
OID20128	LEPR	Cardiometabolic	-3.1867	-2.6109	-2.68665	0.193925	0.259038	1.229609	0.125495	0.184958	1.094901
OID20129	TSHB	Cardiometabolic	-5.669	-5.21285	-5.105	0.000251	0.000893	1.590411	0.000237	0.001093	1.469372
OID20130	S100A11	Cardiometabolic	-2.29	-0.0738	-0.68395	8.89E-05	0.000374	4.912645	0.001475	0.004688	3.088701
OID20131	VSIR	Cardiometabolic	-0.04725	0.26275	0.31925	0.005195	0.011571	1.280139	0.006024	0.015067	1.437837
OID20132	PCDH17	Cardiometabolic	1.27935	0.2328	0.3165	1.39E-05	8.81E-05	-2.01587	5.21E-05	0.000325	-1.86361
OID20133	PDGFRA	Cardiometabolic	0.56085	0.22525	0.27535	0.001297	0.003534	-1.19943	0.002507	0.007207	-1.20535
OID20135	SNX9	Cardiometabolic	-0.57915	0.00715	-0.00525	0.001569	0.004142	1.368552	0.001105	0.003674	1.194287
OID20136	NTRK2	Cardiometabolic	1.21735	1.01155	0.89265	0.047412	0.077237	-1.26326	0.011835	0.026253	-1.10799
OID20137	ACY1	Cardiometabolic	-2.8143	-1.823	-1.84175	0.000805	0.002409	2.083848	0.000392	0.001658	2.103658
OID20141	TGM2	Cardiometabolic	-1.0627	0.7811	0.84285	0.000306	0.001036	2.927055	5.01E-05	0.000314	3.092021
OID20142	DPP7	Cardiometabolic	5.86985	6.48365	6.42375	0.005653	0.01244	1.649753	0.014669	0.030924	1.56401
OID20143	PLIN3	Cardiometabolic	-1.98465	-1.7095	-1.83605	0.009231	0.018946	1.054128	0.120815	0.179264	1.104735
OID20144	PLPBP	Cardiometabolic	-1.0848	0.40165	0.3852	3.78E-05	0.000183	2.543385	0.000447	0.001825	2.427296
OID20145	ICAM5	Cardiometabolic	1.7618	0.52035	0.45805	1.72E-07	5.23E-06	-2.24078	5.76E-08	3.95E-06	-2.29779
OID20146	PRKAR1A	Cardiometabolic	-2.96105	-2.95575	-2.8304	0.312642	0.385036	1.215669	0.40239	0.480421	1.095052
OID20147	IL6ST	Cardiometabolic	1.48325	0.9458	1.07105	0.099585	0.14677	-1.41819	0.030475	0.055291	-1.31148
OID20148	DCTPP1	Cardiometabolic	-1.65735	-0.90335	-1.21105	0.007399	0.015776	1.686462	0.067658	0.108794	1.200762

Supplementary Table 6. (continued)

OID20149	PM20D1	Cardiometabolic	-8.24105	-8.01325	-7.4456	0.218658	0.286158	1.268655	0.46836	0.545922	1.543173
OID20150	LACTB2	Cardiometabolic	0.17515	0.7917	0.7566	0.0002	0.000736	1.467235	0.000633	0.002405	1.388763
OID20151	FADD	Cardiometabolic	-2.76225	-1.87765	-2.5281	0.001776	0.004578	1.672087	0.062591	0.102441	1.149973
OID20152	CGREF1	Cardiometabolic	2.18595	1.45285	1.69525	7.4E-07	1.27E-05	-1.56694	1E-05	0.000103	-1.37307
OID20153	CXCL8	Cardiometabolic	3.38195	3.91985	3.94815	0.006349	0.013696	1.556764	0.003856	0.010301	1.284494
OID20154	LGALS1	Cardiometabolic	-0.71165	-1.6252	-1.6674	0.001638	0.004282	-2.3818	0.013712	0.029463	-1.87664
OID20155	IGFBPL1	Cardiometabolic	3.5638	3.0512	2.85285	9.9E-05	0.000406	-1.65565	0.003013	0.008419	-1.35613
OID20156	MNDA	Cardiometabolic	-2.66915	-2.50195	-2.37805	0.433257	0.506081	1.094977	0.583001	0.649884	1.039976
OID20157	CDH6	Cardiometabolic	-0.1631	-0.4946	-0.3536	0.018362	0.034641	-1.26177	0.092584	0.142178	-1.12276
OID20158	IL19	Cardiometabolic	-0.1562	-0.19465	-0.1844	0.176686	0.239822	-1.00744	0.217306	0.288864	1.048953
OID20159	ACAN	Cardiometabolic	-2.59115	-1.9935	-2.3634	0.029255	0.051149	1.365662	0.995493	0.998161	-1.13002
OID20160	PLXNB3	Cardiometabolic	1.84295	0.8225	0.81085	3.2E-10	1.43E-07	-1.96319	3.25E-10	1.19E-07	-2.14013
OID20161	TYRO3	Cardiometabolic	-1.26035	-1.2746	-1.3705	0.5117	0.584648	1.066068	0.062696	0.102441	-1.14425
OID20162	EIF4EBP1	Cardiometabolic	-0.76635	-0.78745	-0.94565	0.341831	0.41473	1.094825	0.896842	0.922034	-1.13233
OID20163	PDCD6	Cardiometabolic	-4.783	-3.80755	-4.3597	0.001484	0.003957	2.100816	0.027119	0.050298	1.241126
OID20164	APLP1	Cardiometabolic	4.003	5.06305	5.03255	1.12E-05	7.4E-05	2.365426	1.41E-05	0.000124	2.320764
OID20165	CLEC5A	Cardiometabolic	4.1414	4.2438	3.93915	0.230096	0.297229	-1.02229	0.046816	0.079573	-1.02758
OID20166	GSTA1	Cardiometabolic	-5.7216	-4.5786	-4.12235	0.02851	0.050247	2.556642	0.012221	0.027054	4.640724
OID20167	IGSF8	Cardiometabolic	1.89505	1.06695	1.08975	2.01E-08	1.22E-06	-1.77547	2.65E-07	1.04E-05	-1.83668
OID20168	MEP1B	Cardiometabolic	-4.5822	-4.0754	-3.8825	0.034494	0.059087	1.621411	0.021594	0.042339	1.523247
OID20169	SDC1	Cardiometabolic	-0.28695	0.1186	-0.154	0.042201	0.069996	1.098055	0.033553	0.059905	1.123928
OID20170	CDH2	Cardiometabolic	0.88435	0.31835	0.28	0.000701	0.002115	-1.39605	0.000534	0.002139	-1.50139
OID20171	MCFD2	Cardiometabolic	3.4476	4.09325	4.11955	0.000208	0.000757	1.597759	0.000303	0.001358	1.666822
OID20172	FCRL1	Cardiometabolic	-5.1429	-5.31905	-5.39315	0.395066	0.46997	-1.01329	0.859014	0.892334	1.023067
OID20173	GNPMB	Cardiometabolic	0.0788	0.20515	0.32305	0.519993	0.592775	1.11238	0.158233	0.221607	1.116358
OID20174	QDPR	Cardiometabolic	0.13935	1.3201	1.35325	3.6E-07	8.24E-06	1.899013	1.72E-06	3.1E-05	1.976845
OID20175	GHRL	Cardiometabolic	-0.94295	-0.18935	-0.3302	0.001144	0.003172	1.591128	0.001552	0.004883	1.622592
OID20176	PPP1R2	Cardiometabolic	-1.054	-0.49275	-0.69925	0.001052	0.002999	1.614682	0.121569	0.180138	1.302154
OID20177	CD2AP	Cardiometabolic	-3.4592	-2.68985	-2.78585	0.007775	0.016448	1.43485	0.048531	0.082107	1.273104
OID20178	NADK	Cardiometabolic	-2.28855	-2.34755	-2.56565	0.749563	0.79697	-1.13237	0.781885	0.827878	-1.20849
OID20179	CBLIF	Cardiometabolic	-3.0387	-3.48665	-3.00365	0.021337	0.039375	-1.29258	0.511294	0.581763	1.02172
OID20180	TINAGL1	Cardiometabolic	0.7727	-0.4218	-0.4995	3.47E-05	0.00017	-2.12505	7.06E-05	0.000414	-2.05359

Supplementary Table 6. (continued)

OID20181	CANT1	Cardiometabolic	2.29415	2.4346	2.4095	0.724579	0.777322	1.148101	0.909878	0.930211	1.169507
OID20182	SSC4D	Cardiometabolic	-6.1452	-6.3659	-6.43165	0.030145	0.052372	-1.5837	0.103982	0.15748	-1.35952
OID20183	OLR1	Cardiometabolic	3.59345	2.7049	2.62965	2.34E-05	0.000128	-1.9908	9.15E-05	0.000503	-1.71814
OID20184	DIABLO	Cardiometabolic	-0.80575	-0.94675	-1.28865	0.346535	0.41951	1.184066	0.377619	0.457008	-1.01424
OID20185	LRP11	Cardiometabolic	3.3297	0.76465	1.12215	1.37E-07	5E-06	-6.05993	1.4E-07	6.98E-06	-4.24775
OID20186	SEMA3F	Cardiometabolic	3.56895	1.3303	1.7964	5.93E-06	4.89E-05	-4.54837	2.28E-05	0.000176	-2.79607
OID20187	LEP	Cardiometabolic	-2.93065	-4.62285	-3.94635	0.025866	0.046331	-2.36117	0.1159	0.173376	-1.70303
OID20188	LPL	Cardiometabolic	0.85185	0.17235	0.0743	4.3E-06	4.03E-05	-1.72979	2.33E-06	3.69E-05	-1.73333
OID20189	ADGRG2	Cardiometabolic	-3.75865	-3.7664	-3.83005	0.383682	0.459414	-1.18933	0.609186	0.672924	-1.20953
OID20190	SIGLEC7	Cardiometabolic	-2.0063	-1.9577	-1.95845	0.510522	0.58391	-1.05398	0.370019	0.450396	1.063264
OID20191	NPTXR	Cardiometabolic	3.74155	2.70535	2.6873	1.57E-09	2.47E-07	-2.14154	3.25E-09	5.94E-07	-2.09043
OID20192	GRAP2	Cardiometabolic	-3.2926	-2.54345	-2.48315	0.003772	0.008756	1.927526	0.00057	0.002253	2.043015
OID20193	CDHR5	Cardiometabolic	-3.57525	-3.66525	-3.60865	0.219088	0.286379	-1.02772	0.42665	0.503182	-1.00832
OID20194	THOP1	Cardiometabolic	-2.94885	-1.71435	-1.99255	6.19E-06	4.99E-05	2.038418	0.000396	0.001664	1.918129
OID20195	GLRX	Cardiometabolic	-1.72025	-1.06705	-1.44465	0.078607	0.119213	1.214279	0.903943	0.925004	1.178458
OID20196	KITLG	Cardiometabolic	-2.8937	-3.0282	-2.9451	0.036267	0.061452	-1.10792	0.489795	0.564767	-1.02815
OID20197	PILRB	Cardiometabolic	0.33415	-0.02375	0.12695	0.355223	0.428609	-1.10539	0.419489	0.497407	-1.03422
OID20198	CLUL1	Cardiometabolic	4.59345	4.8567	5.0734	0.552572	0.622281	1.229695	0.120682	0.179264	1.295402
OID20199	AHCY	Cardiometabolic	-0.426	0.90675	0.2838	0.001024	0.002927	2.038771	0.126807	0.186142	1.403569
OID20200	FABP2	Cardiometabolic	-0.9443	-0.85315	-0.7279	0.205442	0.272763	1.250019	0.028035	0.051734	1.18173
OID20201	REN	Cardiometabolic	0.5535	-1.17985	-0.6777	1.81E-07	5.23E-06	-2.65222	3.11E-06	4.51E-05	-2.36379
OID20203	RNASE3	Cardiometabolic	-1.9556	-2.5395	-2.68575	0.063104	0.098561	-2.37957	0.07962	0.12489	-2.04798
OID20204	SOST	Cardiometabolic	-3.44365	-3.5435	-3.44765	0.148035	0.20575	-1.1839	0.372902	0.452927	-1.00814
OID20205	NPDC1	Cardiometabolic	4.5501	2.7494	3.297	1.27E-05	8.15E-05	-2.61607	2.23E-05	0.000173	-2.2801
OID20206	CD69	Cardiometabolic	-1.5072	-1.65515	-1.26495	0.605749	0.669801	-1.05727	0.138952	0.199698	1.048589
OID20207	CXCL5	Cardiometabolic	-3.9409	-4.9733	-4.57425	0.050352	0.080684	-1.82887	0.172077	0.238863	-1.68553
OID20208	COL4A1	Cardiometabolic	3.47175	2.7372	2.7165	0.001942	0.004958	-1.55925	0.017789	0.036305	-1.2781
OID20209	LILRA5	Cardiometabolic	-4.69675	-4.42045	-4.38715	0.067847	0.104923	1.191848	0.135909	0.196871	1.169588
OID20210	MARCO	Cardiometabolic	-4.24165	-4.77005	-4.6274	0.078766	0.11929	-1.82456	0.510927	0.581763	-1.63768
OID20211	ANGPTL1	Cardiometabolic	3.50645	2.214	2.61545	6.38E-09	6.37E-07	-2.25566	9.61E-07	2.31E-05	-1.89993
OID20212	PRSS27	Cardiometabolic	1.0384	0.41855	0.36085	0.000208	0.000757	-1.53214	0.000816	0.002871	-1.40342
OID20213	SORT1	Cardiometabolic	2.04395	1.39905	1.44935	8.07E-07	1.28E-05	-1.65353	8.45E-06	9.28E-05	-1.44944

Supplementary Table 6. (continued)

OID20214	SFTPD	Cardiometabolic	-1.1441	-2.569	-2.49815	0.000138	0.00054	-2.68263	0.000423	0.001758	-2.78813
OID20215	DDC	Cardiometabolic	-5.152	-3.9713	-3.98625	0.000565	0.001793	1.630201	9.34E-05	0.000508	2.240313
OID20216	SDC4	Cardiometabolic	1.63005	1.39635	1.4247	0.013959	0.027176	-1.12674	0.023816	0.045716	-1.03987
OID20217	HMOX1	Cardiometabolic	-6.9408	-6.25045	-5.88375	0.473461	0.549066	1.133277	0.237942	0.311407	1.483598
OID20218	SNAP23	Cardiometabolic	-4.7926	-4.2628	-4.5595	0.771979	0.817389	1.534587	0.327861	0.411419	-1.25536
OID20219	SERPINA1 2	Cardiometabolic	-6.6406	-5.7151	-5.7247	0.022797	0.041443	1.767794	0.19098	0.260168	1.847277
OID20220	GH1	Cardiometabolic	-4.54605	-4.58155	-4.46655	0.044469	0.073203	1.115121	0.000808	0.002852	1.130687
OID20221	SPON2	Cardiometabolic	0.2515	0.00045	0.1079	0.021331	0.039375	-1.16724	0.482663	0.559624	-1.03893
OID20222	NECTIN2	Cardiometabolic	0.77875	-0.07335	-0.0166	2.53E-07	6.76E-06	-1.80513	2.56E-06	3.93E-05	-1.63972
OID20223	CHRD12	Cardiometabolic	-3.5096	-3.96675	-3.9748	0.077898	0.118453	-1.33025	0.059381	0.09819	-1.53034
OID20224	ROR1	Cardiometabolic	0.68485	-0.29055	-0.1956	4.37E-09	4.92E-07	-2.08276	3.51E-08	3.21E-06	-2.11462
OID20225	CA13	Cardiometabolic	-2.69135	0.08815	-0.26555	2.16E-07	5.93E-06	6.191762	9.29E-06	9.9E-05	4.367931
OID20226	CCDC80	Cardiometabolic	2.50775	1.16525	1.237	1.18E-06	1.64E-05	-2.43066	1.08E-06	2.38E-05	-2.46229
OID20227	FBP1	Cardiometabolic	-2.22705	-2.4194	-2.6548	0.736224	0.785591	-1.21394	0.212316	0.284296	-1.55251
OID20228	CTSL	Cardiometabolic	2.95515	2.5135	2.46795	0.000303	0.001036	-1.33219	0.002176	0.00637	-1.36183
OID20229	NRCAM	Cardiometabolic	3.3738	3.8806	3.89905	0.001061	0.003017	1.659157	0.003593	0.009764	1.606143
OID20230	HSPB1	Cardiometabolic	-2.09035	-0.11085	0.68015	0.001015	0.002926	4.398769	0.000633	0.002405	7.269973
OID20231	THBD	Cardiometabolic	-3.20575	-2.9078	-2.87535	0.026442	0.047195	1.294684	0.063883	0.10407	1.206979
OID20232	KYAT1	Cardiometabolic	1.96955	2.4123	2.34545	0.00461	0.010414	1.303689	0.070868	0.112772	1.255707
OID20233	FAM3C	Cardiometabolic	2.8006	3.94735	3.84425	2.89E-05	0.000149	2.261845	8.1E-05	0.000463	2.353487
OID20234	TYMP	Cardiometabolic	-3.15685	-3.39935	-2.7864	0.792136	0.833109	-1.32846	0.544896	0.615531	1.061902
OID20235	PCSK9	Cardiometabolic	-2.3093	-2.6694	-2.80075	0.003069	0.007295	-1.47116	0.156336	0.220157	-1.37302
OID20236	IGFBP7	Cardiometabolic	4.0016	5.22055	5.2207	3.4E-06	3.44E-05	2.710859	1.78E-05	0.000147	2.671963
OID20237	UMOD	Cardiometabolic	-3.15355	-2.47715	-2.2766	0.003855	0.008911	1.237304	0.002813	0.007919	1.270767
OID20238	PLTP	Cardiometabolic	1.58075	-0.1272	0.44095	1.53E-06	1.95E-05	-2.69213	5.58E-05	0.000342	-2.38692
OID20239	MFAP5	Cardiometabolic	-0.7269	-3.6572	-3.0168	1.37E-08	1.16E-06	-5.44696	3.15E-07	1.15E-05	-5.17515
OID20240	LDLR	Cardiometabolic	2.7833	2.78925	2.537	0.685529	0.741587	1.009541	0.339675	0.420952	-1.00333
OID20241	CA4	Cardiometabolic	2.6614	3.21545	3.14445	0.047017	0.076708	1.274075	0.011614	0.025866	1.6905
OID20242	EPHB4	Cardiometabolic	2.0822	1.48205	1.52	5.82E-05	0.000261	-1.4651	0.00031	0.001369	-1.47406
OID20243	CDH5	Cardiometabolic	-1.48885	-0.75575	-0.9611	0.243043	0.311754	1.079902	0.03982	0.069181	1.261246
OID20244	CES1	Cardiometabolic	-1.3593	-1.1135	-1.0188	0.192244	0.257419	1.079827	0.122752	0.181647	1.181607
OID20245	FABP4	Cardiometabolic	2.274	2.45905	2.4952	0.038948	0.065091	1.25462	0.009089	0.020965	1.302876

Supplementary Table 6. (continued)

OID20246	IGFBP1	Cardiometabolic	1.68785	1.014	1.4168	0.029773	0.051891	-1.36254	0.233802	0.307442	-1.03764
OID20247	PLAT	Cardiometabolic	3.1782	1.43575	1.9779	4.56E-06	4.07E-05	-2.26475	0.000235	0.00109	-1.94181
OID20248	ICAM2	Cardiometabolic	-0.0621	-0.46795	-0.241	0.000152	0.000582	-1.31098	0.003229	0.008886	-1.16874
OID20249	ADAMTS13	Cardiometabolic	-3.3364	-3.7605	-3.1314	0.005753	0.012584	-1.19478	0.177871	0.246282	-1.05651
OID20250	IL1RL1	Cardiometabolic	-0.39395	1.0682	-0.1772	0.002096	0.005302	1.925789	0.732871	0.785832	1.002394
OID20251	GDF15	Cardiometabolic	4.69195	4.8067	4.76265	0.819809	0.855656	-1.10114	0.198086	0.267855	1.065847
OID20252	PLA2G1B	Cardiometabolic	1.24045	1.9389	2.10095	4.45E-05	0.000208	1.628507	4.01E-06	5.37E-05	1.679638
OID20253	NOTCH3	Cardiometabolic	0.61975	-0.74135	-0.6802	8.65E-07	1.32E-05	-2.87607	4.42E-06	5.58E-05	-2.85067
OID20254	LTBP2	Cardiometabolic	6.80505	7.12925	7.2111	0.001406	0.003794	1.250756	0.001918	0.005802	1.432564
OID20255	TGFBR3	Cardiometabolic	1.80595	-0.7745	-0.2838	1.17E-07	4.46E-06	-4.21768	8.54E-07	2.18E-05	-4.36748
OID20256	VWF	Cardiometabolic	-4.1984	-3.6864	-3.72785	0.030078	0.052339	1.535225	0.091658	0.141151	1.347934
OID20257	PEAR1	Cardiometabolic	-1.25715	-0.89795	-1.15795	0.119236	0.170915	1.102975	0.206742	0.277849	1.147266
OID20258	REG3A	Cardiometabolic	2.63135	3.49675	3.30875	7.53E-06	5.62E-05	2.016915	6.41E-05	0.000378	1.800004
OID20259	XG	Cardiometabolic	2.75805	2.65135	2.87225	0.240941	0.309781	-1.08249	0.847698	0.881621	1.043333
OID20260	ADGRE5	Cardiometabolic	-1.20455	-1.3424	-0.9891	0.01363	0.026869	-1.10026	0.555386	0.62481	1.036629
OID20261	FUCA1	Cardiometabolic	3.29205	3.5481	3.54775	0.931647	0.94281	1.129865	0.34021	0.421139	1.091642
OID20262	CD209	Cardiometabolic	-0.6036	-1.49155	-1.1973	0.003078	0.0073	-2.01342	0.144998	0.206496	-1.48468
OID20263	MB	Cardiometabolic	0.43045	2.52225	2.54785	6.47E-05	0.000285	4.262796	8.82E-06	9.59E-05	3.880386
OID20264	DKK3	Cardiometabolic	2.52515	5.05065	4.90655	1.6E-07	5.23E-06	5.2014	8.24E-07	2.18E-05	5.994336
OID20265	TNFRSF10C	Cardiometabolic	0.33525	0.65665	0.80165	0.255269	0.325157	1.130491	0.074564	0.118141	1.163846
OID20266	REG1B	Cardiometabolic	1.87605	3.0646	2.64535	0.000941	0.002742	2.377425	0.005486	0.013943	1.699959
OID20267	IL2RA	Cardiometabolic	0.2688	0.54755	0.79155	0.020736	0.038568	1.342898	0.006616	0.016252	1.44483
OID20268	PDGFRB	Cardiometabolic	1.9014	1.9775	1.79855	0.085954	0.12858	1.008178	0.07779	0.122369	-1.06751
OID20269	MET	Cardiometabolic	1.68935	1.2394	1.36215	0.000934	0.002741	-1.27655	0.004743	0.012311	-1.25306
OID20270	ESAM	Cardiometabolic	4.8443	4.8731	4.9209	0.343529	0.41633	1.184682	0.807567	0.847714	1.131158
OID20271	CA3	Cardiometabolic	1.63455	2.18615	2.57175	0.015737	0.030208	1.440181	0.045966	0.078249	1.654849
OID20272	TNFSF13B	Cardiometabolic	2.87455	2.7727	3.6339	0.316723	0.387262	1.181607	0.004079	0.010792	1.457807
OID20273	ALCAM	Cardiometabolic	1.47195	1.4711	1.37905	0.711143	0.766275	-1.04012	0.276038	0.354906	-1.01234
OID20274	PLA2G2A	Cardiometabolic	1.6505	0.4827	0.6526	0.020759	0.038568	-1.6362	0.00253	0.007253	-1.85087
OID20275	MSMB	Cardiometabolic	0.18025	0.86665	0.96115	0.000499	0.001617	1.611495	0.00024	0.001102	1.818438
OID20276	CTSB	Cardiometabolic	5.00335	6.03285	6.1357	1.82E-06	2.22E-05	2.203581	9.07E-06	9.76E-05	2.204269
OID20277	ICAM3	Cardiometabolic	-2.3278	-1.9546	-1.8541	0.04338	0.071627	1.179397	0.002011	0.005999	1.461348

Supplementary Table 6. (continued)

OID20278	DPT	Cardiomet abolic	0.7053	1.15475	1.24785	0.835838	0.869082	1.275489	0.236252	0.310293	1.416372
OID20279	CRTAC1	Cardiomet abolic	1.10685	2.5309	2.62045	1.19E-06	1.64E-05	2.683377	2.35E-06	3.69E-05	2.819032
OID20280	CPA1	Cardiomet abolic	1.54865	1.84285	1.98005	0.104825	0.153056	1.164895	0.007683	0.01826	1.239407
OID20281	SERPINA1 1	Cardiomet abolic	-5.4898	-5.83095	-4.9831	0.146339	0.204037	-1.16922	0.042789	0.073756	1.297019
OID20282	CXCL16	Cardiomet abolic	4.44915	4.38385	4.5463	0.651942	0.710856	1.10202	0.605105	0.669763	1.161388
OID20283	SEMA7A	Cardiomet abolic	4.7195	6.15355	6.1951	9.9E-07	1.45E-05	3.075562	4.37E-06	5.58E-05	3.013625
OID20284	LGALS3	Cardiomet abolic	1.45385	1.77015	1.7039	0.009407	0.019235	1.253186	0.035684	0.063195	1.206435
OID20285	MEGF9	Cardiomet abolic	3.3946	3.31065	3.23485	0.267255	0.338462	-1.04395	0.137804	0.198466	-1.01227
OID20286	CORO1A	Cardiomet abolic	-0.85915	-0.83425	-0.97065	0.707223	0.763551	-1.2038	0.6908	0.748193	-1.1303
OID20287	ENG	Cardiomet abolic	-1.2692	-1.5162	-1.51575	0.001096	0.003093	-1.13895	0.186894	0.256087	-1.04247
OID20288	ENPP2	Cardiomet abolic	2.6436	3.71725	3.73615	2.33E-05	0.000128	2.31498	0.00011	0.000579	2.320764
OID20289	CPB1	Cardiomet abolic	1.0698	1.53435	1.71925	0.003211	0.00755	1.358533	0.001938	0.005829	1.428598
OID20290	SELE	Cardiomet abolic	-4.4777	-3.78825	-3.66715	0.053264	0.084514	1.392522	0.013451	0.029188	1.306312
OID20291	PAM	Cardiomet abolic	4.3495	4.78415	4.78325	0.011409	0.023028	1.18218	0.02045	0.040605	1.141357
OID20292	COL6A3	Cardiomet abolic	2.67375	3.11405	2.9665	0.000831	0.002473	1.357592	0.000851	0.002966	1.573525
OID20293	CST6	Cardiomet abolic	2.91135	3.1997	3.25475	0.895037	0.916255	1.023173	0.33802	0.419646	1.18896
OID20294	CSTB	Cardiomet abolic	3.32	4.0211	3.7827	4.85E-06	4.3E-05	1.592618	6.41E-05	0.000378	1.377068
OID20295	DLK1	Cardiomet abolic	0.8631	0.84865	0.9718	0.1767	0.239822	-1.04837	0.717095	0.771932	1.065182
OID20296	GUSB	Cardiomet abolic	5.6087	5.3186	5.2858	0.026477	0.047195	-1.22226	0.013536	0.029314	-1.17862
OID20297	SCARF1	Cardiomet abolic	1.6098	1.92835	1.77845	0.188427	0.253235	1.143733	0.066504	0.107543	1.064296
OID20298	CELA3A	Cardiomet abolic	-1.5249	-2.0275	-1.6309	0.168843	0.231159	-1.16748	0.93342	0.949856	-1.0601
OID20299	CCN3	Cardiomet abolic	4.23515	4.0769	4.20215	0.270109	0.341288	-1.00542	0.998161	0.998161	1.06135
OID20300	HYOU1	Cardiomet abolic	1.3782	0.9525	1.18695	0.009958	0.020249	-1.35637	0.04412	0.075812	-1.1928
OID20301	LILRB2	Cardiomet abolic	-0.2512	0.10315	0.07985	0.410452	0.483558	1.089563	0.067674	0.108794	1.14671
OID20302	ST6GAL1	Cardiomet abolic	0.02665	-0.9458	-0.80685	0.000184	0.000688	-1.94544	0.030886	0.055869	-1.62282
OID20303	TIE1	Cardiomet abolic	0.63035	-0.14555	0.0171	0.149052	0.206902	-1.54601	0.379908	0.458899	-1.35933
OID20304	SIRPA	Cardiomet abolic	4.29015	4.44435	4.3552	0.036998	0.062499	1.272089	0.085198	0.133259	1.163927
OID20305	CASP3	Cardiomet abolic	0.0874	2.3034	1.5338	2.35E-05	0.000128	5.025266	0.002343	0.006824	2.382457
OID20306	PTPRS	Cardiomet abolic	3.621	3.83375	3.91315	0.008162	0.017167	1.258975	0.030155	0.054909	1.31768
OID20307	CNTN1	Cardiomet abolic	4.12805	4.63045	4.57595	0.000941	0.002742	1.463325	0.004579	0.011942	1.697839
OID20308	BPIFB1	Cardiomet abolic	-6.09785	-6.52765	-6.14955	0.397412	0.471739	-1.16429	0.491281	0.56505	-1.12748
OID20309	AZU1	Cardiomet abolic	-2.08095	-2.24835	-1.7782	0.11121	0.16067	-1.53938	0.137914	0.198466	1.041707

Supplementary Table 6. (continued)

OID20310	PTPRF	Cardiometabolic	2.50195	1.44795	1.5452	6.82E-06	5.24E-05	-1.73466	4.63E-05	0.000301	-1.84421
OID20311	NOTCH1	Cardiometabolic	2.45315	1.8254	1.63455	1.35E-06	1.81E-05	-1.49138	6.63E-07	1.87E-05	-1.62721
OID20312	CHIT1	Cardiometabolic	4.23445	4.48805	4.34625	0.895393	0.916255	1.058008	0.025335	0.04788	1.481029
OID20313	PDGFA	Cardiometabolic	3.8659	4.0654	3.85875	0.100537	0.147777	1.222937	0.115849	0.173376	1.165258
OID20314	ACPS	Cardiometabolic	3.6625	3.34575	3.57835	0.314263	0.386406	-1.0072	0.677001	0.735761	1.13418
OID20315	ITGB2	Cardiometabolic	0.2248	0.20575	0.04365	0.477775	0.553373	1.032184	0.076611	0.120687	-1.20752
OID20316	FAS	Cardiometabolic	2.89215	2.40435	2.55555	0.077998	0.118453	-1.24561	0.126097	0.185595	-1.30297
OID20317	PLXNB2	Cardiometabolic	3.28915	3.622	3.5219	0.220094	0.287012	1.09047	0.090721	0.139904	1.018891
OID20318	GAS6	Cardiometabolic	5.3767	5.7414	5.76445	0.007125	0.01525	1.433111	0.000676	0.002516	1.54601
OID20319	EGFR	Cardiometabolic	1.7524	1.55945	1.40215	0.122645	0.17421	-1.13383	0.014673	0.030924	-1.1691
OID20320	QPCT	Cardiometabolic	3.10815	4.20275	4.10325	7.79E-07	1.28E-05	2.328579	4.01E-06	5.37E-05	2.397283
OID20321	RNASET2	Cardiometabolic	4.38115	5.60715	5.57015	7.65E-06	5.64E-05	2.39222	2.99E-05	0.000216	2.501858
OID20322	SPP1	Cardiometabolic	0.71065	2.77015	2.60155	2.87E-05	0.000149	2.896378	0.000235	0.00109	2.868408
OID20323	LILRB1	Cardiometabolic	0.63945	0.48465	0.6797	0.408131	0.481858	-1.0913	0.504681	0.575621	1.04652
OID20324	LILRB5	Cardiometabolic	-1.79775	-1.53385	-1.4387	0.171592	0.234338	1.157891	0.001778	0.005467	1.34742
OID20325	IGFBP2	Cardiometabolic	2.8785	1.18565	1.58195	0.000174	0.000658	-2.50177	0.000415	0.001731	-1.90053
OID20326	PI3	Cardiometabolic	1.56075	1.78595	1.8072	0.000937	0.002742	1.301297	0.000789	0.002821	1.283026
OID20327	RETN	Cardiometabolic	0.5314	0.77575	0.83385	0.040376	0.067172	1.102325	0.029813	0.054376	1.16163
OID20328	CCL15	Cardiometabolic	1.45755	1.14735	1.26115	0.029153	0.051053	-1.28891	0.027474	0.050785	-1.14584
OID20329	PRTN3	Cardiometabolic	-1.51905	-1.66505	-1.47285	0.529893	0.601056	-1.42939	0.434393	0.511215	1.043369
OID20330	FAP	Cardiometabolic	-1.4801	-0.8179	-0.8314	0.000482	0.001571	1.357028	0.000556	0.002205	1.367509
OID20331	PPIB	Cardiometabolic	4.38555	5.034	5.02455	7.04E-07	1.25E-05	1.495316	1.1E-07	6.37E-06	1.659214
OID20332	COL18A1	Cardiometabolic	1.226	2.09425	2.04765	1.05E-05	7.12E-05	2.105116	1.82E-05	0.000149	2.226689
OID20333	AMY2A	Cardiometabolic	1.3792	1.3717	1.43715	0.789108	0.832148	-1.06161	0.69779	0.754108	1.063227
OID20334	CCL16	Cardiometabolic	0.30785	-0.7732	-0.66045	0.000195	0.000725	-2.11558	0.000335	0.001441	-1.97178
OID20335	COL1A1	Cardiometabolic	0.56095	1.44215	1.42915	2.47E-05	0.000133	2.208627	0.000282	0.001283	1.812021
OID20336	BLMH	Cardiometabolic	0.0951	0.62675	0.4966	0.035602	0.0607	1.518924	0.325697	0.40964	1.215037
OID20337	PRCP	Cardiometabolic	2.91555	3.14345	3.22525	0.001235	0.00338	1.241471	0.000176	0.000864	1.322163
OID20338	PGLYRP1	Cardiometabolic	1.6995	2.06305	2.13615	0.016633	0.031651	1.16542	0.006509	0.016023	1.457807
OID20339	CD59	Cardiometabolic	2.48935	3.86345	3.81905	2.24E-06	2.59E-05	2.768971	1.58E-05	0.000135	3.045122
OID20340	AMY2B	Cardiometabolic	1.674	1.6115	1.51195	0.321238	0.392346	-1.09981	0.271175	0.349882	1.03154
OID20341	TCN2	Cardiometabolic	2.79645	2.46025	3.1853	0.473557	0.549066	1.10784	0.171749	0.238709	1.258888

Supplementary Table 6. (continued)

OID20342	SERPINE1	Cardiometabolic	1.5863	2.62275	2.31915	1.79E-05	0.000104	1.845997	8.66E-05	0.000488	1.536928
OID20343	MCAM	Cardiometabolic	5.1513	4.94705	4.89935	0.10038	0.147744	-1.03605	0.023566	0.045317	-1.11903
OID20344	DEFA1_DEFA1B	Cardiometabolic	5.8592	5.79415	5.95685	0.110878	0.160401	-1.11092	0.277487	0.356352	1.024592
OID20345	VASN	Cardiometabolic	4.565	4.9156	4.87845	4.46E-06	4.05E-05	1.42568	9.16E-05	0.000503	1.273324
OID20346	ART3	Cardiometabolic	3.9926	4.2965	4.2838	0.121179	0.172733	1.278587	0.087533	0.136134	1.166996
OID20347	CNDP1	Cardiometabolic	4.0504	4.71405	4.76585	0.001024	0.002927	1.471768	0.001242	0.004071	1.682143
OID20348	IL18BP	Cardiometabolic	4.83705	5.1006	5.22875	0.018976	0.0356	1.116087	0.000658	0.002467	1.303599
OID20349	F9	Cardiometabolic	0.4529	0.1619	0.5609	0.09086	0.134905	-1.27514	0.488588	0.564705	1.166552
OID20350	ANPEP	Cardiometabolic	1.0734	1.84055	1.69185	0.000585	0.001839	1.523564	0.000439	0.00181	1.525255
OID20351	SSC5D	Cardiometabolic	2.4033	1.19695	1.6052	3.15E-05	0.000159	-2.14712	0.000356	0.00152	-1.65267
OID20352	ITIH3	Cardiometabolic	0.6621	1.032	1.0776	0.304431	0.376849	1.079864	0.33137	0.415348	1.048335
OID20353	CDH1	Cardiometabolic	3.8441	4.20005	4.37905	0.002204	0.005488	1.259324	0.011661	0.025918	1.192426
OID20354	NCAM1	Cardiometabolic	5.83545	6.13515	6.15895	0.001711	0.004433	1.387656	0.02288	0.044386	1.451053
OID20355	LBP	Cardiometabolic	-0.2335	-2.2774	-1.85665	1.6E-05	9.65E-05	-3.24013	0.000309	0.001369	-3.08677
OID20356	OSMR	Cardiometabolic	3.80045	3.99085	4.01555	0.049212	0.07923	1.170967	0.014603	0.030894	1.188177
OID20357	HYAL1	Cardiometabolic	3.59935	2.02735	2.5792	1.71E-05	0.000101	-2.31049	0.000785	0.002816	-2.39214
OID20358	CTSD	Cardiometabolic	2.88425	5.30915	5.1402	1.42E-06	1.87E-05	4.921336	1.29E-05	0.00012	5.71994
OID20359	TNC	Cardiometabolic	5.26285	3.63875	3.7516	4.48E-09	4.92E-07	-2.9673	8.13E-08	5.25E-06	-2.96339
OID20360	CD163	Cardiometabolic	4.20375	4.8824	4.49375	4.44E-05	0.000208	1.725124	0.149782	0.212756	1.248763
OID20361	SPARCL1	Cardiometabolic	4.6388	5.90145	5.93975	1.15E-06	1.64E-05	2.58874	6.06E-06	7.19E-05	2.803444
OID20362	NID1	Cardiometabolic	4.2103	3.87435	3.83985	0.00099	0.002861	-1.20966	0.005751	0.014484	-1.08478
OID20363	AXL	Cardiometabolic	5.4685	5.8894	5.7247	0.000251	0.000893	1.51803	0.001713	0.005299	1.449444
OID20364	PRSS2	Cardiometabolic	5.44125	5.94835	6.0387	0.012742	0.025485	1.499883	0.001782	0.005467	1.512987
OID20365	CTSZ	Cardiometabolic	5.1498	6.34465	6.27395	7.89E-06	5.72E-05	2.435554	7.26E-05	0.000424	2.439017
OID20366	CHL1	Cardiometabolic	4.94225	5.39585	5.46335	0.000505	0.001631	1.454527	0.002045	0.006069	1.518556
OID20367	C1QTNF1	Cardiometabolic	6.2567	4.983	5.3464	2.58E-06	2.83E-05	-2.1114	0.000163	0.000811	-1.74279
OID20368	THBS4	Cardiometabolic	2.4982	1.61675	1.62495	2.21E-06	2.58E-05	-1.51645	1.58E-05	0.000135	-1.63376
OID20369	TFRC	Cardiometabolic	0.2211	0.35925	0.47285	0.134152	0.188844	1.275842	0.031563	0.056813	1.309395
OID20370	KIT	Cardiometabolic	1.87215	2.1318	2.08775	0.120167	0.172026	1.181197	0.337316	0.419449	1.224973
OID20371	TFF3	Cardiometabolic	6.65475	7.30635	7.3981	0.003603	0.0084	1.52425	0.001876	0.00569	1.514561
OID20372	F7	Cardiometabolic	1.36295	1.36405	1.25365	0.027978	0.049628	-1.1719	0.232879	0.306595	-1.0357
OID20373	RARRES2	Cardiometabolic	5.16925	5.53795	5.55685	0.013885	0.027127	1.381227	0.021942	0.042946	1.34574

Supplementary Table 6. (continued)

OID20374	APOM	Cardiometabolic	-0.65755	-0.36785	0.3839	0.000512	0.00165	1.585898	1.04E-05	0.000105	2.063009
OID20375	TIMD4	Cardiometabolic	-0.65525	-0.6787	-0.51465	0.048974	0.079078	-1.37402	0.634301	0.695072	-1.00933
OID20376	ITGB1	Cardiometabolic	1.19965	1.28075	1.2426	0.621273	0.684898	-1.05948	0.459789	0.536502	-1.00187
OID20377	CD55	Cardiometabolic	6.3024	6.86915	6.92805	0.00014	0.000543	1.701904	0.00061	0.002356	1.86995
OID20378	CD14	Cardiometabolic	5.59685	5.7712	5.77415	0.839867	0.872445	-1.05041	0.350988	0.431079	1.062601
OID20379	SELP	Cardiometabolic	4.0772	4.3818	4.26325	0.000152	0.000582	1.667978	0.000822	0.002885	1.664397
OID20380	CD46	Cardiometabolic	7.25245	7.1737	7.12705	0.455167	0.529421	1.001596	0.134527	0.195644	-1.01168
OID20381	GP1BA	Cardiometabolic	-1.16945	-0.2449	-0.01785	0.000236	0.000853	2.012864	0.000207	0.000979	1.743516
OID20382	FCN2	Cardiometabolic	-2.8608	-1.86	-1.8691	0.002828	0.006781	1.358157	0.001947	0.005829	1.712317
OID20383	HSPG2	Cardiometabolic	5.705	6.641	6.67665	6.56E-06	5.14E-05	1.949021	4.97E-05	0.000313	2.129626
OID20384	PCOLCE	Cardiometabolic	3.7901	4.5169	4.65165	0.003888	0.00897	1.838017	0.004878	0.012602	2.055301
OID20385	IL6R	Cardiometabolic	2.965	3.269	3.19765	0.406543	0.481409	-1.02076	0.086217	0.134469	1.118527
OID20386	SOD1	Cardiometabolic	3.73125	5.42175	5.3259	2.65E-05	0.000141	3.131057	0.000201	0.000956	3.113204
OID20387	FCGR3B	Cardiometabolic	1.67225	2.47365	2.4362	0.000295	0.001019	1.49516	2.87E-05	0.000209	1.663821
OID20388	TFPI	Cardiometabolic	3.74955	2.41915	2.4627	7.11E-05	0.000309	-2.28802	0.000117	0.000602	-2.23605
OID20389	AOC3	Cardiometabolic	1.985	2.73405	2.67305	4.73E-05	0.000218	1.434452	6.55E-06	7.57E-05	1.595159
OID20390	NRP1	Cardiometabolic	3.7178	4.15565	4.12915	0.002148	0.005403	1.409907	0.002469	0.007135	1.416765
OID20391	FCGR2A	Cardiometabolic	3.4728	3.45195	3.63685	0.942555	0.951218	1.02445	0.313996	0.396284	1.076539
OID20392	LCN2	Cardiometabolic	4.64975	5.53615	5.53355	3.06E-06	3.23E-05	2.13739	7.73E-06	8.6E-05	2.113084
OID20393	CR2	Cardiometabolic	-1.42455	-1.1031	-1.03005	0.562914	0.630693	1.051829	0.143131	0.204632	1.166835
OID20394	ADA2	Cardiometabolic	4.93755	4.95555	5.26325	0.273864	0.34513	1.062159	0.028155	0.051869	1.282314
OID20395	CCL18	Cardiometabolic	1.9845	-0.5529	0.27055	0.000662	0.002029	-3.1428	0.002629	0.007459	-2.43142
OID20396	VCAM1	Cardiometabolic	2.3887	2.3568	2.6981	0.172114	0.234759	-1.16091	0.306974	0.388763	1.202428
OID20397	CD93	Cardiometabolic	3.4242	3.31615	3.535	0.024077	0.043409	-1.09126	0.422209	0.499016	-1.04616
OID20398	IGFBP3	Cardiometabolic	1.63405	0.75055	1.1796	0.052747	0.084131	-1.69261	0.069391	0.110904	-1.28766
OID20399	CHI3L1	Cardiometabolic	5.25425	6.66085	6.76455	0.000108	0.000439	2.528795	0.000807	0.002852	2.378249
OID20400	CST3	Cardiometabolic	-0.363	2.17575	1.90015	7.22E-06	5.47E-05	5.032063	5.34E-05	0.000331	4.868239
OID20401	CCL14	Cardiometabolic	5.77295	6.5745	6.58355	1.91E-07	5.38E-06	1.677078	1.97E-07	8.18E-06	1.82798
OID20402	PROC	Cardiometabolic	1.54985	0.8396	1.5852	0.028266	0.049897	-1.28953	0.421708	0.498961	1.184066
OID20403	ANG	Cardiometabolic	1.4712	2.49175	2.57615	1.24E-05	8.04E-05	2.801598	8.9E-05	0.000499	2.680868
OID20404	FETUB	Cardiometabolic	1.8962	1.8226	2.1258	0.190877	0.256214	-1.05019	0.830509	0.868475	1.054128
OID20405	COMP	Cardiometabolic	2.50455	2.99315	2.94815	0.001135	0.003154	1.319737	0.006357	0.015757	1.34355

Supplementary Table 6. (continued)

OID20406	DPP4	Cardiometabolic	-0.341	0.7413	0.89215	0.008943	0.018491	1.383574	0.00262	0.007453	1.785094
OID20407	ANGPTL3	Cardiometabolic	2.27425	3.0448	2.9107	4.58E-07	1.01E-05	1.419468	4.71E-07	1.48E-05	1.612166
OID20408	PTGDS	Cardiometabolic	0.3557	2.10685	2.10855	2.41E-06	2.75E-05	3.152836	9.6E-06	0.0001	3.126178
OID20409	CA1	Cardiometabolic	0.62345	2.38135	1.78685	0.01127	0.022789	2.420408	0.137053	0.198006	1.500351
OID20410	C2	Cardiometabolic	0.67835	1.55205	1.65405	1.08E-06	1.56E-05	1.860126	4.28E-07	1.43E-05	2.005275
OID20411	ICAM1	Cardiometabolic	3.60225	3.8634	3.93415	0.125625	0.178212	1.185873	0.00875	0.020355	1.25867
OID20412	CCL5	Cardiometabolic	-4.03585	-2.8665	-2.8601	0.001129	0.003147	2.015866	0.001688	0.005237	2.408859
OID20413	IGFBP6	Cardiometabolic	4.5702	6.02715	5.85675	5.89E-06	4.89E-05	2.937421	4.69E-05	0.000303	2.974609
OID20414	EFEMP1	Cardiometabolic	3.7855	5.61875	5.54455	3.12E-06	3.27E-05	3.239452	2.33E-05	0.000177	3.302137
OID20415	REG1A	Cardiometabolic	5.0667	5.99585	5.83845	0.000129	0.000512	1.904154	0.000925	0.003215	1.70214
OID20416	GGH	Cardiometabolic	4.6099	5.0561	5.18975	0.002183	0.005448	1.426223	0.001023	0.003487	1.637619
OID20417	TGFBI	Cardiometabolic	3.5308	4.4623	4.55525	0.021846	0.040046	1.287748	0.005519	0.013994	1.595435
OID20418	TIMP1	Cardiometabolic	4.5699	6.15955	6.1266	3.86E-05	0.000184	3.427009	0.000478	0.001929	3.567961
OID20421	PAPPA	Inflammation	2.82195	2.73145	2.53825	0.225493	0.292579	-1.14124	0.196767	0.266399	-1.3223
OID20422	IL20RA	Inflammation	0.7313	0.70235	0.7751	0.718075	0.772987	1.061975	0.608298	0.672619	-1.12249
OID20423	EIF5A	Inflammation	-0.404	0.52125	0.55715	0.031899	0.055157	1.922855	0.190659	0.260054	1.999307
OID20425	RAB37	Inflammation	-1.4137	-0.5069	-0.3424	0.010021	0.020338	2.277262	0.015125	0.031693	2.552834
OID20429	LILRB4	Inflammation	-1.88105	-1.66415	-1.57035	0.571596	0.639117	1.043514	0.090448	0.139876	1.282714
OID20430	IL13	Inflammation	-5.16445	-4.5561	-3.81865	0.073816	0.11304	1.549551	0.001477	0.004688	2.693627
OID20434	LTO1	Inflammation	-0.91665	-0.3177	0.18285	0.022147	0.040461	2.180335	0.009168	0.02106	2.698112
OID20436	LAP3	Inflammation	-3.9373	-1.52915	-1.2883	0.000181	0.000679	3.285698	0.000363	0.001544	4.568277
OID20437	AMBN	Inflammation	-1.5813	0.27585	0.12855	0.000622	0.001939	3.059615	0.00283	0.007926	2.85997
OID20438	LRRN1	Inflammation	-1.6395	-2.1041	-1.86525	0.002358	0.00578	-1.18214	0.003838	0.010277	-1.0295
OID20442	NBN	Inflammation	-2.6927	-1.54655	-1.85785	0.013102	0.026015	2.178825	0.126433	0.185841	1.90554
OID20443	FCRL3	Inflammation	-2.3414	-1.494	-1.7759	0.042654	0.070639	1.396872	0.105461	0.15928	1.219128
OID20448	NCLN	Inflammation	-0.22555	0.3515	0.1798	0.068122	0.105202	2.028973	0.187051	0.256087	2.008544
OID20457	SCGN	Inflammation	-1.59	-1.18945	-1.35925	0.036046	0.061267	1.377498	0.215366	0.287329	1.124279
OID20458	CXCL14	Inflammation	1.23575	0.5367	0.365	0.16054	0.220341	-1.31217	0.005424	0.013852	-1.51987
OID20459	MVK	Inflammation	-3.6765	-2.2896	-2.56465	0.000177	0.000667	3.691993	0.000946	0.003266	2.813274
OID20461	IL3RA	Inflammation	-0.7875	-0.31925	0.0827	0.036139	0.06133	1.490349	0.024229	0.046267	2.008544
OID20462	SIT1	Inflammation	-0.93915	0.0516	0.2112	0.020838	0.038606	1.40976	0.012971	0.028314	2.078078
OID20465	NRTN	Inflammation	-0.4911	1.1144	0.36805	0.000423	0.001394	3.201615	0.015816	0.032725	2.267339

Supplementary Table 6. (continued)

OID20467	PADI2	Inflammat ion	0.04295	-0.14685	-0.53925	0.205181	0.272746	-1.11973	0.004349	0.011396	-1.45675
OID20469	IL17A	Inflammat ion	-3.41015	-1.55815	-2.17885	0.00067	0.002048	3.327296	0.006268	0.015588	2.555579
OID20475	SPRY2	Inflammat ion	0.16375	-0.0101	-0.04565	0.72381	0.777322	1.023421	0.551219	0.6215	-1.31731
OID20480	TNFRSF13 C	Inflammat ion	-0.88415	-1.06365	-1.05495	0.274658	0.345372	-1.04156	0.203844	0.27429	-1.02947
OID20481	IL17D	Inflammat ion	2.99095	0.7337	1.3505	1.15E-07	4.46E-06	-4.54427	8.48E-07	2.18E-05	-3.19164
OID20482	MILR1	Inflammat ion	-2.34815	-2.649	-2.24995	0.544789	0.61541	-1.1113	0.937323	0.952064	1.081175
OID20483	PRDX3	Inflammat ion	-0.72515	0.92195	0.317	0.000674	0.002054	2.594488	0.014326	0.03051	1.52119
OID20485	IRAK1	Inflammat ion	-2.6334	-1.6866	-1.7051	0.022106	0.040454	1.607646	0.042132	0.072851	1.539114
OID20486	IL12RB1	Inflammat ion	-2.77555	-2.11915	-2.3846	0.02894	0.050761	1.636655	0.095849	0.146373	1.26164
OID20487	IL1RL2	Inflammat ion	-4.34245	-4.1196	-4.10525	0.914885	0.929272	1.12783	0.72502	0.779698	1.178703
OID20488	IL1A	Inflammat ion	-0.813	1.06085	1.6632	0.005765	0.012584	2.695681	0.007586	0.018108	2.729052
OID20490	FGF5	Inflammat ion	-3.59125	-3.2666	-3.1278	0.150858	0.208881	1.421732	0.543092	0.614124	1.243581
OID20491	CSF3	Inflammat ion	-4.08325	-3.50525	-2.72395	0.01386	0.027127	1.858257	0.000696	0.002551	2.532391
OID20498	IL15RA	Inflammat ion	-3.0176	-1.98695	-2.2524	0.043036	0.071165	1.657203	0.349442	0.429661	1.241126
OID20501	FABP9	Inflammat ion	-3.6842	-3.17775	-3.0697	0.049759	0.079877	1.243107	0.06744	0.108736	1.185339
OID20504	ITM2A	Inflammat ion	-0.6425	-1.42805	-1.0485	0.000486	0.00158	-1.26835	0.089404	0.138651	-1.36051
OID20506	IFNLR1	Inflammat ion	-0.70005	-0.82785	-0.8497	0.183832	0.248275	-1.05636	0.232198	0.306067	-1.12818
OID20508	SULT2A1	Inflammat ion	-2.3611	-1.78845	-1.72815	0.01333	0.026372	1.838781	0.004015	0.010696	2.008544
OID20509	SPINK4	Inflammat ion	4.4171	4.7346	4.4366	0.328865	0.400326	1.304458	0.476399	0.554116	1.264047
OID20510	NUB1	Inflammat ion	-3.80645	-2.7302	-2.7283	0.003823	0.008856	1.845933	0.009874	0.022354	1.75747
OID20511	PSPN	Inflammat ion	-4.55435	-4.574	-4.0938	0.049685	0.079874	1.479746	0.029633	0.054229	1.79185
OID20512	ALDH3A1	Inflammat ion	1.0631	5.8351	4.94285	6.7E-05	0.000294	26.24735	0.002125	0.006256	15.70661
OID20513	BCL2L11	Inflammat ion	0.92765	1.3868	1.29915	0.020249	0.037749	1.374732	0.038595	0.067479	1.388378
OID20520	HLA-DRA	Inflammat ion	1.766	0.78865	1.27365	5.19E-06	4.56E-05	-2.10089	0.000444	0.001818	-1.72273
OID20522	EPO	Inflammat ion	0.4091	-1.16515	-0.50245	2.75E-05	0.000144	-2.08204	0.000298	0.00134	-1.75115
OID20523	CCL7	Inflammat ion	-3.63765	-2.4932	-3.2355	0.286286	0.357613	1.538474	0.40199	0.480421	1.079153
OID20525	EDAR	Inflammat ion	-3.45525	-3.03245	-2.995	0.211499	0.278449	1.441929	0.298435	0.379262	1.417796
OID20527	DPP10	Inflammat ion	2.34085	1.7245	1.58135	0.006738	0.014478	-1.24229	0.00077	0.002782	-1.2967
OID20530	LSP1	Inflammat ion	-2.41055	-2.974	-3.44125	0.00315	0.007437	-1.56255	0.000968	0.003323	-1.77732
OID20531	STX8	Inflammat ion	-0.67765	-0.676	-0.9579	0.501545	0.57544	1.002568	0.110171	0.166164	-1.06463
OID20533	TRIM21	Inflammat ion	-4.2484	-4.22595	-4.93335	0.803776	0.843734	1.063928	0.06225	0.102321	-1.26822
OID20535	IL7	Inflammat ion	2.56075	1.77945	1.98425	0.000136	0.000533	-1.72937	0.001864	0.00567	-1.52536

Supplementary Table 6. (continued)

OID20537	SIGLEC10	Inflammat ion	-1.98085	-2.05965	-2.21035	0.166273	0.227925	-1.02218	0.893459	0.92028	-1.17243
OID20538	ISM1	Inflammat ion	-4.8989	-6.30475	-6.4541	7.71E-07	1.28E-05	-3.34928	1.26E-07	6.94E-06	-3.52238
OID20540	BTN3A2	Inflammat ion	-1.22555	-2.02055	-1.66745	0.054045	0.085507	-1.51761	0.454713	0.531144	-1.34471
OID20542	SCRN1	Inflammat ion	-0.26965	-1.22795	-1.23905	0.002323	0.005719	-1.81623	0.000612	0.002356	-2.2931
OID20548	CEACAM2 1	Inflammat ion	-4.6111	-4.114	-4.09805	0.037887	0.063511	1.648381	0.165879	0.231136	1.162677
OID20550	COL9A1	Inflammat ion	-1.6017	-1.56335	-1.2537	0.176291	0.239822	1.076054	0.203435	0.274075	1.045614
OID20556	CNTNAP2	Inflammat ion	6.5171	6.34205	6.29315	0.13603	0.191244	-1.00979	0.013438	0.029188	-1.14175
OID20560	SEPLG	Inflammat ion	-4.1718	-4.2851	-4.1837	0.671349	0.728083	-1.05625	0.780227	0.827719	1.053361
OID20561	CXADR	Inflammat ion	-0.3948	-0.7592	-0.71665	0.02269	0.041317	-1.23705	0.018965	0.038208	-1.19615
OID20562	IL15	Inflammat ion	-0.2888	-0.4342	-0.3078	0.200803	0.26725	-1.27859	0.586655	0.652632	-1.20556
OID20563	IL6	Inflammat ion	-0.11635	0.1358	0.79905	0.407312	0.481409	1.214532	0.072083	0.114541	1.563631
OID20564	FCAR	Inflammat ion	-2.3787	-1.88535	-2.24425	0.037107	0.062539	1.153366	0.032594	0.058477	1.154445
OID20565	CD83	Inflammat ion	-0.2024	-0.46755	-0.5227	0.595636	0.659948	-1.05008	0.335807	0.418414	1.047137
OID20566	NCR1	Inflammat ion	-4.22865	-3.54145	-3.71125	0.13183	0.186053	1.364384	0.284564	0.364588	1.241126
OID20567	NCF2	Inflammat ion	-6.99905	-7.2407	-7.0723	0.411646	0.484446	-1.1113	0.820043	0.859167	-1.093
OID20568	NPPC	Inflammat ion	3.51165	3.63245	3.33445	0.550721	0.621472	1.11632	0.213857	0.286011	-1.09331
OID20569	CCL28	Inflammat ion	-0.6838	-0.00035	-0.1038	0.021708	0.039913	1.59859	0.020164	0.040181	1.560924
OID20570	PTX3	Inflammat ion	-2.7987	-2.56285	-2.92645	0.893042	0.915556	-1.02559	0.563751	0.63263	1.083463
OID20571	PTH1R	Inflammat ion	-3.3569	-2.82925	-2.76895	0.077785	0.118453	1.58447	0.202049	0.272543	1.299809
OID20572	EGLN1	Inflammat ion	-3.2779	-3.13575	-3.3408	0.056108	0.088516	1.1776	0.135801	0.196871	1.241126
OID20573	CLEC4A	Inflammat ion	-4.84415	-4.6899	-4.32595	0.66044	0.717273	1.175154	0.52235	0.5925	1.556872
OID20574	OSM	Inflammat ion	-5.4422	-4.70315	-4.54605	0.002242	0.005567	1.868979	9.32E-05	0.000508	2.313537
OID20576	IL4R	Inflammat ion	-3.58585	-3.8316	-3.9528	0.041506	0.068946	-1.3583	0.066823	0.107899	-1.08523
OID20580	LY75	Inflammat ion	-3.31365	-3.47485	-3.4566	0.225431	0.292579	-1.12218	0.666126	0.725587	-1.09194
OID20584	CD4	Inflammat ion	-3.09095	-3.3417	-3.12605	0.113532	0.163808	-1.35712	0.867687	0.897946	-1.13508
OID20585	IL17RB	Inflammat ion	-1.6646	-2.89405	-2.8304	1.59E-05	9.63E-05	-2.1536	1.38E-05	0.000123	-2.30561
OID20586	LTA	Inflammat ion	-4.81555	-4.80475	-4.3939	0.182725	0.247084	-1.08621	0.016895	0.034709	1.388378
OID20589	HEXIM1	Inflammat ion	-3.36845	-2.8563	-2.92855	0.101507	0.148805	1.421191	0.159319	0.222844	1.219128
OID20593	MICB_MI CA	Inflammat ion	-2.05595	-1.92615	-2.11885	0.141464	0.198375	-1.32708	0.584813	0.651242	-1.16894
OID20595	CD200R1	Inflammat ion	-7.08035	-6.3784	-6.5463	0.039656	0.066073	1.1776	0.070291	0.112179	1.241126
OID20596	KRT19	Inflammat ion	-3.40505	-2.189	-2.74595	7.83E-05	0.000337	2.40219	0.025621	0.048033	1.70102
OID20597	PLXNA4	Inflammat ion	0.8233	-0.69455	-0.93195	2.14E-05	0.000119	-2.5169	3.79E-06	5.2E-05	-2.57844

Supplementary Table 6. (continued)

OID20598	PTPRM	Inflammat ion	-4.7736	-4.52055	-4.56115	0.697651	0.753957	1.091226	0.875985	0.90568	1.436542
OID20599	CD70	Inflammat ion	-6.37215	-5.77165	-5.44345	0.219688	0.286822	-1.02883	0.066414	0.107543	1.79036
OID20600	TGFA	Inflammat ion	0.21635	0.2434	0.17345	0.109243	0.158873	1.287034	0.288087	0.368671	-1.08192
OID20601	ILSRA	Inflammat ion	-5.3181	-5.73685	-5.5641	0.207369	0.274434	-1.48324	0.935914	0.951512	-1.21887
OID20603	CD200	Inflammat ion	1.756	0.932	1.0417	0.000815	0.00243	-1.35693	0.000324	0.001399	-1.46739
OID20604	GZMB	Inflammat ion	-5.78185	-2.89355	-3.82955	3.17E-07	7.73E-06	4.405635	2.58E-06	3.93E-05	6.391701
OID20606	LIFR	Inflammat ion	-3.79	-4.8413	-4.4315	2.27E-05	0.000126	-1.94929	0.004066	0.010785	-1.5495
OID20607	CD84	Inflammat ion	-5.0646	-4.85755	-5.02745	0.809904	0.848209	1.154325	0.744813	0.797081	1.195157
OID20609	CLEC4D	Inflammat ion	-0.4408	-0.63305	-0.7436	0.749791	0.79697	-1.01621	0.402538	0.480421	-1.1475
OID20610	CCL3	Inflammat ion	-4.4018	-4.3362	-4.11385	0.552133	0.622281	-1.02979	0.361921	0.442527	1.012414
OID20611	TNFSF10	Inflammat ion	-6.47025	-6.9297	-7.05145	0.064499	0.100454	-1.73345	0.012507	0.027411	-1.57309
OID20612	TLR3	Inflammat ion	1.97395	1.85655	1.8531	0.73497	0.785591	-1.14088	0.899524	0.923063	1.031683
OID20614	PCDH1	Inflammat ion	-1.4962	-1.40155	-1.50705	0.595169	0.659948	1.04623	0.421616	0.498961	1.087338
OID20615	HPCAL1	Inflammat ion	-4.5505	-4.79755	-4.9292	0.21593	0.283602	-1.23602	0.195839	0.26547	-1.34471
OID20617	CKAP4	Inflammat ion	-0.7355	-2.71205	-2.4907	6.45E-06	5.13E-05	-3.38792	3.7E-06	5.14E-05	-3.70173
OID20619	IDS	Inflammat ion	0.61845	1.01975	1.17605	0.01527	0.029414	1.528377	0.032827	0.058799	1.675626
OID20620	DFFA	Inflammat ion	-3.64985	-4.3469	-4.15265	0.909923	0.925813	-1.23876	0.700083	0.755842	-1.25518
OID20621	TGFB1	Inflammat ion	-5.6586	-5.55995	-5.15995	0.643345	0.705687	1.119768	0.008459	0.019846	1.291099
OID20622	CXCL17	Inflammat ion	-2.6134	-3.23185	-3.24645	0.951232	0.958214	-1.12043	0.569976	0.636657	-1.05863
OID20623	NUDC	Inflammat ion	-5.85625	-6.0086	-5.7745	0.987272	0.987272	-1.1113	0.386906	0.466326	1.020517
OID20624	TNFSF12	Inflammat ion	2.8452	2.22175	2.0978	0.001383	0.00374	-1.66342	0.000154	0.000768	-1.49319
OID20625	EIF4G1	Inflammat ion	-4.88855	-4.45155	-4.33455	0.075555	0.115542	1.448992	0.227328	0.30161	1.516978
OID20626	BSG	Inflammat ion	-0.0844	-0.3676	-0.5196	0.085333	0.128175	-1.10723	0.008483	0.019861	-1.13579
OID20627	DNAJA2	Inflammat ion	-3.49605	-3.04495	-3.11985	0.048154	0.078099	1.288149	0.413044	0.49035	1.134534
OID20628	CD244	Inflammat ion	-5.02685	-4.85525	-5.02505	0.528414	0.599998	1.160382	0.789576	0.834349	1.008492
OID20629	KLRB1	Inflammat ion	-5.4875	-5.05125	-4.9577	0.157776	0.21709	1.142703	0.035052	0.062175	1.140606
OID20630	SERPINB8	Inflammat ion	-2.94295	-1.99055	-2.37955	1.26E-05	8.11E-05	1.528165	0.001635	0.005116	1.779905
OID20631	CXCL8	Inflammat ion	0.1215	0.60075	0.6379	0.035197	0.060104	1.296345	0.016914	0.034709	1.229695
OID20632	KLRD1	Inflammat ion	-7.28635	-7.23065	-6.7525	0.654045	0.712442	-1.19433	0.482663	0.559624	1.298189
OID20633	IL16	Inflammat ion	-5.7923	-5.42175	-5.3788	0.266441	0.33782	1.089752	0.216276	0.287844	1.275931
OID20634	NFASC	Inflammat ion	1.56225	1.74475	1.77655	0.008877	0.018391	1.274958	0.006275	0.015588	1.355476
OID20635	CD79B	Inflammat ion	-5.4918	-4.9476	-4.736	0.052896	0.084174	1.196027	0.008991	0.020783	1.32464

Supplementary Table 6. (continued)

OID20636	CLEC7A	Inflammat ion	-0.41005	-0.59905	-0.57725	0.120914	0.172645	-1.14969	0.389241	0.468113	-1.10726
OID20637	CD22	Inflammat ion	-7.2086	-7.10405	-6.91885	0.533655	0.604699	1.04968	0.73497	0.787314	1.233194
OID20638	LAMP3	Inflammat ion	-7.7052	-6.6222	-6.8675	0.001485	0.003957	1.939184	0.007755	0.018392	1.801252
OID20641	GMPR	Inflammat ion	-2.7476	-1.08225	-1.0081	0.003219	0.007553	2.449946	0.001948	0.005829	2.695401
OID20642	TPSAB1	Inflammat ion	-6.61895	-6.27055	-5.86405	0.142975	0.200239	1.273148	0.016084	0.033133	2.433445
OID20643	PPP1R9B	Inflammat ion	-4.27765	-4.3047	-4.2659	0.658158	0.715502	1.521242	0.92463	0.941784	-1.29092
OID20645	ADA	Inflammat ion	-4.40305	-3.64875	-3.74165	8.01E-05	0.000344	1.547136	0.00078	0.002809	1.526471
OID20646	TNFRSF11 A	Inflammat ion	-2.77375	-2.864	-2.7778	0.635403	0.699771	-1.01248	0.67977	0.737537	-1.02151
OID20647	CD160	Inflammat ion	-5.73765	-5.4174	-5.27805	0.287076	0.357786	1.209701	0.012956	0.028314	1.388233
OID20649	CD6	Inflammat ion	-8.0597	-7.75905	-7.4901	0.284762	0.356114	1.711724	0.075836	0.119943	1.56857
OID20650	VEGFA	Inflammat ion	0.9764	-0.12615	0.0236	5.56E-05	0.000251	-2.01224	0.000319	0.001393	-1.78151
OID20651	ADAM23	Inflammat ion	-2.0002	-3.74925	-3.68225	1.39E-09	2.47E-07	-3.40428	1.65E-08	2.26E-06	-2.98225
OID20652	IL18R1	Inflammat ion	-5.2108	-5.08895	-5.3255	0.175144	0.238595	1.140092	0.334142	0.416918	-1.0572
OID20653	TNFRSF4	Inflammat ion	-5.16505	-4.37915	-4.38605	0.001189	0.003279	1.496041	0.001069	0.003566	1.309213
OID20654	C1QA	Inflammat ion	-3.46565	-4.5033	-4.0893	5.42E-05	0.000247	-1.94767	0.024581	0.046615	-1.59649
OID20656	HGF	Inflammat ion	3.45475	1.93805	2.22985	7.13E-08	3.26E-06	-2.62934	2.84E-06	4.23E-05	-2.68068
OID20657	MERTK	Inflammat ion	-4.51565	-5.9948	-5.467	9.91E-05	0.000406	-2.03941	0.003195	0.008838	-1.89612
OID20658	AGRP	Inflammat ion	-1.8363	-3.57595	-4.0345	2.05E-05	0.000116	-3.13453	1.05E-06	2.38E-05	-3.7218
OID20660	CTSO	Inflammat ion	1.73265	1.1885	1.3656	0.016895	0.031985	-1.29563	0.047687	0.080802	-1.18954
OID20661	FLT3LG	Inflammat ion	-1.5365	-1.5026	-1.36205	0.789571	0.832148	1	0.405197	0.483068	1.121166
OID20662	VEGFD	Inflammat ion	-8.64545	-9.7392	-9.1256	0.001855	0.004769	-1.67725	0.123422	0.182393	-1.41069
OID20663	GZMA	Inflammat ion	-5.317	-4.95785	-4.18775	0.016305	0.031136	1.307173	0.002034	0.006052	2.159803
OID20664	CLSTN2	Inflammat ion	3.5487	3.64245	3.73995	0.086399	0.128895	1.122138	0.009977	0.022541	1.27253
OID20665	FASLG	Inflammat ion	-6.338	-5.53315	-5.92815	0.006912	0.014823	1.501704	0.003204	0.00884	1.585348
OID20666	IL12B	Inflammat ion	-4.98965	-3.8454	-3.52945	9.27E-05	0.000387	1.709116	4.18E-06	5.53E-05	1.898553
OID20667	CDSN	Inflammat ion	-4.6989	-4.12625	-4.40775	0.437747	0.510782	1.1776	0.349237	0.429661	1.265493
OID20668	CCL11	Inflammat ion	-6.8499	-7.8764	-7.54475	0.001759	0.004545	-1.8377	0.00961	0.021892	-1.54355
OID20669	KYNU	Inflammat ion	-5.87685	-4.6439	-4.2448	0.031045	0.053851	1.90587	0.009507	0.021701	2.846916
OID20670	LY9	Inflammat ion	-7.0326	-6.495	-6.26935	0.079892	0.120662	1.263828	0.05018	0.084376	1.659444
OID20671	CCL20	Inflammat ion	-5.00875	-5.1361	-5.55015	0.900389	0.92051	-1.11199	0.238235	0.311407	-1.45538
OID20672	MMP1	Inflammat ion	-3.2775	-3.02005	-2.8872	0.003192	0.007522	1.550464	0.000594	0.002309	1.552722
OID20673	PGF	Inflammat ion	-3.20005	-4.53755	-4.5167	3.47E-05	0.00017	-2.8567	0.000473	0.001916	-2.01768

Supplementary Table 6. (continued)

OID20674	CCL25	Inflammat ion	-6.3784	-6.50165	-6.3352	0.122392	0.174076	-1.26387	0.366093	0.446633	-1.17952
OID20675	CXCL9	Inflammat ion	0.29195	0.83035	0.94135	0.035902	0.061117	1.249846	0.022847	0.044386	1.369263
OID20676	SMPDL3A	Inflammat ion	1.31175	1.443	1.52255	0.557539	0.625996	1.138512	0.964567	0.972539	1.041852
OID20677	EPHA1	Inflammat ion	-3.90355	-4.31265	-3.94275	0.001584	0.004171	-1.41603	0.423125	0.49956	-1.04522
OID20678	IFNGR1	Inflammat ion	1.60025	1.38405	1.291	0.021738	0.039913	-1.0896	0.018292	0.036947	-1.13391
OID20679	CTSC	Inflammat ion	2.7979	2.17315	2.48675	1.4E-05	8.81E-05	-1.56879	0.140887	0.20195	-1.19897
OID20680	CD276	Inflammat ion	-1.77405	-3.26765	-2.65155	3.04E-06	3.23E-05	-2.64368	0.000509	0.002045	-1.80801
OID20681	DBNL	Inflammat ion	-3.1078	-2.34505	-2.54665	2.73E-05	0.000144	1.553853	0.000957	0.003292	1.415341
OID20684	TIMP3	Inflammat ion	5.1115	0.91805	0.9212	3.91E-10	1.43E-07	-15.9341	5.49E-10	1.51E-07	-14.8023
OID20685	PLA2G4A	Inflammat ion	-3.28975	-2.822	-2.89805	0.009006	0.018553	1.345554	0.027215	0.050391	1.27306
OID20686	CCL21	Inflammat ion	-1.32875	-0.7936	-0.6153	0.000131	0.000516	1.497701	3.68E-06	5.14E-05	1.604362
OID20687	MMP10	Inflammat ion	0.8338	0.7445	0.8891	0.023394	0.042318	-1.4711	0.15745	0.221074	-1.06186
OID20688	SCGB1A1	Inflammat ion	-0.28725	-0.29435	-0.26405	0.013366	0.026396	1.248287	0.001312	0.004267	1.499415
OID20689	ENPP7	Inflammat ion	-3.5575	-3.28295	-2.92725	0.007956	0.016767	1.407905	0.000707	0.002578	1.597815
OID20690	SIGLEC1	Inflammat ion	-2.6691	-2.62595	-2.27505	0.14944	0.207178	1.030361	0.056352	0.093749	1.222005
OID20691	F2R	Inflammat ion	-0.36675	0.24985	0.1319	0.392982	0.467998	1.376925	0.156112	0.220157	1.266942
OID20692	CD48	Inflammat ion	-2.8348	-2.80705	-2.5472	0.795905	0.836272	-1.14429	0.026497	0.049479	1.128495
OID20693	CCL23	Inflammat ion	-3.3866	-3.898	-4.08235	0.013699	0.026947	-1.46932	0.025295	0.04788	-1.60943
OID20694	IL18	Inflammat ion	-5.2804	-3.74495	-4.18775	0.000244	0.000877	2.069741	0.003727	0.010031	2.321086
OID20695	CCL4	Inflammat ion	-0.4932	-0.2603	-0.19405	0.083096	0.125156	1.251016	0.018015	0.03663	1.338391
OID20696	IL10RB	Inflammat ion	-0.83555	-0.9011	-0.8229	0.128568	0.181684	-1.04359	0.580742	0.648024	-1.02569
OID20697	CXCL10	Inflammat ion	-1.7184	-2.955	-2.70695	0.301761	0.373966	-3.06216	0.677463	0.735761	-2.40761
OID20698	EGF	Inflammat ion	-2.08515	-2.25895	-1.88865	0.810357	0.848209	1.006781	0.377926	0.457008	1.18649
OID20699	CRLF1	Inflammat ion	-0.41285	-0.13145	-0.13555	0.004341	0.009931	1.310348	0.001924	0.005803	1.249759
OID20700	IL1RN	Inflammat ion	-2.3713	1.07015	0.03545	4.54E-05	0.000211	8.301373	0.000936	0.003242	4.554207
OID20701	CRIM1	Inflammat ion	4.7787	4.16155	4.1432	0.000481	0.001571	-1.39857	0.000542	0.002163	-1.31376
OID20702	TNFRSF13 B	Inflammat ion	-3.21095	-3.34595	-3.23565	0.910636	0.925813	-1.12596	0.592319	0.6576	-1.37545
OID20703	ANGPTL4	Inflammat ion	-0.3679	-0.6302	-0.3791	0.918071	0.931647	-1.23093	0.040607	0.070436	1.194577
OID20704	CST7	Inflammat ion	-2.37185	-2.44515	-2.11865	0.843929	0.874684	-1.08828	0.031211	0.056272	1.125331
OID20705	ERBB3	Inflammat ion	-2.38155	-2.70945	-2.5654	0.005435	0.012032	-1.2438	0.010491	0.023509	-1.16143
OID20706	NELL2	Inflammat ion	1.7861	2.31375	2.606	0.000459	0.001504	1.577948	0.000121	0.000618	1.775346
OID20707	MANF	Inflammat ion	-0.58525	-0.2936	-0.08415	0.307931	0.380324	1.24509	0.014538	0.030816	1.449143

Supplementary Table 6. (continued)

OID20708	LY6D	Inflammat ion	-0.5523	-0.2455	-0.13355	0.004493	0.010193	1.270988	0.000425	0.001762	1.402159
OID20709	CCN2	Inflammat ion	1.02845	1.23085	1.1272	0.907881	0.925583	1.173648	0.760012	0.809402	-1.20748
OID20710	GL0D4	Inflammat ion	0.49755	1.2134	1.0219	6.4E-07	1.2E-05	1.726799	0.000308	0.001369	1.449495
OID20711	MGLL	Inflammat ion	-4.3945	-3.93475	-4.0127	0.004778	0.010728	1.551754	0.017941	0.036548	1.234391
OID20712	DNER	Inflammat ion	2.52385	2.24865	2.32195	0.000396	0.001317	-1.19557	9.93E-05	0.000527	-1.20932
OID20713	BTN2A1	Inflammat ion	3.2577	3.57135	3.51965	0.054011	0.085507	1.179152	0.093776	0.143807	1.319325
OID20714	SHMT1	Inflammat ion	-2.35445	-0.13095	-0.6341	0.000273	0.000951	4.817885	0.002616	0.007453	3.62241
OID20715	FGF19	Inflammat ion	-1.3271	-3.109	-2.6289	3.65E-06	3.61E-05	-3.31659	1.64E-05	0.000137	-3.2578
OID20716	CD58	Inflammat ion	-2.29935	-2.4639	-2.3129	0.374532	0.450423	-1.08136	0.960597	0.969426	1.103893
OID20717	PTPN6	Inflammat ion	-4.82805	-4.38035	-4.4158	0.232928	0.300182	1.406929	0.244354	0.317891	1.41539
OID20718	HSPA1A	Inflammat ion	-0.82135	-0.3163	-0.50595	0.003108	0.007354	1.36273	0.104182	0.157565	1.216132
OID20719	CSF1	Inflammat ion	0.57215	0.8596	0.65585	0.523984	0.596201	1.140052	0.859968	0.892481	1.178825
OID20720	SPINT2	Inflammat ion	4.26135	4.2602	4.44115	0.757847	0.8032	1.007444	0.619925	0.681359	1.129826
OID20721	CKMT1A_ CKMT1B	Inflammat ion	-4.57135	-2.77975	-3.18025	0.002488	0.006044	3.473042	0.019069	0.038321	2.849483
OID20722	FIS1	Inflammat ion	-2.82055	-3.19195	-3.06365	0.608961	0.672	-1.29025	0.345162	0.425829	-1.34471
OID20723	SCGB3A2	Inflammat ion	-3.8009	-4.4886	-4.62225	0.0003	0.001032	-1.47258	0.005011	0.012915	-1.57549
OID20724	CD40	Inflammat ion	0.32075	0.4896	0.3983	0.752759	0.798619	1.027366	0.452273	0.528856	1.078181
OID20725	OMD	Inflammat ion	0.77955	0.53835	0.5661	0.009006	0.018553	-1.24225	0.098901	0.150199	-1.06677
OID20726	ANGPTL2	Inflammat ion	-2.66165	-4.1967	-4.12505	2.05E-05	0.000116	-2.84248	3.88E-05	0.00026	-2.65461
OID20727	GAL	Inflammat ion	-5.9339	-5.6074	-5.36655	0.274093	0.34513	1.2444	0.131373	0.191564	1.356087
OID20728	SCG3	Inflammat ion	3.51375	3.23445	3.73445	0.106316	0.155027	-1.30767	0.800106	0.842297	1.09509
OID20729	PDLIM7	Inflammat ion	-2.12755	-1.34335	-1.28735	0.007467	0.015889	1.50906	0.017283	0.035338	1.66048
OID20730	SMOC2	Inflammat ion	1.82225	1.28105	1.24845	4.66E-05	0.000216	-1.7796	0.00236	0.006854	-1.59698
OID20731	TREM2	Inflammat ion	0.67075	-1.7062	-1.41815	4.9E-07	1.03E-05	-4.81805	6.53E-06	7.57E-05	-4.58461
OID20732	MZB1	Inflammat ion	-3.91975	-4.36565	-3.8989	0.575734	0.642435	-1.19951	0.063099	0.102946	1.167725
OID20733	TNFSF13	Inflammat ion	3.44195	1.5952	1.9535	2.46E-08	1.42E-06	-2.96997	5.72E-07	1.7E-05	-2.78466
OID20734	ROBO1	Inflammat ion	1.4582	-0.14515	-0.0925	3.5E-07	8.17E-06	-2.63253	3.12E-06	4.51E-05	-2.40719
OID20735	TNFRSF11 B	Inflammat ion	1.8147	1.18975	1.53125	0.00087	0.002575	-1.54002	0.137505	0.198397	-1.26313
OID20736	ENPP5	Inflammat ion	2.12375	2.58015	2.5162	0.027204	0.048332	1.36155	0.149291	0.212333	1.291995
OID20737	LAIR1	Inflammat ion	2.5166	2.902	2.6849	0.049085	0.079141	1.188836	0.028274	0.052002	1.197396
OID20738	COLEC12	Inflammat ion	2.52945	3.41115	3.45325	2.35E-05	0.000128	1.995223	4.59E-05	0.0003	2.224683
OID20739	SIRPB1	Inflammat ion	-1.0664	-1.20855	-0.93885	0.153418	0.212157	-1.09889	0.801728	0.843197	-1.06958

Supplementary Table 6. (continued)

OID20740	ANGPT1	Inflammat ion	-2.32015	-3.0096	-2.83985	0.001124	0.00314	-1.48437	0.008705	0.020293	-1.40776
OID20741	PDGFB	Inflammat ion	-5.2421	-4.62915	-5.2267	0.422392	0.496559	1.570039	0.805813	0.846682	-1.06245
OID20742	CRKL	Inflammat ion	-2.05	-1.96635	-2.00655	0.276236	0.346636	-1.05343	0.486324	0.563274	1.030397
OID20743	EPCAM	Inflammat ion	-3.60675	-2.73325	-2.87805	0.000685	0.002075	1.850032	0.006639	0.01627	1.750176
OID20744	DNPH1	Inflammat ion	-3.02495	-2.26805	-2.42125	0.009471	0.019329	1.757774	0.039208	0.068334	1.537728
OID20745	CCL17	Inflammat ion	-4.8509	-5.7834	-5.5895	0.300342	0.372628	-1.53204	0.786541	0.832005	-1.53251
OID20746	MEGF10	Inflammat ion	2.26645	0.74705	0.7964	6.44E-08	3.08E-06	-2.51612	5.35E-08	3.91E-06	-2.15995
OID20747	CRHBP	Inflammat ion	-3.84865	-5.06	-4.72325	5.84E-06	4.89E-05	-2.02434	0.000171	0.000842	-1.84114
OID20748	LGALS4	Inflammat ion	-1.72585	-2.29145	-1.9857	0.002778	0.00669	-1.43968	0.185526	0.254671	-1.24259
OID20749	LHPP	Inflammat ion	1.44595	2.18145	2.24725	7.43E-06	5.58E-05	1.626138	0.000545	0.002167	1.413969
OID20750	TPP1	Inflammat ion	2.9673	4.29665	4.27625	2.06E-06	2.46E-05	2.817177	6.09E-06	7.19E-05	3.114715
OID20751	CRELD2	Inflammat ion	0.1946	-0.48705	-0.54625	0.000131	0.000516	-1.51556	0.000224	0.001046	-1.4102
OID20752	CTRC	Inflammat ion	-4.9503	-5.98095	-5.84	0.000392	0.001307	-2.04068	0.002705	0.007656	-1.63399
OID20753	MEPE	Inflammat ion	-4.2819	-4.88765	-4.82165	0.000249	0.00089	-1.8264	0.000287	0.001304	-1.72692
OID20754	CDON	Inflammat ion	0.99665	0.92015	0.97065	0.489804	0.563737	1.145438	0.847735	0.881621	1.02406
OID20755	ADGRE2	Inflammat ion	-4.41015	-4.4614	-4.42405	0.222039	0.289203	-1.12159	0.27403	0.352738	1.000243
OID20756	AGER	Inflammat ion	-2.507	-2.8818	-2.89745	0.046317	0.075791	-1.15621	0.451344	0.528333	1.005456
OID20757	IL1R2	Inflammat ion	-3.42955	-2.9026	-3.20455	0.000292	0.001011	1.594827	0.210418	0.282443	1.22731
OID20758	ESM1	Inflammat ion	2.26295	-0.48675	-0.27255	1.48E-08	1.16E-06	-5.81972	8.72E-08	5.32E-06	-4.80588
OID20759	SPON1	Inflammat ion	3.94565	4.4125	4.40565	0.428111	0.502207	1.308986	0.343477	0.424226	1.292666
OID20760	ATP5IF1	Inflammat ion	-1.3241	-0.94005	-0.8822	0.005574	0.012314	1.652672	0.036813	0.064673	1.516452
OID20761	SKAP2	Inflammat ion	-4.1964	-4.27615	-4.42045	0.902198	0.920645	-1.03839	0.896499	0.922034	-1.01614
OID20762	CXCL1	Inflammat ion	0.7878	-0.03965	-0.2048	0.005888	0.012802	-1.80344	0.014338	0.03051	-2.04202
OID20763	PRSS8	Inflammat ion	-0.90915	-1.1232	-0.98865	0.004639	0.010447	-1.21159	0.036683	0.064617	-1.13135
OID20764	PLAUR	Inflammat ion	0.0718	0.05835	0.2959	0.286837	0.357786	-1.02619	0.732158	0.785832	1.027259
OID20765	CCL22	Inflammat ion	-8.43875	-8.68215	-7.9745	0.777746	0.82191	-1.1113	0.975215	0.981472	-1.34471
OID20766	PRELP	Inflammat ion	3.0049	2.29405	2.7959	0.00268	0.006495	-1.52737	0.170648	0.23748	-1.24608
OID20767	MATN2	Inflammat ion	0.9945	0.15665	0.29315	6.79E-05	0.000297	-1.74158	0.007263	0.017566	-1.5042
OID20768	LTBR	Inflammat ion	-0.43485	-0.70025	-0.66735	0.028694	0.05041	-1.10347	0.155403	0.219604	-1.02211
OID20769	LAMA4	Inflammat ion	-1.2289	-1.04735	-0.7481	0.28379	0.355709	-1.15261	0.290281	0.370615	1.196234
OID20770	FST	Inflammat ion	-2.22985	-3.0033	-2.8604	0.001424	0.003822	-1.23795	0.00137	0.004412	-1.32492
OID20771	CHRD1	Inflammat ion	2.841	3.1653	2.99565	0.019	0.0356	1.275842	0.023337	0.045033	1.108301

Supplementary Table 6. (continued)

OID20772	CCL24	Inflammat ion	-3.77495	-4.82765	-4.49125	0.000948	0.002753	-1.63824	0.002722	0.007683	-1.81623
OID20773	LGMN	Inflammat ion	3.02685	2.15705	2.5284	0.050409	0.080684	-1.17723	0.777	0.825093	-1.08309
OID20774	DAG1	Inflammat ion	1.76305	1.19805	1.526	5.62E-05	0.000253	-1.34406	0.009811	0.022257	-1.27766
OID20775	PNLIPRP2	Inflammat ion	-4.52355	-5.2995	-5.1224	0.064419	0.100454	-1.61748	0.376802	0.456654	-1.22086
OID20776	OSCAR	Inflammat ion	-1.86955	-2.59635	-2.38935	9.43E-05	0.000391	-1.64627	0.020076	0.04008	-1.26861
OID20777	PON3	Inflammat ion	-6.64505	-5.96135	-5.8916	0.086237	0.128827	1.225652	0.012385	0.027252	1.264923
OID20778	FABP1	Inflammat ion	-0.96915	-1.45205	-0.97255	0.025668	0.046051	-1.62068	0.413091	0.49035	-1.2571
OID20779	MPIG6B	Inflammat ion	-4.82645	-4.2244	-4.68845	0.090442	0.13456	1.176621	0.892704	0.92028	1.090054
OID20780	B4GALT1	Inflammat ion	0.58715	-2.04545	-1.55585	1.18E-08	1.08E-06	-4.74481	1.99E-07	8.18E-06	-4.25527
OID20781	LGALS9	Inflammat ion	0.22905	0.50345	0.4289	0.001217	0.003341	1.126424	0.00329	0.009032	1.191724
OID20782	FSTL3	Inflammat ion	0.606	0.30055	0.07905	0.097558	0.143976	-1.13567	0.030991	0.055967	-1.21167
OID20783	TNFRSF14	Inflammat ion	3.1142	3.1249	2.9764	0.583473	0.649091	1.052558	0.479943	0.557648	-1.00793
OID20784	REG4	Inflammat ion	-2.32335	-1.80585	-1.8878	0.315158	0.386499	1.077098	0.037945	0.066555	1.279695
OID20785	WFIKN2	Inflammat ion	2.0899	1.2088	1.56635	0.010478	0.021226	-1.52832	0.053482	0.089403	-1.23119
OID20786	AGRN	Inflammat ion	4.5325	3.27285	3.6447	2.02E-05	0.000115	-2.0824	0.006694	0.016333	-1.83045
OID20787	TFF2	Inflammat ion	-0.7966	-0.53835	-0.3188	0.19838	0.264668	1.139499	0.019091	0.038321	1.276594
OID20788	CXCL3	Inflammat ion	-3.4606	-5.04965	-4.46195	0.002857	0.006833	-3.25284	0.100607	0.152577	-1.73418
OID20789	NME3	Inflammat ion	-2.7056	-2.6618	-2.5473	0.155956	0.215396	-1.06014	0.367879	0.448314	-1.02473
OID20792	NDRG1	Neurology	-1.318	-0.8489	-1.12425	0.003596	0.0084	1.266064	0.183023	0.252281	1.108378
OID20793	ODAM	Neurology	-1.2412	-1.65055	-1.69865	0.391959	0.467287	-1.43212	0.195672	0.26547	-1.39441
OID20797	COPE	Neurology	-0.77175	-0.24605	-0.18575	0.056845	0.089421	1.549551	0.020769	0.041089	2.308011
OID20799	SLC16A1	Neurology	-1.1484	-0.43535	-0.61965	0.557582	0.625996	1.394792	0.809727	0.849169	1.461196
OID20801	CALB2	Neurology	0.8629	0.05475	-0.2186	0.000412	0.001362	-1.52124	9.53E-05	0.000515	-2.09333
OID20806	AMFR	Neurology	-1.6693	-0.8547	-1.09005	0.000762	0.002292	2.073905	0.030516	0.055291	1.291995
OID20807	MMP13	Neurology	-2.574	-1.98895	-2.2931	0.071878	0.11038	1.500091	0.211263	0.283232	1.298009
OID20808	CDHR1	Neurology	5.02495	3.4523	3.4331	5.32E-07	1.06E-05	-2.51324	1.17E-06	2.48E-05	-2.88146
OID20809	IL34	Neurology	5.17975	5.30255	5.09385	0.396657	0.471352	-1.13634	0.144473	0.206014	-1.28842
OID20810	ANXA5	Neurology	-0.31975	0.1931	-0.6564	0.138564	0.194557	1.37102	0.499458	0.571255	-1.25941
OID20816	TDGF1	Neurology	-1.782	-3.02435	-2.9928	0.028578	0.050286	-2.35487	0.051846	0.086911	-1.74993
OID20824	LIF	Neurology	4.14795	5.4645	5.7343	5.46E-05	0.000248	2.387995	3.85E-08	3.25E-06	3.335031
OID20830	MAPT	Neurology	2.869	3.174	2.75615	0.298645	0.371362	1.176621	0.915186	0.933898	1.065995
OID20840	EPHA10	Neurology	2.1906	0.5198	0.64305	2.55E-06	2.83E-05	-2.611	9.56E-06	0.0001	-2.18185
OID20841	GHRHR	Neurology	-1.4195	-0.7143	-0.9962	0.101951	0.149256	1.549551	0.143673	0.205141	1.239192
OID20846	FKBP7	Neurology	-0.4188	-0.5087	-0.39305	0.59013	0.655833	-1.07531	0.370407	0.450396	1.129826
OID20847	PTS	Neurology	0.72705	1.0137	1.00845	0.725642	0.777322	1.029433	0.755449	0.805323	1.252578
OID20849	NXPH1	Neurology	2.7541	0.30695	0.74465	1.2E-09	2.47E-07	-4.09922	4.11E-09	6.44E-07	-4.53468
OID20855	ILKAP	Neurology	-1.17365	-0.5555	-0.44215	0.003703	0.008613	1.912885	0.005712	0.014419	1.584415
OID20861	GDNF	Neurology	-3.616	-2.97815	-2.21985	0.009902	0.020171	1.892312	0.003065	0.008521	2.735207
OID20862	SLIT2	Neurology	2.66085	0.23175	0.5142	5.54E-07	1.09E-05	-5.38557	3.82E-05	0.000258	-3.99045
OID20863	NOS1	Neurology	-2.02095	-2.48715	-2.5225	0.192675	0.257682	-1.68395	0.302325	0.383761	-1.25937
OID20866	MUC13	Neurology	-4.5923	-3.47645	-3.6708	0.001337	0.003635	2.251677	0.003614	0.009798	2.008544
OID20871	NEFL	Neurology	0.24245	0.515	-0.5245	0.59946	0.663516	-1.17707	0.076088	0.120036	-1.89842
OID20872	FHIT	Neurology	-1.2801	-1.24845	-0.6229	0.185487	0.249895	1.152327	0.1841	0.25331	1.526471

Supplementary Table 6. (continued)

OID20873	KIRREL2	Neurology	1.35775	-0.09285	0.0769	2.11E-05	0.000118	-2.48045	0.000117	0.000602	-2.37528
OID20874	CLSTN1	Neurology	0.71895	2.2011	2.025	3.74E-05	0.000182	2.788714	2.75E-05	0.000203	2.654058
OID20876	GSTP1	Neurology	0.3658	1.5496	0.8951	0.000371	0.001247	2.438595	0.022166	0.043254	1.613004
OID20877	MAX	Neurology	-1.10005	-0.3301	-0.6158	0.033018	0.056646	1.603751	0.080776	0.126522	1.193418
OID20880	PTK7	Neurology	0.41365	-0.50185	-0.33275	0.001081	0.003066	-1.44358	0.001027	0.003491	-2.03962
OID20889	ISLR2	Neurology	-0.3371	-1.62895	-1.72565	2.59E-05	0.000139	-2.43058	0.00106	0.003547	-1.90752
OID20891	ECE1	Neurology	-2.44055	-2.5139	-2.11865	0.875396	0.899986	-1.05493	0.214657	0.286731	-1.06201
OID20895	AKT1S1	Neurology	-2.597	-2.456	-2.6239	0.120711	0.172579	1.019492	0.487355	0.563873	-1.10194
OID20897	DPEP1	Neurology	-5.93105	-5.5146	-5.55085	0.02013	0.03759	1.714158	0.024091	0.046083	1.94174
OID20900	IL18RAP	Neurology	-7.8752	-7.07245	-6.2581	0.008768	0.018233	1.928461	0.007638	0.018193	2.765902
OID20901	CRIP2	Neurology	-4.4805	-3.1433	-3.70025	0.000903	0.002665	1.587658	0.013873	0.029635	1.302425
OID20902	PPP3R1	Neurology	-2.0973	-1.93655	-2.28995	0.842919	0.874684	-1.06319	0.651046	0.710585	-1.17153
OID20903	DARS1	Neurology	-2.68565	-2.1607	-2.38255	0.264504	0.335752	1.493451	0.750742	0.801863	1.33256
OID20904	MAD1L1	Neurology	-3.39275	-2.6656	-2.84745	0.001412	0.003801	1.57538	0.025521	0.048033	1.422669
OID20906	SKAP1	Neurology	-5.4609	-4.07545	-4.24805	0.016459	0.031375	2.397698	0.078415	0.123175	1.78912
OID20909	GGT5	Neurology	-5.40185	-4.90615	-4.7406	0.64635	0.708276	-1.03462	0.096241	0.146768	1.291636
OID20910	CLPP	Neurology	-1.6946	-0.66565	-0.49335	0.000646	0.001999	2.325595	0.001805	0.005519	2.532391
OID20911	IL6	Neurology	-0.15095	0.24515	0.58915	0.365829	0.440439	1.189372	0.070571	0.112464	1.318274
OID20912	PTPRN2	Neurology	2.30355	1.77815	1.76235	3E-05	0.000152	-1.32777	4.71E-05	0.000303	-1.39508
OID20913	NMNAT1	Neurology	-0.9151	-0.80455	-0.7306	0.961942	0.967228	1.002012	0.923058	0.941056	1.042321
OID20915	CSF2RA	Neurology	-2.21255	-2.6693	-2.4739	0.023563	0.042553	-1.20752	0.395756	0.474522	-1.16239
OID20916	PSG1	Neurology	-4.10165	-3.1411	-3.1061	0.000186	0.000694	1.520136	2.63E-05	0.000196	1.655653
OID20917	DRAXIN	Neurology	-0.6729	-2.77095	-2.68245	7.92E-06	5.72E-05	-3.75341	1.69E-05	0.000141	-3.81676
OID20918	EBAG9	Neurology	-3.5369	-2.8317	-2.90395	0.011491	0.023151	1.400459	0.036722	0.064617	2.057083
OID20919	ULBP2	Neurology	-0.1135	-0.2461	0.00625	0.210292	0.277525	-1.06884	0.997143	0.998161	1.016598
OID20921	ABHD14B	Neurology	-3.7946	-3.12785	-3.2287	0.001629	0.00427	1.587493	0.003491	0.009512	1.505142
OID20922	PHOSPHO 1	Neurology	-3.1549	-3.06115	-3.1299	0.725006	0.777322	1.070437	0.594058	0.658864	1.105884
OID20923	CRADD	Neurology	-5.0573	-4.3966	-5.59995	0.108472	0.157961	1.427806	0.821489	0.859862	-1.06847
OID20924	SPOCK1	Neurology	5.27125	6.05975	5.87245	0.000202	0.000739	1.544243	0.000224	0.001046	1.454981
OID20926	WWP2	Neurology	-4.65965	-4.0528	-3.95165	0.069696	0.10733	1.286053	0.057867	0.096125	1.500663
OID20927	APRT	Neurology	-5.20815	-3.81365	-4.4396	0.009161	0.018837	2.559656	0.085825	0.134048	1.590687
OID20931	NAAA	Neurology	0.1826	-2.14835	-1.5515	5.7E-06	4.85E-05	-3.36814	5.46E-05	0.000337	-3.26278
OID20932	SERPINB9	Neurology	-0.46115	-0.19405	-0.01505	0.032016	0.055186	1.215584	0.007493	0.017925	1.446333
OID20933	GFRA3	Neurology	-3.8672	-4.54805	-4.4389	0.00026	0.00092	-1.42306	0.010381	0.023348	-1.57037
OID20934	DKK4	Neurology	-0.02165	0.01615	0.1124	0.859363	0.886824	1.162596	0.312599	0.394976	-1.00125
OID20935	EFNA4	Neurology	-0.03335	-1.7292	-1.3796	1.41E-05	8.86E-05	-2.88526	0.000183	0.000886	-2.44765
OID20936	VWC2	Neurology	0.5452	-1.86955	-1.73785	9.84E-07	1.45E-05	-4.20425	4.5E-07	1.45E-05	-4.57335
OID20937	FKBP5	Neurology	-3.4194	-2.39275	-2.5036	3.52E-05	0.000172	1.985772	0.000297	0.00134	1.873908
OID20938	RSP01	Neurology	1.0112	0.02375	0.1323	3.83E-05	0.000184	-2.07247	2.5E-05	0.000189	-1.8214
OID20939	WFIKKN1	Neurology	-4.96305	-5.60715	-5.25275	0.047521	0.077301	-1.48854	0.036657	0.064617	-1.57009
OID20940	CDCP1	Neurology	-0.8785	-0.6725	-0.6979	0.132301	0.186479	1.142782	0.088889	0.138048	1.132962
OID20941	IFNGR2	Neurology	-1.16875	-1.3983	-1.33685	0.294206	0.366257	-1.12471	0.227718	0.30161	-1.04265
OID20943	SCARB2	Neurology	0.1936	-1.03335	-0.9756	9.94E-08	4.37E-06	-2.22677	1.61E-07	7.38E-06	-2.42477
OID20944	GPC5	Neurology	0.89535	-0.52745	-0.3208	1.5E-07	5.14E-06	-2.6899	1.43E-06	2.71E-05	-2.73938
OID20945	LY96	Neurology	-4.44565	-4.47495	-4.41845	0.269932	0.341288	1.033508	0.387489	0.466517	1.098702
OID20946	FLRT2	Neurology	-2.7984	-3.0189	-2.7807	0.039644	0.066073	-1.19036	0.611893	0.674557	-1.07427
OID20947	TMPRSS5	Neurology	0.38835	-0.25125	-0.2808	5.12E-05	0.000235	-1.56884	0.000217	0.001025	-1.38939
OID20948	SETMAR	Neurology	-5.60225	-5.05065	-5.08375	0.029741	0.051891	1.285964	0.325498	0.40964	1.328778
OID20949	GLB1	Neurology	0.7992	0.2771	0.29455	0.004107	0.009453	-1.44328	0.003447	0.009415	-1.46586
OID20950	RHOC	Neurology	-4.1319	-3.78135	-3.66905	0.116529	0.167691	1.253403	0.044896	0.076905	1.311666
OID20951	MDGA1	Neurology	2.53625	1.10345	1.14665	3.94E-06	3.8E-05	-2.83049	1.05E-05	0.000105	-2.72915
OID20952	TXLNA	Neurology	-3.07605	-2.8802	-3.13715	0.774238	0.818992	-1.17789	0.94494	0.956262	-1.0284
OID20953	TNFSF14	Neurology	-5.67865	-4.78665	-5.36385	0.000596	0.001866	2.352427	0.007474	0.017925	1.543922
OID20955	STAMBP	Neurology	-4.89375	-4.43555	-4.40075	0.004446	0.010107	1.373827	0.034016	0.060632	1.442329
OID20956	DNMBP	Neurology	-3.70645	-3.13165	-3.22975	9.31E-05	0.000387	1.450198	0.000725	0.002627	1.359051
OID20957	TNR	Neurology	3.6787	4.22395	4.10465	0.028187	0.04985	1.302741	0.026941	0.050053	1.284093
OID20958	CC2D1A	Neurology	-3.05645	-2.56045	-2.79205	0.004516	0.010225	1.303012	0.090357	0.139876	1.254489

Supplementary Table 6. (continued)

OID20959	MITD1	Neurology	-4.07995	-2.64585	-3.31925	7.18E-06	5.47E-05	2.500905	0.010398	0.023348	1.627209
OID20960	FKBP4	Neurology	-3.4451	-3.20285	-3.2106	0.048863	0.079015	1.144645	0.020886	0.041246	1.219509
OID20961	NPM1	Neurology	-2.20415	-1.07025	-1.36775	0.005434	0.012032	1.971304	0.038125	0.066765	2.253082
OID20962	PRDX1	Neurology	1.06825	2.0415	2.04965	6.28E-05	0.00028	1.854782	0.000799	0.002849	1.689856
OID20963	LPO	Neurology	-3.8529	-4.53555	-4.48765	0.844413	0.874684	-1.71642	0.749734	0.801566	1.010241
OID20964	TNFRSF6B	Neurology	-2.9825	-4.77675	-3.80345	0.004387	0.009993	-2.42469	0.090602	0.139904	-3.06556
OID20965	LAIR2	Neurology	-3.21805	-3.4405	-3.42475	0.387157	0.46307	1.200429	0.284139	0.364469	1.188795
OID20966	CD274	Neurology	-1.4888	-1.8502	-1.92665	0.005737	0.012584	-1.20506	0.065359	0.106003	-1.18267
OID20967	TNFRSF10 A	Neurology	-2.8504	-2.30125	-2.3012	0.001124	0.00314	1.21934	0.00044	0.00181	1.4799
OID20968	FRZB	Neurology	0.0495	0.12445	0.3244	0.968232	0.971772	1.035301	0.847898	0.881621	1.142386
OID20969	PSME1	Neurology	-4.2402	-3.6572	-3.71825	0.012855	0.025664	1.395469	0.029712	0.054282	1.338855
OID20970	LAYN	Neurology	2.1421	1.8541	1.7526	0.847748	0.877311	1.0475	0.701196	0.756299	-1.10581
OID20971	SNCG	Neurology	-2.0454	-2.2233	-2.21895	0.147043	0.20463	-1.05205	0.251105	0.325902	-1.17259
OID20973	CCS	Neurology	-4.7676	-4.2597	-4.5454	0.013746	0.026953	1.421979	0.242205	0.315469	1.166511
OID20974	ADGRB3	Neurology	-0.4112	-1.12065	-1.3188	8.95E-07	1.35E-05	-1.38319	2.01E-07	8.18E-06	-1.48519
OID20975	HNMT	Neurology	-4.08215	-5.5563	-5.4383	0.002788	0.006699	-2.20083	0.003655	0.009885	-2.06194
OID20976	CX3CL1	Neurology	-3.5282	-3.5549	-3.8368	0.829185	0.8638	-1.02073	0.664962	0.725053	-1.13171
OID20978	AFP	Neurology	-6.9247	-6.06135	-6.35	0.02085	0.038606	1.758749	0.119651	0.178018	1.39944
OID20979	MATN3	Neurology	-1.8429	-2.76235	-2.7607	0.048046	0.078039	-1.65893	0.039061	0.068185	-1.36495
OID20980	OBP2B	Neurology	-1.36535	-1.90095	-1.91265	6.31E-05	0.00028	-1.44033	2.09E-05	0.000167	-1.7483
OID20981	TNFRSF10 B	Neurology	-1.12925	-1.0757	-1.0641	0.012246	0.024627	1.309395	0.030346	0.055166	1.264792
OID20982	EPHB6	Neurology	-0.96435	-2.6837	-2.791	2.8E-07	7.16E-06	-3.57031	1.05E-06	2.38E-05	-3.36778
OID20983	CALCA	Neurology	-3.63115	-3.88575	-4.0288	0.085251	0.128175	-1.30351	0.012265	0.027096	-1.4289
OID20984	SFRP1	Neurology	4.1552	1.5413	1.9412	2.05E-06	2.46E-05	-5.43489	0.000125	0.000633	-3.06227
OID20985	TNFRSF9	Neurology	-3.5754	-2.73065	-2.68805	0.008867	0.018391	1.533417	0.000682	0.002529	1.752665
OID20986	RBKS	Neurology	-0.6247	-0.3104	-0.45515	0.336857	0.409147	1.144248	0.55299	0.622752	1.134377
OID20988	CD63	Neurology	-3.78065	-4.0863	-4.5287	0.005024	0.011235	-1.29016	4.4E-06	5.58E-05	-1.82267
OID20989	PSME2	Neurology	-4.35415	-3.7289	-3.50825	0.004357	0.009946	2.000901	0.003378	0.009248	1.643931
OID20990	ASGR1	Neurology	-5.1694	-4.59805	-4.60305	0.016075	0.030803	1.640289	0.007159	0.017352	1.481851
OID20992	FUT8	Neurology	-3.2097	-4.17805	-4.11175	0.000797	0.00239	-1.75747	0.000575	0.002265	-1.9445
OID20994	BMP4	Neurology	-5.2224	-5.24355	-5.13005	0.307861	0.380324	-1.18715	0.792017	0.835383	1.105118
OID20995	CST5	Neurology	-1.13295	-1.34935	-1.19155	0.433153	0.506081	-1.01491	0.298211	0.379262	-1.07814
OID20996	ASAH2	Neurology	-1.0442	-3.2211	-2.5941	4E-06	3.8E-05	-2.83825	5.3E-06	6.54E-05	-3.35753
OID20997	CXCL8	Neurology	-0.02325	0.38015	0.50545	0.062724	0.098106	1.187107	0.024013	0.046015	1.121671
OID20998	BCAN	Neurology	1.8278	1.4782	1.2413	0.006037	0.013074	-1.26484	0.004175	0.01102	-1.25959
OID20999	ACVRL1	Neurology	-3.5669	-3.2006	-3.1648	0.066073	0.102469	1.16538	0.022369	0.043548	1.129043
OID21001	ADAM22	Neurology	2.0141	0.96725	0.8021	6.51E-06	5.14E-05	-2.06229	3.16E-06	4.51E-05	-2.21039
OID21004	CCL2	Neurology	1.04685	1.7064	1.79585	0.038069	0.063719	1.219128	0.000692	0.00255	1.819069
OID21005	CDH15	Neurology	-2.75645	-3.11735	-2.667	0.002408	0.005863	-1.51819	0.252068	0.326766	-1.13379
OID21006	CNTN5	Neurology	1.86275	0.93815	0.972	1.1E-07	4.46E-06	-1.94524	1.39E-07	6.98E-06	-1.82273
OID21007	ROBO2	Neurology	1.7695	0.53425	0.72945	2.66E-06	2.86E-05	-2.10577	1.11E-05	0.00011	-2.09732
OID21008	CDH3	Neurology	1.9649	1.20925	1.1319	1.13E-05	7.4E-05	-1.58112	1.22E-06	2.48E-05	-1.66077
OID21009	DDR1	Neurology	2.4806	2.58705	2.61215	0.27141	0.342538	1.163483	0.293794	0.374229	1.163685
OID21010	CD300C	Neurology	-0.7302	-0.7854	-0.96885	0.210967	0.278081	-1.16955	0.861455	0.89318	-1.04138
OID21011	PVR	Neurology	-0.3353	-1.74415	-1.6147	1.65E-08	1.17E-06	-2.45879	2.46E-08	3.01E-06	-2.20473
OID21012	TXNDC5	Neurology	0.11755	-0.23725	-0.16875	0.067571	0.104644	-1.17911	0.110869	0.166904	-1.25946
OID21013	FUT3_FUT 5	Neurology	4.04945	4.12705	4.6054	0.252165	0.322326	1.305453	0.015771	0.032725	1.369928
OID21014	STC1	Neurology	2.421	0.25065	0.75095	0.052792	0.084131	-1.94895	0.442742	0.519926	-1.84395
OID21015	RWDD1	Neurology	-0.15715	-0.04215	-0.07805	0.308973	0.381183	1.105922	0.178737	0.246859	1.120467
OID21016	VTA1	Neurology	-4.59425	-3.7128	-3.96945	0.0019	0.004864	1.515664	0.023487	0.045244	1.45902
OID21017	EZR	Neurology	-0.87515	0.56455	0.18655	3E-07	7.49E-06	2.45105	1.27E-05	0.000119	2.071822
OID21018	CD8A	Neurology	-6.9647	-6.38325	-5.8791	0.034551	0.059092	1.494228	0.0137	0.029463	2.301062
OID21019	MMP8	Neurology	-4.26595	-3.93255	-3.80875	0.000596	0.001866	1.588428	0.00581	0.014597	1.533151
OID21020	SEMA4D	Neurology	-0.13735	-0.622	-0.62255	0.002348	0.005767	-1.2588	0.001251	0.00409	-1.29424
OID21021	PDCD5	Neurology	-2.22535	-2.44955	-2.6404	0.228154	0.295069	-1.17475	0.882781	0.911847	-1.1765

Supplementary Table 6. (continued)

OID21022	NSFL1C	Neurology	-0.83945	-0.50355	-0.6516	0.005597	0.01234	1.294594	0.116655	0.174032	1.085831
OID21023	SERPINB6	Neurology	3.0121	2.99915	3.12415	0.263294	0.334603	1.111764	0.114133	0.171199	1.212429
OID21024	CXCL13	Neurology	-4.2618	-5.223	-5.13985	0.330839	0.402283	-1.82576	0.494377	0.567216	-2.09333
OID21025	CD34	Neurology	-5.4629	-5.3437	-4.98795	0.206889	0.274353	1.378071	0.231299	0.305615	1.368741
OID21026	VCAN	Neurology	0.7032	1.0537	0.71315	0.44997	0.523932	1.200346	0.943142	0.955323	1.011362
OID21027	PPCDC	Neurology	-1.96405	-1.96485	-2.15745	0.398079	0.472021	1.110416	0.902929	0.924828	-1.14457
OID21028	SMPD1	Neurology	2.30465	0.8966	1.22725	1.73E-05	0.000101	-1.95511	0.001663	0.005172	-1.92532
OID21029	BST2	Neurology	-2.67445	-2.34365	-2.34235	0.909622	0.925813	-1.09555	0.633106	0.694455	1.013327
OID21030	CCL19	Neurology	-5.92305	-6.9964	-6.4075	0.037452	0.062974	-3.1341	0.238666	0.311599	-2.54788
OID21031	SULT1A1	Neurology	-3.54015	-3.12545	-3.29935	0.983579	0.984998	-1.12097	0.61679	0.679038	-1.27606
OID21032	SUMF2	Neurology	2.7002	3.22815	3.0335	0.001712	0.004433	1.46871	0.007484	0.017925	1.330621
OID21033	PRL	Neurology	-4.423	-4.89705	-4.29715	0.002399	0.005853	-1.54034	0.690954	0.748193	-1.06925
OID21034	LBR	Neurology	-1.6785	-0.82075	-0.9272	0.002398	0.005853	1.772456	0.019671	0.039413	1.376591
OID21035	LAMP2	Neurology	0.5902	0.7627	0.7967	0.901556	0.920645	1.020977	0.445598	0.52272	1.148937
OID21036	ANXA3	Neurology	-6.2941	-4.6269	-5.03195	5.23E-05	0.000239	3.205835	0.000578	0.002266	2.322534
OID21037	ENO1	Neurology	-0.10895	1.54115	1.10505	6.36E-05	0.000281	2.846226	0.000141	0.000708	2.348029
OID21038	CHGB	Neurology	4.6656	5.14525	5.12775	6E-06	4.92E-05	1.70734	1.34E-05	0.000123	1.753576
OID21039	ADAM8	Neurology	-2.69115	-2.4268	-2.3979	0.738474	0.787228	1.017021	0.157766	0.221235	1.105577
OID21040	CD74	Neurology	0.4341	0.2067	0.47985	0.879601	0.903463	-1.15053	0.015192	0.031773	1.193336
OID21041	LGALS8	Neurology	0.4047	0.9501	0.78205	0.000126	0.000502	1.472584	0.001325	0.004291	1.368315
OID21042	CXCL11	Neurology	-4.5138	-4.55415	-4.9692	0.217691	0.285232	-1.02836	0.408835	0.48635	-1.14461
OID21043	FABP5	Neurology	-1.06435	0.74315	1.22325	0.0079	0.016682	2.401357	0.005321	0.01365	2.420324
OID21044	GOLM2	Neurology	0.10445	-1.26585	-0.9505	1.18E-07	4.46E-06	-2.64313	1.08E-06	2.38E-05	-2.05951
OID21045	RGMB	Neurology	2.09405	0.26055	0.436	1.8E-07	5.23E-06	-3.37889	5.54E-06	6.68E-05	-2.88966
OID21046	ENO2	Neurology	4.0769	4.42055	4.4929	0.006695	0.014413	1.271693	0.002306	0.006735	1.433359
OID21047	OXT	Neurology	-1.1299	-1.60115	-1.2527	0.349735	0.422452	-1.17601	0.541795	0.613289	1.162072
OID21048	SCARF2	Neurology	-1.95305	-1.52655	-1.4098	0.000408	0.001354	1.301207	0.000113	0.000591	1.379313
OID21049	TNFRSF8	Neurology	-3.70995	-3.6974	-3.7694	0.649161	0.709939	1.064592	0.797571	0.840434	1.0511
OID21050	THY1	Neurology	2.6497	2.69185	2.7325	0.480737	0.555047	1.034583	0.497903	0.570666	1.110031
OID21051	BST1	Neurology	-3.0593	-2.75425	-2.65165	0.01676	0.031838	1.235975	0.001056	0.003545	1.348916
OID21052	TNFRSF21	Neurology	1.2686	0.5103	0.3958	8.23E-07	1.29E-05	-1.56303	9.54E-07	2.31E-05	-1.41962
OID21053	CPPED1	Neurology	-0.95475	0.66505	0.1741	0.000916	0.002697	3.073324	0.010304	0.023231	2.712833
OID21054	CPM	Neurology	0.5352	0.40905	0.27005	0.071996	0.110407	-1.12459	0.03427	0.060986	-1.09897
OID21055	NCAN	Neurology	2.9275	4.0488	3.97635	0.00228	0.005638	2.101836	0.000393	0.001658	2.334559
OID21056	CTSS	Neurology	0.3024	0.4112	0.6186	0.10085	0.148039	1.022535	0.002169	0.006367	1.266371
OID21057	NTRK3	Neurology	-0.1054	-0.96595	-0.96185	8.06E-06	5.75E-05	-1.47565	1.22E-06	2.48E-05	-1.56765
OID21058	SCARA5	Neurology	-3.26845	-3.116	-2.98895	0.984101	0.984998	1.077024	0.07312	0.11602	1.255098
OID21059	LRPAP1	Neurology	1.63565	1.14975	1.2092	0.008409	0.017653	-1.37316	0.046907	0.079605	-1.37869
OID21060	CPA2	Neurology	-0.57835	-0.847	-0.90775	0.008438	0.017681	-1.19222	0.00242	0.007012	-1.32588
OID21061	CLEC1B	Neurology	-4.3263	-4.34735	-4.101	0.807983	0.84734	-1.09699	0.978404	0.98378	1.128182
OID21062	FOLR2	Neurology	-0.9449	-0.96305	-0.1725	0.118044	0.169428	1.053215	0.000808	0.002852	1.667631
OID21063	MSR1	Neurology	-0.00865	-0.62885	-0.9052	1.17E-05	7.65E-05	-1.77571	0.000166	0.00082	-1.57975
OID21064	JAM2	Neurology	4.5153	4.18945	4.1703	0.012918	0.025695	-1.22999	0.013704	0.029463	-1.0949
OID21065	RGMA	Neurology	-0.4528	-1.9173	-1.61415	7.29E-07	1.27E-05	-2.8297	3.03E-05	0.000217	-2.18079
OID21066	DKK1	Neurology	0.8707	0.6271	0.54035	0.861002	0.887681	-1.02968	0.153542	0.217535	-1.75443
OID21067	BIN2	Neurology	-1.8662	-0.89225	-1.4866	0.000126	0.000502	2.256442	0.015297	0.03187	1.372922
OID21068	CCN5	Neurology	0.9083	-0.31355	-0.0146	2.9E-05	0.000149	-1.9668	0.000667	0.002491	-1.98412
OID21069	CGA	Neurology	1.14785	0.89095	1.01865	0.964992	0.969407	-1.04801	0.151524	0.214952	1.03976
OID21070	APP	Neurology	6.36165	7.43995	7.54365	5.81E-06	4.89E-05	1.98364	1.82E-06	3.21E-05	2.013981
OID21071	ITGAM	Neurology	0.06265	0.3397	0.07285	0.014189	0.027574	1.175521	0.561688	0.631253	-1.09863
OID21072	APOH	Neurology	1.9318	2.32965	2.3097	0.937955	0.948319	-1.05052	0.752786	0.803264	1.148141
OID21073	MIF	Neurology	4.91065	5.8846	5.8256	2.83E-08	1.52E-06	2.163849	1.5E-07	7.18E-06	2.080312
OID21074	NPTX1	Neurology	5.459	4.63615	4.67175	0.0002	0.000736	-1.79061	0.001111	0.003674	-1.59654
OID21075	GGT1	Neurology	0.892	0.8046	0.8454	0.814785	0.852032	-1.05588	0.592287	0.6576	-1.03283
OID21076	BCAM	Neurology	4.20585	4.1972	4.1587	0.487741	0.56229	1.046629	0.473458	0.55128	1.091188
OID21077	STIP1	Neurology	0.1226	0.57465	0.48555	0.082968	0.125136	1.28187	0.237666	0.311407	1.237776
OID21078	AHSP	Neurology	-1.8156	-1.55855	-1.90255	0.20745	0.274434	-1.15221	0.641527	0.70229	-1.17952
OID21079	DSC2	Neurology	3.7782	3.83365	3.83395	0.520433	0.592775	1.025907	0.498832	0.571134	1.130883

Supplementary Table 6. (continued)

OID21080	CD164	Neurology	7.32675	8.2426	8.31725	6.67E-06	5.19E-05	2.089996	3.54E-05	0.000246	2.291989
OID21081	NRP2	Neurology	1.83155	0.4068	0.6398	3.01E-05	0.000153	-2.28105	0.000321	0.001393	-2.18192
OID21082	SIGLEC5	Neurology	2.24035	1.46395	1.83885	0.012618	0.025328	-1.45433	0.08671	0.135046	-1.47264
OID21083	MMP3	Neurology	2.03895	4.0783	2.47995	0.005941	0.012893	3.914288	0.231755	0.305849	1.310757
OID21084	WARS	Neurology	-0.01795	0.0596	0.02765	0.853105	0.882024	-1.50948	0.551313	0.6215	-1.1959
OID21085	NID2	Neurology	4.686	4.72155	5.0341	0.502935	0.576433	-1.0121	0.048956	0.08257	1.172916
OID21086	HARS1	Neurology	0.5842	1.0251	1.11555	0.005753	0.012584	1.264135	0.013804	0.029604	1.184436
OID21087	PEBP1	Neurology	6.50525	7.2594	7.2492	6.47E-07	1.2E-05	1.628733	1.4E-06	2.71E-05	1.692786
OID21088	SERPINB1	Neurology	3.8225	4.73955	4.07145	0.000209	0.000758	2.045424	0.005425	0.013852	1.358675
OID21089	IL17RA	Neurology	0.9167	0.90915	0.89545	0.708736	0.764432	1.014768	0.94853	0.959011	1.000555
OID21090	CLEC14A	Neurology	1.67465	0.72265	0.8214	1.2E-06	1.64E-05	-1.77154	2.26E-06	3.69E-05	-1.79957
OID21091	GP6	Neurology	-1.6048	-1.6552	-1.20965	0.83251	0.866442	-1.05662	0.189296	0.258516	1.116513
OID21092	CNTN3	Neurology	0.3438	0.2318	0.20695	0.013719	0.026947	-1.13875	0.095476	0.146006	-1.04932
OID21093	INHBC	Neurology	-2.223	-3.65985	-2.94815	0.000373	0.001247	-2.38064	0.045161	0.077128	-1.656
OID21094	PIK3IP1	Neurology	5.41135	5.8989	5.95445	0.001566	0.004142	1.355805	0.003674	0.009912	1.478362
OID21095	NCAM2	Neurology	1.75265	0.24225	0.35615	1.93E-08	1.22E-06	-2.48423	4.16E-08	3.26E-06	-2.62551
OID21096	CA6	Neurology	-3.59335	-3.2149	-3.0576	0.865439	0.890583	1.299944	0.696589	0.753551	1.562547
OID21097	SPINK1	Neurology	3.05765	3.80745	3.85245	0.00136	0.003687	1.4081	0.000638	0.002414	1.414459
OID21098	SUSD2	Neurology	-0.24875	-0.01265	-0.06715	0.230857	0.297863	1.236318	0.095025	0.14552	1.184846
OID21099	MESD	Neurology	4.35025	3.42865	3.4863	0.000197	0.000727	-1.88524	0.000244	0.001115	-1.93657
OID21100	MPO	Neurology	-3.6103	-3.01795	-2.8635	0.001652	0.004309	1.678357	0.000825	0.002885	1.687632
OID21101	GUCA2A	Neurology	2.54175	2.5602	2.7167	0.736012	0.785591	1.012871	0.913992	0.933547	1.006397
OID21102	PIGR	Neurology	1.38645	1.5777	1.6335	0.128222	0.181662	1.200512	0.133812	0.194862	1.160945
OID21103	MMP9	Neurology	-2.04115	-0.3686	-0.76585	0.001106	0.003107	2.257381	0.000982	0.003358	1.89146
OID21104	THBS2	Neurology	2.34375	2.19095	2.29875	0.650817	0.710351	-1.10336	0.599539	0.664272	1.218917
OID21105	PLA2G7	Neurology	-2.6454	-3.66675	-3.61775	0.002769	0.006683	-2.06859	0.035973	0.063605	-2.11741
OID21106	NOMO1	Neurology	2.35645	1.11465	1.2859	1.8E-07	5.23E-06	-2.2061	1.96E-06	3.32E-05	-1.98618
OID21107	TNXB	Neurology	-0.0238	-0.70475	-0.43365	0.00116	0.003208	-1.38819	0.020562	0.040754	-1.25553
OID21108	CD109	Neurology	-0.41915	-0.29035	-0.3583	0.543783	0.614906	-1.0864	0.790276	0.834349	1.039615
OID21109	SPINT1	Neurology	2.06845	1.89205	2.0968	0.157416	0.216867	-1.11736	0.610329	0.673509	1.001596
OID21110	CD99	Neurology	2.688	3.28715	3.18955	0.000627	0.001951	1.218833	0.001034	0.003495	1.345507
OID21111	FCRL5	Neurology	-6.24025	-5.6193	-5.21155	0.02604	0.046566	2.256442	0.00111	0.003674	2.008544
OID21112	CLEC11A	Neurology	1.3528	-1.77955	-0.8648	3.35E-06	3.44E-05	-5.04533	0.000179	0.000869	-3.63965
OID21113	DSG2	Neurology	1.86035	2.1761	2.23565	0.109904	0.159622	1.14021	0.045861	0.078192	1.241126
OID21114	SOD2	Neurology	1.94785	2.10425	2.18135	0.032235	0.055477	1.213185	0.025452	0.048018	1.247725
OID21115	NUDT5	Neurology	4.13835	3.97865	3.9479	0.225963	0.292579	-1.01157	0.326102	0.40968	-1.12264
OID21116	IL1R1	Neurology	2.19465	1.6797	1.8388	3.81E-05	0.000183	-1.5132	0.001656	0.005165	-1.33284
OID21117	CNTN4	Neurology	4.81255	4.8968	4.87615	0.527727	0.599839	1.065293	0.448207	0.52522	1.059953
OID21118	TMSB10	Neurology	4.9713	6.09695	6.00765	2.42E-05	0.000131	2.005553	0.000178	0.000869	1.979999
OID21119	CD177	Neurology	-0.76575	-0.3837	-0.2216	0.156321	0.215628	1.242977	0.00748	0.017925	1.36822
OID21120	TREML2	Neurology	-2.16695	-1.69275	-1.85465	0.543013	0.614668	1.018291	0.273186	0.352064	1.169871
OID21121	MYOC	Neurology	8.2328	8.0403	8.2921	0.732236	0.783621	-1.06382	0.192315	0.261663	1.301567
OID21122	CD99L2	Neurology	4.65405	5.291	5.28035	0.000139	0.00054	1.642508	0.000464	0.001887	1.614626
OID21123	OGN	Neurology	3.52745	3.85375	4.1673	0.357993	0.431477	1.226672	0.042504	0.073379	1.327397
OID21124	PLAU	Neurology	2.17215	0.85165	1.31095	0.002914	0.006955	-1.94963	0.068719	0.109991	-1.6052
OID21125	EFNA1	Neurology	4.66345	5.0931	5.1235	4.52E-06	4.07E-05	1.426965	1.92E-06	3.32E-05	1.469882
OID21126	IL1RAP	Neurology	-0.2627	-0.37215	-0.1321	0.008608	0.017968	-1.07882	0.564067	0.63263	1.04888
OID21127	B4GAT1	Neurology	4.76285	5.9123	5.97875	8.44E-06	5.9E-05	2.611991	1.39E-05	0.000123	1.778795
OID21128	ALDH1A1	Neurology	0.55795	0.59305	1.25205	0.076042	0.116126	1.571944	0.055404	0.092312	1.655251
OID21129	PILRA	Neurology	3.4424	3.18275	3.38765	0.719653	0.773926	-1.12183	0.345705	0.42602	1.056945
OID21130	CD300LG	Neurology	-0.41125	0.0078	-0.00095	0.145027	0.202669	1.066179	0.009751	0.022167	1.184107
OID21131	PECAM1	Neurology	-0.2224	-0.0757	-0.12035	0.606443	0.669894	1.096306	0.49146	0.56505	1.033437
OID21132	GNLY	Neurology	-0.9498	-2.53615	-2.29505	0.0042	0.009627	-2.69279	0.008794	0.020372	-2.54074
OID21133	FCER2	Neurology	0.1392	0.0019	0.21715	0.825623	0.860906	-1.00598	0.049096	0.082681	1.088167
OID21134	MFGE8	Neurology	6.8164	6.6397	6.65385	0.3147	0.386499	1.24104	0.568513	0.635669	1.212009
OID21135	TXNRD1	Neurology	2.6228	2.70875	2.66905	0.008515	0.017808	1.170764	0.216242	0.287844	-1.00497
OID21136	IL7R	Neurology	-1.8892	-1.97365	-1.7751	0.076559	0.116752	-1.15926	0.832231	0.869448	-1.07177
OID21137	HAVCR2	Neurology	4.0934	3.23005	3.4331	0.000111	0.00045	-1.50822	0.00209	0.00617	-1.32726

Supplementary Table 6. (continued)

OID21138	ARSA	Neurology	5.12345	5.6496	5.74105	0.002246	0.005567	1.457504	0.002822	0.007926	1.815919
OID21139	ITGA5	Neurology	-3.28495	-3.18815	-3.07195	0.576377	0.642499	1.069362	0.30566	0.387546	1.039543
OID21140	FYB1	Neurology	-3.0707	-3.58945	-3.49915	0.429488	0.502749	-1.41593	0.520502	0.591015	-1.4082
OID21141	BLVRB	Neurology	0.12485	1.203	0.92355	0.001215	0.003341	1.888316	0.062555	0.102441	1.514037
OID21142	RELT	Neurology	7.03895	5.70565	6.0281	0.000583	0.001839	-2.15823	0.000697	0.002551	-1.60798
OID21143	DBI	Neurology	5.29325	5.8902	5.768	2.82E-05	0.000148	1.473707	0.000101	0.000531	1.372209
OID21144	VSIG4	Neurology	3.41215	4.32885	3.826	4.3E-05	0.000203	1.736943	0.025635	0.048033	1.319462
OID21145	TNFRSF1B	Neurology	2.91535	2.916	2.8639	0.562548	0.630693	1.025587	0.863677	0.894639	-1.05172
OID21146	IGF2R	Neurology	1.18715	0.85105	0.9559	0.001437	0.003848	-1.25671	0.023068	0.044618	-1.16078
OID21147	TIMP4	Neurology	2.85125	1.4661	1.56065	3.32E-05	0.000166	-2.57041	0.000186	0.00089	-2.46776
OID21148	SPINK5	Neurology	-0.68965	-0.30295	-0.16175	1.04E-05	7.12E-05	1.334963	2.92E-07	1.1E-05	1.504308
OID21149	CA2	Neurology	3.48325	3.47575	3.6376	0.11398	0.164239	1.464289	0.309302	0.39126	1.289311
OID21150	CTRB1	Neurology	0.59755	0.8892	1.25795	0.478521	0.553653	1.063706	0.00668	0.016333	1.35327
OID21151	F11R	Neurology	1.4677	1.58565	1.62055	0.254393	0.324418	-1.00274	0.230414	0.304812	1.112342
OID21152	LILRA2	Neurology	-0.91395	-1.03235	-1.11875	0.007508	0.015914	-1.23038	0.018159	0.036788	-1.12998
OID21153	PAMR1	Neurology	0.3813	-2.0812	-1.38275	3.89E-07	8.71E-06	-6.19584	1.14E-05	0.000111	-4.01862
OID21154	TFF1	Neurology	3.5362	4.50825	4.40095	0.002154	0.005403	1.365425	0.00258	0.007376	1.6383
OID21155	TNFRSF1A	Neurology	4.7799	4.68335	4.65215	0.444671	0.518311	1.053945	0.646129	0.706624	-1.0049
OID21156	CLPS	Neurology	-0.9214	-2.3324	-2.1661	1.12E-05	7.4E-05	-2.95806	5.95E-05	0.000359	-3.64066
OID21157	STC2	Neurology	3.3009	3.42855	3.51415	0.002174	0.005438	1.159216	0.001534	0.004839	1.241514
OID21158	IGFBP4	Neurology	4.7061	1.89495	2.812	0.001883	0.00483	-2.51411	0.012365	0.027252	-1.8569
OID21159	GRN	Neurology	1.89455	1.85315	2.08635	0.200584	0.26725	1.033687	0.00064	0.002415	1.03742
OID21160	PARK7	Neurology	4.2802	4.6736	4.7602	8.65E-05	0.000365	1.397308	0.000321	0.001393	1.502641
OID30420	DAND5	Inflammation	-0.2923	0.19855	0.59315	0.095776	0.141538	1.718143	0.026908	0.050053	2.470324
OID30430	FOLH1	Inflammation	-0.70795	-0.91495	-0.7263	0.789707	0.832148	-1.03552	0.504848	0.575621	-1.10995
OID30442	ST8SIA1	Inflammation	3.79095	2.4374	2.66795	1.79E-12	1.97E-09	-2.86126	9.72E-11	1.07E-07	-2.62133
OID30447	UNC5D	Inflammation	2.86345	2.03565	2.16325	3.47E-07	8.17E-06	-1.62586	4.05E-07	1.39E-05	-1.64103
OID30449	GAD2	Inflammation	0.80465	0.81605	1.01985	0.636857	0.700671	-1.01428	0.631477	0.693362	1.153805
OID30451	NME1	Inflammation	0.284	0.8236	0.7865	0.217665	0.285232	1.256012	0.053495	0.089403	2.071822
OID30454	FGF12	Inflammation	0.12685	0.1199	0.44285	0.409216	0.48262	-1.1113	0.127425	0.18655	1.113846
OID30460	PGR	Inflammation	-0.52765	0.196	0.26315	0.01681	0.031878	1.214995	0.06815	0.109239	1.954094
OID30464	KLRF1	Inflammation	-3.13895	-2.0554	-1.9909	0.024892	0.044733	1.549551	0.013199	0.028754	2.182528
OID30477	SMPD3	Inflammation	-0.35405	0.05195	0.24525	0.065793	0.102179	1.237947	0.009218	0.02109	1.615522
OID30479	TRAF3	Inflammation	-1.1907	-0.41135	-0.58115	0.0377	0.063295	1.799879	0.015107	0.031693	1.525783
OID30480	TERF1	Inflammation	1.5341	1.1723	1.06405	0.022417	0.040886	-1.13002	0.028819	0.052916	-1.33904
OID30481	HS6ST2	Inflammation	2.37125	1.5329	1.3656	1.13E-09	2.47E-07	-1.76804	2.58E-10	1.19E-07	-1.89343
OID30483	PRSS22	Inflammation	-1.32415	-1.28435	-1.0206	0.377591	0.453605	-1.0147	0.021101	0.041595	1.297739
OID30491	ADAM12	Inflammation	-1.57545	-1.1426	-0.97345	0.023339	0.042287	1.331405	0.005611	0.014196	1.839993
OID30509	NAGA	Inflammation	2.5785	2.15095	2.05935	0.001093	0.003093	-1.35336	0.001468	0.004687	-1.22846
OID30510	SLITRK1	Inflammation	-0.1413	-0.9791	-0.95345	3.84E-06	3.73E-05	-2.00555	4.82E-05	0.000308	-1.75583
OID30515	C1QL2	Inflammation	4.26365	3.16365	3.21715	1.44E-07	5.09E-06	-1.95051	1.59E-06	2.91E-05	-1.9288
OID30517	TYRP1	Inflammation	5.12255	4.61865	4.6991	0.000166	0.000629	-1.29769	0.001566	0.004914	-1.18079

Supplementary Table 6. (continued)

OID30525	BNIP3L	Inflammat ion_II	1.7138	1.445	1.1758	0.065063	0.101189	-1.17198	0.034987	0.062162	-1.282
OID30528	ADAMTS1	Inflammat ion_II	4.0651	2.7937	2.95045	0.000162	0.000618	-2.5295	0.023081	0.044618	-1.53677
OID30529	S100A13	Inflammat ion_II	0.0268	1.3065	1.1855	2.65E-05	0.000141	2.163849	3.54E-05	0.000246	2.237209
OID30531	SEMA6C	Inflammat ion_II	-1.2821	-0.87355	-0.81705	0.031957	0.055171	1.1776	0.092145	0.141701	1.229098
OID30536	STAT2	Inflammat ion_II	-2.3597	-1.77395	-1.24235	0.005076	0.011329	1.956602	0.004483	0.011719	1.973354
OID30538	MTDH	Inflammat ion_II	-3.1534	-1.07865	-1.8894	0.000309	0.001042	2.738527	0.054433	0.090833	2.071822
OID30539	ERMAP	Inflammat ion_II	-2.4715	-2.95535	-2.9907	0.052353	0.083673	-1.27235	0.188112	0.257219	-1.1319
OID30542	GIPR	Inflammat ion_II	-0.0836	0.0974	0.21455	0.566994	0.634617	1.1776	0.227688	0.30161	1.241126
OID30545	NECTIN1	Inflammat ion_II	0.82805	0.7915	0.6334	0.089752	0.133715	-1.1113	0.020251	0.040282	1.032971
OID30547	LMOD1	Inflammat ion_II	-2.29495	-1.8896	-1.604	0.110059	0.159637	1.331359	0.004809	0.012453	1.338299
OID30554	VEGFB	Inflammat ion_II	-1.2623	-1.7684	-1.8974	0.017653	0.033361	-1.45665	0.004346	0.011396	-1.47734
OID30559	TOP2B	Inflammat ion_II	-2.9385	-3.0256	-2.29305	0.949555	0.957402	-1.11069	0.441511	0.519036	1.594606
OID30564	PPL	Inflammat ion_II	-2.2201	-0.8408	-0.8741	0.001498	0.003983	1.728475	0.005128	0.013187	2.407774
OID30565	NXPH3	Inflammat ion_II	2.87555	2.3775	2.5267	0.000533	0.001712	-1.37817	0.003804	0.010213	-1.1911
OID30573	PDIA3	Inflammat ion_II	0.4403	-0.05475	0.2559	0.005877	0.012802	-1.2902	0.156565	0.220157	-1.2174
OID30574	GHR	Inflammat ion_II	-3.10665	-2.45985	-2.67785	0.004115	0.009453	1.651584	0.039421	0.068597	1.526471
OID30575	BMP10	Inflammat ion_II	-5.31525	-4.8891	-5.057	0.102263	0.149514	1.190692	0.49282	0.566021	1.065995
OID30576	CD300A	Inflammat ion_II	-0.8317	-0.88415	-0.88325	0.891994	0.915335	1.062785	0.504748	0.575621	1.11265
OID30578	ERP29	Inflammat ion_II	-1.51185	-0.96725	-0.63095	0.028194	0.04985	1.311302	0.004231	0.011141	1.786703
OID30579	HEG1	Inflammat ion_II	-2.133	-2.5309	-2.5118	0.000659	0.002028	-1.3169	0.000615	0.002359	-1.30387
OID30580	DTD1	Inflammat ion_II	1.1256	1.5006	1.5734	0.057898	0.090817	1.250712	0.01086	0.024286	1.478158
OID30581	CST1	Inflammat ion_II	-1.0227	-0.9302	-0.29025	0.637795	0.700999	1.088506	0.900723	0.92343	1.466523
OID30582	DIPK2B	Inflammat ion_II	-1.1255	-3.2034	-2.5197	6.84E-07	1.25E-05	-3.74444	3.32E-05	0.000234	-3.22746
OID30583	MXRA8	Inflammat ion_II	0.586	0.522	0.6385	0.790919	0.832626	-1.12479	0.342461	0.423448	1.065293
OID30585	ACHE	Inflammat ion_II	4.8647	4.95085	4.994	0.186401	0.250819	1.106152	0.56534	0.633411	1.026974
OID30586	JAM3	Inflammat ion_II	5.6882	5.4718	5.4862	0.146431	0.204037	-1.09684	0.064956	0.105505	-1.14211
OID30587	COL5A1	Inflammat ion_II	1.54525	1.01065	1.2183	0.018469	0.034785	-1.35238	0.134909	0.195939	-1.41451
OID30588	ADGRD1	Inflammat ion_II	-2.51155	-3.03585	-2.80235	0.002044	0.005195	-1.58058	0.015826	0.032725	-1.39581
OID30589	TNFRSF17	Inflammat ion_II	-2.94645	-3.31775	-3.0004	0.67954	0.735833	-1.25028	0.429655	0.506182	-1.00521
OID30590	APPL2	Inflammat ion_II	-2.3923	-2.43395	-2.22665	0.261956	0.333288	1.112997	0.178134	0.246336	1.124123
OID30591	PAXX	Inflammat ion_II	-3.3618	-4.3911	-4.27395	0.004643	0.010447	-2.03313	0.011532	0.025737	-1.71
OID30592	HMCN2	Inflammat ion_II	-5.9459	-6.1592	-6.509	4.38E-06	4.05E-05	-1.99868	7.36E-06	8.34E-05	-1.79683

Supplementary Table 6. (continued)

OID30593	CELSR2	Inflammat ion_II	2.46845	-0.3161	0.2005	0.00027	0.000948	-3.63751	0.001425	0.004563	-2.22075
OID30594	GLA	Inflammat ion_II	-0.0639	-0.82135	-0.70365	0.048532	0.078596	-1.55882	0.127105	0.186329	-1.1743
OID30595	MENT	Inflammat ion_II	-4.6658	-6.0769	-5.3456	5.19E-07	1.06E-05	-2.56641	0.000127	0.00064	-1.69484
OID30596	PSTPIP2	Inflammat ion_II	-4.0026	-3.89195	-3.92915	0.383424	0.459414	1.075383	0.176471	0.244652	1.008527
OID30597	GNPDA2	Inflammat ion_II	-1.7769	-0.97085	-1.18255	1.52E-05	9.32E-05	1.67702	0.000658	0.002467	1.293742
OID30598	MFAP4	Inflammat ion_II	2.1257	2.55275	2.59535	0.20851	0.275504	1.361078	0.195461	0.26547	1.545207
OID30599	SEMA3G	Inflammat ion_II	-1.22385	-1.96795	-1.56385	0.00162	0.004256	-1.49231	0.154109	0.218056	-1.17887
OID30600	GAPDH	Inflammat ion_II	-1.7166	-0.07715	-0.4381	0.037136	0.062539	2.731039	0.118342	0.176308	2.562408
OID30601	RLN2	Inflammat ion_II	1.04395	-2.76615	-2.18285	6.24E-06	5E-05	-10.8849	1.37E-05	0.000123	-7.75002
OID30602	DDI2	Inflammat ion_II	-3.828	-2.93965	-3.7199	0.024739	0.044531	1.31736	0.400394	0.479425	1.235547
OID30603	UROD	Inflammat ion_II	-2.17215	-0.61185	-0.72685	1.42E-05	8.87E-05	2.87398	8.31E-05	0.000473	2.270484
OID30604	SERPINI1	Inflammat ion_II	3.248	4.59785	4.51985	1.04E-05	7.12E-05	2.627972	2.67E-05	0.000198	2.664196
OID30605	YWHAQ	Inflammat ion_II	-1.2075	-0.00325	-0.42695	0.002071	0.005251	2.198927	0.016943	0.034709	1.595767
OID30606	CPA4	Inflammat ion_II	-1.28635	-1.29705	-1.21855	0.5781	0.643766	1.094256	0.239449	0.31225	1.066216
OID30607	PRR4	Inflammat ion_II	-1.12585	0.40625	0.43285	0.11023	0.159673	1.066586	0.025591	0.048033	1.954839
OID30608	LEG1	Inflammat ion_II	-4.2408	-3.92235	-4.04555	0.939982	0.949494	1.511153	0.617195	0.679038	1.021578
OID30609	TPD52L2	Inflammat ion_II	-4.3508	-4.3192	-4.44895	0.312798	0.385036	-1.03108	0.64717	0.707058	1.241126
OID30610	PCBD1	Inflammat ion_II	0.71285	1.32245	1.63485	0.000268	0.000948	1.527847	3.29E-05	0.000233	1.794087
OID30611	GP5	Inflammat ion_II	-6.2746	-5.5849	-5.7566	0.022857	0.041483	1.267952	0.043796	0.075373	1.323171
OID30613	DNAJB6	Inflammat ion_II	0.6605	0.67625	0.58885	0.752797	0.798619	1.048371	0.760991	0.809659	1.086132
OID30614	NHLRC3	Inflammat ion_II	3.5274	3.39745	3.39775	0.254364	0.324418	-1.09426	0.141829	0.203036	-1.03908
OID30615	BMPER	Inflammat ion_II	-0.34785	-1.15125	-0.97145	0.000192	0.000714	-1.71677	0.014527	0.030816	-1.38867
OID30616	TBCA	Inflammat ion_II	-0.05035	-0.29075	-0.46565	0.428642	0.502293	-1.03372	0.980356	0.984841	-1.26427
OID30617	ANXA1	Inflammat ion_II	-2.52555	0.08365	-0.20585	0.000151	0.000582	3.340815	0.00059	0.002307	5.770315
OID30618	C1QTNF5	Inflammat ion_II	1.86805	1.4496	1.62705	0.044327	0.07308	-1.4343	0.333524	0.416621	-1.05172
OID30619	CRELD1	Inflammat ion_II	-1.5074	-3.20205	-3.0408	1.94E-09	2.67E-07	-2.77233	9.25E-10	2.03E-07	-3.29825
OID30620	C1QTNF9	Inflammat ion_II	-3.36615	-3.50515	-3.3717	0.28432	0.355967	-1.04236	0.116281	0.17371	1.03724
OID30622	EPHA4	Inflammat ion_II	-0.68565	-1.8448	-1.78825	5.14E-07	1.06E-05	-2.13909	3.71E-07	1.32E-05	-2.36723
OID30623	GMPR2	Inflammat ion_II	-0.2317	1.7519	1.02535	3.38E-05	0.000168	4.061464	0.00115	0.003793	2.638833
OID30624	ST13	Inflammat ion_II	-4.42365	-3.79235	-4.1649	0.026805	0.047701	1.383526	0.059941	0.098821	1.403812
OID30625	INHBB	Inflammat ion_II	-2.8684	-3.7201	-3.80015	0.061027	0.095589	-1.66238	0.022178	0.043254	-1.8501
OID30626	FUOM	Inflammat ion_II	-4.41175	-4.0629	-3.9359	0.09538	0.141143	1.153526	0.006052	0.015104	1.437289

Supplementary Table 6. (continued)

OID30627	SPINK2	Inflammat ion_II	-0.9846	-1.65065	-1.39565	0.000843	0.002502	-1.5481	0.006433	0.015908	-1.41363
OID30628	CTSE	Inflammat ion_II	-2.6738	-2.8525	-2.90475	0.184372	0.248698	-1.24151	0.950017	0.959631	-1.02958
OID30630	SLC9A3R1	Inflammat ion_II	-1.17385	-1.4244	-1.73535	0.274914	0.345372	-1.62119	0.045167	0.077128	-1.56417
OID30631	GCHFR	Inflammat ion_II	0.6734	1.49945	1.7019	1.49E-06	1.95E-05	2.046913	8.6E-05	0.000487	1.898618
OID30632	TP53I3	Inflammat ion_II	-1.32615	1.3898	0.8212	1.52E-06	1.95E-05	3.480151	1.37E-05	0.000123	3.488363
OID30633	MYOM3	Inflammat ion_II	-5.2204	-4.64555	-4.59795	0.018695	0.035149	1.423162	0.021276	0.041866	1.446283
OID30634	PHYKPL	Inflammat ion_II	-1.83775	-1.02315	-1.1767	0.000106	0.000434	1.769081	0.001404	0.004507	1.639209
OID30635	RIDA	Inflammat ion_II	0.91495	0.58595	0.6407	0.000347	0.001168	-1.12553	0.002075	0.006141	-1.25666
OID30638	MOCS2	Inflammat ion_II	-0.6614	-0.71235	-0.5809	0.650831	0.710351	-1.05522	0.355585	0.436237	-1.02605
OID30639	GIMAP7	Inflammat ion_II	-2.2948	-2.3526	-2.10265	0.643076	0.705687	-1.09293	0.997632	0.998161	1.136738
OID30640	VT11A	Inflammat ion_II	2.14825	1.61045	1.57115	3.95E-05	0.000188	-1.40644	2.03E-06	3.37E-05	-1.57778
OID30641	CD72	Inflammat ion_II	-3.32355	-3.69875	-3.098	0.863383	0.889301	-1.1113	0.033329	0.059602	1.238935
OID30642	ACYP1	Inflammat ion_II	1.9826	2.10395	2.11665	0.180699	0.244948	1.248504	0.333031	0.416478	1.154846
OID30643	NAGPA	Inflammat ion_II	-2.83825	-3.311	-3.26705	0.003556	0.008324	-1.30554	0.008011	0.018956	-1.16663
OID30646	FGFR4	Inflammat ion_II	-3.17065	-3.8324	-3.7284	3.78E-06	3.7E-05	-1.69144	0.00032	0.001393	-1.35079
OID30647	RABEP1	Inflammat ion_II	-1.35185	-1.2563	-0.86105	0.14508	0.202669	1.176621	0.024364	0.046363	1.528747
OID30648	SFRP4	Inflammat ion_II	0.07465	-0.0299	0.0078	0.480557	0.555047	1.035229	0.666774	0.725587	-1.02562
OID30649	CACYBP	Inflammat ion_II	-2.4691	-2.27205	-2.2959	0.223433	0.290675	1.006432	0.773893	0.82259	1.033401
OID30650	SUSD5	Inflammat ion_II	0.6541	-0.70585	-0.63335	6.08E-07	1.17E-05	-2.62733	5.12E-07	1.56E-05	-2.31362
OID30652	KLK7	Inflammat ion_II	-2.4771	-1.5163	-1.4367	0.002954	0.007035	1.732313	0.003058	0.008521	2.002011
OID30653	ARHGAP4 5	Inflammat ion_II	-1.79	-1.21735	-1.4156	0.117029	0.16819	1.1776	0.408562	0.48635	1.223234
OID30654	UBXN1	Inflammat ion_II	-1.7617	-0.80255	-1.10655	0.000676	0.002055	1.754123	0.002507	0.007207	2.0114
OID30655	ADD1	Inflammat ion_II	-2.45525	-2.28915	-2.4715	0.815771	0.852251	1.183041	0.899403	0.923063	1.097979
OID30656	SPART	Inflammat ion_II	-0.27905	-1.12235	-0.9867	0.000579	0.001832	-1.6517	0.00922	0.02109	-1.67446
OID30658	MARS1	Inflammat ion_II	-0.3586	-0.69015	-0.6528	0.574095	0.641258	-1.1113	0.361793	0.442527	-1.1414
OID30659	PTRHD1	Inflammat ion_II	-1.77155	1.05175	0.74015	7.61E-06	5.64E-05	6.359662	0.000722	0.002624	3.198953
OID30660	CHAD	Inflammat ion_II	-2.7962	-1.8228	-1.58495	0.00063	0.001954	1.872739	1.14E-05	0.000111	2.168955
OID30661	DNAJB2	Inflammat ion_II	-0.05225	-0.6054	-0.6089	0.014567	0.028259	-1.39566	0.007097	0.017241	-1.36084
OID30662	SDK2	Inflammat ion_II	-0.9598	-1.19765	-1.46515	0.079061	0.119571	-1.22137	0.026889	0.050053	-1.35858
OID30665	OLFM4	Inflammat ion_II	-2.1912	-1.80235	-2.08985	0.128395	0.181673	1.216722	0.363357	0.443789	-1.03491
OID30666	PENK	Inflammat ion_II	3.20685	1.2219	1.07115	1.11E-05	7.4E-05	-2.63381	3.72E-05	0.000254	-2.72225
OID30668	PRDX2	Inflammat ion_II	0.7968	0.55175	0.45945	0.425572	0.499762	-1.14231	0.965551	0.972638	-1.14878

Supplementary Table 6. (continued)

OID30669	MDH1	Inflammat ion_II	7.30185	7.42265	7.37115	0.248621	0.318166	1.158935	0.513478	0.583642	1.064075
OID30670	TXN	Inflammat ion_II	5.01285	5.6418	5.2276	0.000126	0.000502	1.760945	0.021343	0.041922	1.181566
OID30671	CGB3_CG B5_CGB8	Inflammat ion_II	4.51605	2.53785	2.74135	0.007315	0.015626	-1.8551	0.00041	0.001717	-2.27995
OID30672	RNASE1	Inflammat ion_II	3.0689	5.29865	4.97265	3.52E-06	3.51E-05	4.019873	1.22E-05	0.000117	4.115592
OID30674	NRGN	Inflammat ion_II	1.00265	1.3865	1.03975	0.085865	0.12858	1.028648	0.567522	0.635209	1.016845
OID30675	ABO	Inflammat ion_II	3.50005	3.0889	3.0667	0.169536	0.231819	-1.35411	0.064465	0.104863	-1.41025
OID30676	PI16	Inflammat ion_II	1.23595	2.0206	1.9862	4.46E-06	4.05E-05	1.911096	1.26E-05	0.000119	1.842545
OID30677	LCAT	Inflammat ion_II	0.2083	-0.48175	-0.3401	0.000441	0.00145	-1.50927	0.013627	0.029454	-1.26313
OID30678	NEXN	Inflammat ion_II	0.5921	1.30425	1.14845	0.00027	0.000948	1.846445	0.001033	0.003495	1.644558
OID30679	ASGR2	Inflammat ion_II	2.66535	1.28075	1.30705	1.65E-07	5.23E-06	-2.69746	7.61E-05	0.00044	-2.07527
OID30680	GPI	Inflammat ion_II	6.2055	6.6561	6.4781	6.93E-05	0.000302	1.383814	0.005454	0.013894	1.448239
OID30681	CSF1R	Inflammat ion_II	4.2377	4.6164	4.59625	0.021652	0.039889	1.240696	0.112731	0.169327	1.209156
OID30683	SOD3	Inflammat ion_II	4.4333	5.9661	6.00705	7.03E-07	1.25E-05	3.316818	1.26E-06	2.51E-05	3.622034
OID30684	F10	Inflammat ion_II	1.3661	1.6038	1.56785	0.921974	0.933881	1.092513	0.336103	0.418414	1.174503
OID30685	SHBG	Inflammat ion_II	1.7073	1.8606	1.70365	0.054314	0.085809	-1.27143	0.391942	0.470845	1.073856
OID30686	APOA2	Inflammat ion_II	-2.9282	-2.69755	-2.865	0.492363	0.566089	-1.08798	0.781844	0.827878	1.034404
OID30687	TCN1	Inflammat ion_II	5.0403	4.82625	4.7858	0.40707	0.481409	1.002672	0.395867	0.474522	-1.10168
OID30688	PEPD	Inflammat ion_II	3.27965	3.53645	3.6937	0.085833	0.12858	1.273633	0.07592	0.119943	1.143138
OID30689	RNASE6	Inflammat ion_II	8.2307	8.66345	8.93555	2.66E-06	2.86E-05	1.424346	1.41E-06	2.71E-05	1.529966
OID30690	FCN1	Inflammat ion_II	-3.37545	-4.29525	-3.04155	0.648703	0.709939	-2.17943	0.250663	0.325713	-1.28886
OID30691	TGOLN2	Inflammat ion_II	4.33885	4.96815	4.8483	0.000276	0.00096	1.602862	0.001359	0.00439	1.689504
OID30692	VASP	Inflammat ion_II	-0.26015	0.0077	-0.06455	0.046312	0.075791	1.1776	0.060309	0.09928	1.011783
OID30693	PNLIP	Inflammat ion_II	2.9978	2.3188	2.53925	0.000538	0.001722	-1.60103	0.048741	0.082334	-1.2485
OID30694	MRC1	Inflammat ion_II	3.218	3.4426	3.4552	0.016212	0.031011	1.405857	0.024485	0.046514	1.16538
OID30696	ACE	Inflammat ion_II	3.8301	3.8656	3.8009	0.973669	0.976336	-1.00166	0.983867	0.987464	-1.00692
OID30697	CR1	Inflammat ion_II	0.6946	0.67095	0.74275	0.554059	0.623317	-1.06925	0.383168	0.462328	1.078181
OID30698	C9	Inflammat ion_II	1.595	2.0421	2.6594	0.505953	0.579287	1.413675	0.01384	0.029622	1.687632
OID30699	PGA4	Inflammat ion_II	4.37665	4.12005	4.4284	0.191221	0.256363	-1.14572	0.26632	0.344023	1.008178
OID30700	GSR	Inflammat ion_II	3.7806	4.4051	4.45575	2.86E-05	0.000149	1.671334	0.003156	0.008751	1.399877
OID30701	APOF	Inflammat ion_II	-1.2824	-2.7079	-2.1926	0.009325	0.019103	-1.47903	0.321839	0.405716	-1.21008
OID30702	FGL1	Inflammat ion_II	-2.97685	-2.9969	-3.23845	0.390925	0.46656	-1.19823	0.256811	0.332522	-1.16866
OID30703	DBH	Inflammat ion_II	-0.2963	0.54545	0.8353	0.000686	0.002075	1.819006	0.000309	0.001369	1.724765

Supplementary Table 6. (continued)

OID30704	PON1	Inflammation_II	-0.17125	-0.2547	-0.2148	0.857083	0.885303	1.011468	0.183122	0.252281	1.110878
OID30705	RNASE4	Inflammation_II	4.0091	4.14805	4.38535	0.592314	0.657594	1.134652	0.489146	0.564755	1.297334
OID30706	AHSG	Inflammation_II	0.5989	1.45195	1.3854	0.01562	0.030036	1.700725	0.01603	0.033085	1.724885
OID30707	AMY1A_A MY1B_A MY1C	Inflammation_II	4.34595	4.50845	4.44505	0.488035	0.56229	-1.04073	0.490185	0.564767	1.053325
OID30708	APOL1	Inflammation_II	-4.21585	-4.7216	-4.737	0.000141	0.000544	-1.85221	0.024323	0.046363	-1.43779
OID30709	AGT	Inflammation_II	0.51455	1.06315	1.26485	0.015035	0.029014	1.309395	0.009131	0.021018	1.475905
OID30710	CTBS	Inflammation_II	4.9689	5.45765	5.72825	0.000306	0.001036	1.628507	0.000689	0.002547	1.866842
OID30711	VNN1	Inflammation_II	2.1588	2.6093	2.5854	0.046579	0.076107	1.269579	0.007069	0.01721	1.482621
OID30712	CAT	Inflammation_II	0.4906	1.2862	1.102	0.012641	0.025329	1.573907	0.110966	0.166904	1.3764
OID30713	MST1	Inflammation_II	1.67685	1.5732	1.80795	0.323432	0.394587	-1.07449	0.843679	0.879733	1.117248
OID30714	GSN	Inflammation_II	1.3843	1.20555	1.3051	0.516259	0.589244	-1.10623	0.539346	0.611147	1.005212
OID30715	CRISP3	Inflammation_II	0.69045	1.6452	1.6972	0.014626	0.028324	1.741765	0.067773	0.108794	2.217448
OID30716	CFHR5	Inflammation_II	-1.19015	-2.2878	-2.0146	0.000241	0.000866	-2.53178	0.001859	0.00567	-2.23101
OID30717	BCHE	Inflammation_II	1.8851	1.70915	1.9003	0.655178	0.712969	-1.0279	0.836074	0.872633	1.005038
OID30718	HGFAC	Inflammation_II	0.8557	1.39775	1.33855	0.000304	0.001036	1.296435	9.14E-05	0.000503	1.397502
OID30719	C8B	Inflammation_II	4.92	5.7457	5.86875	0.046222	0.075791	1.523881	0.018305	0.036947	1.617146
OID30720	CFHR2	Inflammation_II	8.1911	8.94265	8.7562	0.000956	0.002771	1.442779	0.001207	0.003968	1.646954
OID30721	C1RL	Inflammation_II	7.5192	7.8596	8.0802	0.744873	0.793278	1.201678	0.130473	0.190528	1.210791
OID30722	ITIH1	Inflammation_II	5.65505	6.25015	6.27345	3.31E-05	0.000166	1.641085	1.2E-05	0.000116	1.661746
OID30723	F2	Inflammation_II	2.95335	3.9319	4.04825	4.11E-05	0.000195	1.970484	3.95E-05	0.000263	1.96149
OID30724	SERPINA7	Inflammation_II	7.4142	8.39515	8.60835	8.37E-06	5.89E-05	1.91926	3.13E-05	0.000223	2.05281
OID30725	B2M	Inflammation_II	6.27155	7.67055	7.44825	7.69E-05	0.000332	2.911273	0.00062	0.002373	2.947312
OID30726	SAA4	Inflammation_II	6.1123	6.726	6.8927	3.42E-05	0.000169	1.523353	1.96E-06	3.32E-05	1.736401
OID30727	APOE	Inflammation_II	5.20535	7.36205	7.2309	7.95E-07	1.28E-05	4.13432	2.35E-06	3.69E-05	3.966867
OID30728	LYVE1	Inflammation_II	7.34655	7.96915	8.13255	9.84E-05	0.000406	1.590136	0.000117	0.000602	1.818186
OID30729	QSOX1	Inflammation_II	7.6191	8.3032	8.46025	0.020106	0.03759	1.619951	0.00596	0.014941	1.515612
OID30730	PZP	Inflammation_II	2.48655	2.95125	3.17315	0.00168	0.004372	1.737003	0.000126	0.000636	2.047197
OID30731	C1S	Inflammation_II	6.2831	7.50435	7.5971	1.67E-06	2.06E-05	2.375284	7.76E-06	8.6E-05	2.709262
OID30732	CLU	Inflammation_II	3.92495	5.58665	5.8021	5.3E-06	4.62E-05	3.171466	3.82E-05	0.000258	3.479307
OID30733	APOA4	Inflammation_II	5.65425	6.0662	6.1224	0.380873	0.457048	1.291323	0.059067	0.097821	1.249672
OID30734	CPB2	Inflammation_II	6.30845	7.54815	7.66805	7.89E-07	1.28E-05	2.182906	2.85E-06	4.23E-05	2.499605

Supplementary Table 6. (continued)

OID30735	ATRN	Inflammat ion_II	7.39785	8.0551	8.08395	0.000654	0.002016	1.597427	0.001765	0.005443	1.802876
OID30736	ECM1	Inflammat ion_II	7.0941	7.33225	7.5877	0.316098	0.386929	1.227183	0.164975	0.230168	1.219255
OID30737	PROS1	Inflammat ion_II	6.86255	7.35735	7.4334	0.388601	0.464291	1.278631	0.130489	0.190528	1.34048
OID30738	CFD	Inflammat ion_II	7.31725	8.55935	8.64215	5.36E-06	4.63E-05	2.449437	2.17E-05	0.00017	2.668816
OID30739	F12	Inflammat ion_II	6.36115	7.12405	7.27005	0.000303	0.001036	1.670697	0.0002	0.000954	1.574125
OID30740	IGLC2	Inflammat ion_II	6.31975	7.38455	7.36605	7.88E-06	5.72E-05	1.864708	1.58E-05	0.000135	2.065226
OID30741	AFM	Inflammat ion_II	5.72645	6.59335	6.79375	8.09E-05	0.000346	1.860577	9E-05	0.000502	2.117556
OID30742	PGLYRP2	Inflammat ion_II	3.62295	4.52955	4.54915	0.002306	0.00569	1.889298	0.001041	0.003506	2.097761
OID30743	SERPINA6	Inflammat ion_II	4.74265	5.92315	5.86445	1.72E-05	0.000101	2.326966	6.2E-05	0.000372	2.490612
OID30744	C5	Inflammat ion_II	5.1532	5.68205	5.80695	0.000117	0.000471	1.596209	2.31E-05	0.000177	1.594053
OID30745	SELENOP	Inflammat ion_II	6.90525	6.8029	7.00985	0.50141	0.57544	-1.00993	0.889331	0.917749	1.153326
OID30746	APOD	Inflammat ion_II	5.59645	6.33935	6.54765	8.29E-05	0.000353	1.674929	0.000344	0.001474	1.814787
OID30747	LPA	Inflammat ion_II	-1.47705	-1.04705	-0.57475	0.012902	0.025695	1.284538	0.007357	0.017754	1.588538
OID30748	ADIPOQ	Inflammat ion_II	4.344	5.0964	5.45835	8.16E-06	5.78E-05	1.783672	5.36E-06	6.54E-05	1.823361
OID30749	APOC1	Inflammat ion_II	2.21105	3.1054	3.12075	0.000551	0.001754	1.750843	9.61E-05	0.000517	1.878655
OID30750	SERPINA1	Inflammat ion_II	-0.0077	1.32345	1.30585	3.41E-06	3.44E-05	2.485352	2.11E-05	0.000167	2.57273
OID30751	LRG1	Inflammat ion_II	2.30225	3.63535	3.64675	1.92E-05	0.000111	2.607378	0.000117	0.000602	2.501598
OID30752	SELL	Inflammat ion_II	6.78055	7.85255	7.91785	8.4E-07	1.3E-05	1.931404	1.55E-06	2.89E-05	2.239536
OID30753	C1R	Inflammat ion_II	5.29925	6.58575	6.6755	6.06E-06	4.93E-05	2.365754	2.77E-05	0.000203	2.846522
OID30754	BTD	Inflammat ion_II	4.3173	5.95055	5.87605	2.53E-06	2.83E-05	3.267755	1.06E-05	0.000105	3.401332
OID30755	CFB	Inflammat ion_II	4.82225	6.15725	6.26655	2.6E-07	6.79E-06	2.434373	6.44E-07	1.86E-05	2.616068
OID30756	CLEC3B	Inflammat ion_II	6.5119	7.21905	7.24675	0.008704	0.018135	1.557789	0.008374	0.019688	1.669019
OID30757	SERPINF1	Inflammat ion_II	3.7767	5.11	4.90995	1.33E-05	8.48E-05	2.557085	5.81E-05	0.000353	2.562941
OID30758	CFHR4	Inflammat ion_II	5.10995	4.3681	4.67905	0.00614	0.013271	-1.41172	0.044613	0.076539	-1.15081
OID30759	MBL2	Inflammat ion_II	5.94885	6.37395	6.43645	0.005216	0.011594	1.394889	0.029486	0.05405	1.450651
OID30760	KLKB1	Inflammat ion_II	4.83675	5.72045	5.85705	1.65E-06	2.06E-05	1.663532	1.19E-06	2.48E-05	1.90356
OID30761	PPBP	Inflammat ion_II	5.87045	5.74015	5.82645	0.181444	0.245654	-1.43998	0.289659	0.37025	-1.31185
OID30762	ITIH4	Inflammat ion_II	3.4822	3.93215	4.0339	0.031308	0.054221	1.253186	0.020044	0.04008	1.26659
OID30763	SERPINA5	Inflammat ion_II	6.1802	7.35875	7.43585	5.53E-06	4.74E-05	2.24848	7.29E-06	8.34E-05	2.458196
OID30764	PF4	Inflammat ion_II	-0.4101	-0.55	-0.50355	0.956014	0.962148	-1.47928	0.938398	0.952274	-1.1103
OID30765	LGALS3BP	Inflammat ion_II	4.9376	6.47555	6.55705	9.35E-06	6.5E-05	2.337149	1.87E-05	0.000151	2.773101
OID30766	C7	Inflammat ion_II	6.30895	7.27745	7.41375	1.56E-05	9.5E-05	1.831658	1.63E-05	0.000137	1.950576

Supplementary Table 6. (continued)

OID30767	SERPIND1	Inflammat ion_II	3.7737	4.1817	4.32445	0.068716	0.10597	1.549551	0.008166	0.019241	1.331682
OID30768	ORM1	Inflammat ion_II	0.07575	1.70595	1.66105	6.73E-06	5.2E-05	3.078548	3.65E-05	0.000252	2.702791
OID30769	APOA1	Inflammat ion_II	2.6766	3.84175	3.9129	1.7E-05	0.000101	2.085148	4.33E-05	0.000285	2.344695
OID30770	HRG	Inflammat ion_II	3.53935	5.07115	5.3094	8.02E-06	5.75E-05	2.358468	1.54E-05	0.000134	2.584885
OID30771	A1BG	Inflammat ion_II	3.7678	4.77065	4.87535	1.75E-05	0.000102	2.231633	6.33E-05	0.000378	2.044503
OID30772	CFI	Inflammat ion_II	5.53665	6.6334	6.58405	8.62E-05	0.000365	2.273713	0.000184	0.000888	2.066802
OID30773	F11	Inflammat ion_II	5.84805	6.73465	6.8758	4.89E-07	1.03E-05	1.758567	3.01E-08	3.19E-06	1.965232
OID30774	SERPINA4	Inflammat ion_II	3.45425	5.1965	5.01595	2.2E-06	2.58E-05	3.124553	1.93E-05	0.000155	3.172895
OID30775	CD5L	Inflammat ion_II	4.74135	5.0994	5.30765	0.071876	0.11038	1.351068	0.124371	0.183548	1.460639
OID30776	C3	Inflammat ion_II	1.7051	2.86045	2.7215	9.11E-05	0.000382	2.303456	0.000595	0.002309	2.022865
OID30777	SERPING1	Inflammat ion_II	0.0637	1.66765	1.73435	2.02E-05	0.000115	2.868109	9.64E-05	0.000517	3.127912
OID30778	APCS	Inflammat ion_II	2.66155	1.71635	2.546	0.053081	0.084345	-1.22837	0.332374	0.416131	-1.23478
OID30779	SERPINF2	Inflammat ion_II	5.29035	6.6359	6.71025	1.51E-05	9.32E-05	2.452495	4.13E-05	0.000273	2.492338
OID30780	GC	Inflammat ion_II	-0.15535	1.48365	1.2456	9.91E-06	6.84E-05	2.908247	8.01E-05	0.000461	2.972033
OID30781	F13B	Inflammat ion_II	4.72805	3.73305	4.41325	0.002704	0.006539	-1.9373	0.025758	0.048182	-1.26396
OID30782	TTR	Inflammat ion_II	-0.30215	1.86985	1.6858	3.22E-06	3.33E-05	4.081643	5.32E-06	6.54E-05	4.157453
OID30784	CFP	Inflammat ion_II	3.85925	4.77815	4.74225	5.88E-08	2.93E-06	1.643589	3.2E-08	3.19E-06	1.800565
OID30785	PLG	Inflammat ion_II	2.51485	4.09275	4.08145	1.63E-06	2.06E-05	2.605481	1.37E-05	0.000123	2.678082
OID30786	SERPINA3	Inflammat ion_II	0.33555	1.94225	1.81845	1.45E-05	9.02E-05	2.807819	4.91E-05	0.000312	2.702323
OID30787	FN1	Inflammat ion_II	2.3007	3.4611	3.54865	2.92E-05	0.000149	2.489662	2.51E-05	0.000189	2.431337
OID30788	FGA	Inflammat ion_II	2.9047	1.5603	2.0004	0.002138	0.005397	-1.58058	0.062688	0.102441	-1.87164
OID30789	SERPINC1	Inflammat ion_II	0.44765	2.1224	1.81175	4.46E-06	4.05E-05	3.19264	3.73E-05	0.000254	3.273196
OID30790	CFH	Inflammat ion_II	4.36005	5.42175	5.40985	1.95E-05	0.000112	2.269147	7.44E-05	0.000432	2.146297

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Reviewer #1 (Public Review):

Summary:

This study analyzed biomarker data from 28 subjects with geographic atrophy (GA) in a Phase I/II clinical trial of PPY988, a subretinal AAV2 complement factor I (CFI) gene therapy, to evaluate pharmacokinetics and pharmacodynamics. Post-treatment, a 2-fold increase in the vitreous humor (VH) FI was observed, correlating with a reduction in FB breakdown product Ba but minimal changes in other complement factors. The aqueous humor (AH) was found to be an unreliable proxy for VH in assessing complement activation. In vitro assays showed that the increase in FI had a minor effect on the complement amplification loop compared to the more potent C3 inhibitor pegcetacoplan. These findings suggest that PPY988 may not provide enough FI protein to effectively modulate complement activation and slow GA progression, highlighting the need for a thorough biomarker review to determine optimal dosing in future studies.

Strengths:

This manuscript provides critical data on the efficacy of gene therapy for the eye, specifically introducing complement FI expression. It presents the results from a halted clinical trial, making sharing this data essential for understanding the outcomes of this gene therapy approach. The findings offer valuable insights and lessons for future gene therapy attempts in similar contexts.

Weaknesses:

No particular weaknesses. The study was carefully performed and limitations are discussed.

I have just some concerns about the methodology used. The authors use the MILLIPLEX assays, which allow for multiplexed detection of complement proteins and they mention extensive validation. How are the measurements with this assay correlating with gold standard methods? Is the specificity and the expected normal ranges preserved with this assay? This also stands for the Olink assay. Some of the proteins are measured by both assay and/or by standard ELISA. How do these measurements correlate?

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Reviewer #2 (Public Review):

Summary:

The results presented demonstrate that AAV2-CFI gene therapy delivers long-term and marginally higher FI protein in vitreous humor that results in a concomitant reduction in the FB activation product Ba. However, the lack of clinical efficacy in the phase I/II study, possibly due to lower in vitro potency when compared to currently approved pegcetacoplan, raises important considerations for the utility of this therapeutic approach. Despite the early termination of the PPY988 clinical development program, the study achieved significant milestones, including the implementation of subretinal gene therapy delivery in older adults, complement biomarker comparison between serial vitreous humor and aqueous humor samples and vitreous humor proteomic assessment via Olink.

Strengths:

Long-term augmentation of FI protein in vitreous humor over 96 weeks and reduction of FB breakdown product Ba in vitreous humor suggests modulation of the complement system. Developed a novel in vitro assay suggesting FI's ability to reduce C3 convertase activity is weaker than pegcetacoplan and FH and may suggest a higher dose of FI will be required for clinical efficacy. Warn of the poor correlation between vitreous humor and aqueous humor biomarkers and suggest aqueous humor may not be a reliable proxy for vitreous humor with regard to complement activation/inhibition studies.

Weaknesses:

The vitrectomy required for the subretinal route of administration causes a long-term loss of total protein and may influence the interpretation of complement biomarker results even with normalization. The modified in vitro assay of complement activation suggests a several hundred-fold increase in FI protein is required to significantly affect C3a levels. Interestingly, the in vitro assay demonstrates 100% inhibition of C3a with pegcetacoplan and FH therapeutics, but only a 50% reduction with FI even at the highest concentrations tested. This observation suggests FI may not be rate-limiting for negative complement regulation under the in vitro conditions tested and potentially in the eye. It is unclear if pharmacokinetic and pharmacodynamic properties in aqueous humor and vitreous humor compartments are reliable predictors of FI level/activity after subretinal delivery AAV2-CFI gene therapy.

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Reviewer #3 (Public Review):**Summary:**

The manuscript by Hallam et al describes the analysis of various biomarkers in patients undergoing complement factor I supplementation treatment (PPY988 gene therapy) as part of the FOCUS Phase I/II clinical trial. The authors used validated methods (multiplexed assays and OLINK proteomics) for measuring multiple soluble complement proteins in the aqueous humour (AH) and vitreous humour (VH) of 28 patients over a series of time points, up to and including 96 weeks. Based on biomarker comparisons, the levels of FI synthesised by PPY988 were believed to be insufficient to achieve the desired level of complement inhibition. Subsequent comparative experiments showed that PPY988-delivered FI was much less efficacious than Pegceptacoplan (FDA-approved complement inhibitor under the name SYFORVE) when tested in an artificial VH matrix.

Strengths:

The manuscript is well written with data clearly presented and appropriate statistics used for the analysis itself. It's great to see data from real clinical samples that can help support future studies and therapeutic design. The identification that complement biomarker levels present in the AH do not represent the levels found in the VH is an important finding for the field, given the number of complement-targeting therapies in development and the desperate need for good biomarkers for target engagement. This study also provides a wealth of baseline complement protein measurements in both human AH and VH (and companion measurements in plasma) that will prove useful for future studies.

Weaknesses:

Perhaps the conclusions drawn regarding the lack of observed efficacy are not fully justified. The authors focus on the hypothesis that not enough FI was synthesised in these patients receiving the PPY988 gene therapy, suggesting a delivery/transduction/expression issue. But beyond rare CFI genetic variants, most genetic associations with AMD imply that it is a FI-cofactor disease. A hypothesis supported by the authors' own experiments when they supplement their artificial VH matrix with FH and achieve a significantly greater breakdown of C3b than achieved with PPY988 treatment alone. Justification around doubling FI levels driving complement turnover refers to studies conducted in blood, which has an entirely different complement protein profile than VH. In Supplemental Table 5 we see there is approx. 10-fold more FH than FI (533ug/ml vs 50ug/ml respectively) so increasing FI levels will have a direct effect. Yet in Supplemental Table 3 we see there is more FI than FH in VH (608ng/ml vs 466ng/ml respectively). Therefore, adding more FI without more co-factors would have a very limited effect. Surely this demonstrates that the study was delivering the wrong payload, i.e. FI, which hit a natural ceiling of endogenous co-factors within the eye?

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